

**URBAN INFRASTRUCTURE SECURITY FOCUSING
EARTHQUAKE: BANGLADESH PERSPECTIVE**



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03 August 2026

NATIONAL DEFENCE COLLEGE

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DEDICATED

To

My Parents, My Supervisor, Liberation War Freedom Fighters and My Family

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ALL PRAISES ARE OF ALMIGHTY ALLAH (SUBHANAHU WA TA'ALA). My sincere gratitude to ALMIGHTY ALLAH (SUBHANAHU WA TA'ALA) for being able to complete this research work effectively.

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I remember Corporal Mozazzem, National Defence college and Lance Corporal Mithu, Padma Bridge Rail Link Project.

ABSTRACT

Urban infrastructure mainly include airport, building, bridges, electricity, gas supply line, roads, water supply line etc. Earthquake resistant building structures are analyzed as per Bangladesh National Building Code. Electric, gas and water once installed inside a building, the combined analysis remain absent. Therefore, the cascading effect remains once gas or electricity disrupt. Base isolation allows the building to move more gently during seismic shaking by separating the structure from the ground motion. Electric, gas, and water systems in base isolation are safer during earthquakes because the building moves more gently. Urban infrastructure security focusing on earthquakes includes proactive measures such as educating people on earthquake resistant building, electricity, gas, water, properly designed earthquake resistant structure having Dynamic time history analysis, the structural retrofitting of buildings, soil-structure interaction and liquefaction mitigation, and earthquake resilient utility networks. It also includes structural innovations like base isolation.

Therefore, in this study online/virtual **Bangladesh Earthquake Engineering University** is formed/launched. The seismic analysis of structural model focusing earthquake were also done.

The primary objective of earthquake resistant design is to prevent building collapse during earthquake thus minimizing the risk of death or injury to people in or around those building. The standardized codes have gone through a number of modifications as the aftermath of large earthquakes made these revisions essential. Though code provisions promise better safety in future buildings provided the code specifications are followed properly, yet the stability of the already constructed structures following old codes of practice are in real threat. Findings of Urban Resilient Project (Rajuk) that Total Building vulnerability due to Madhupur fault, Magnitude 6.9 is 864,619-1,391,685 (40%-64%) of the 2.15 million buildings (Rajuk, Project Director Presentation at Ministry 03 Oct 2024).

Urban Resilience project surveyed 3252 important buildings, using the Rapid Visual Assessment methodology. Then 578 of the buildings prioritized in the Rapid Visual Assessment stage, went through Preliminary Engineering Analysis. Finally, 187 of the buildings, went through in-depth Detail Engineering Analysis. Out of 187 buildings 42 buildings were earmarked to be demolished. The Gas supply line study revealed that total constructed gas pipe line (As on June 30,2025): 13,438.69 km

(Abandoned/Unused 448.21 km pipe line) and total number of customers (As on June 30,2025): 27,78,712. Initially titas gas used pipe of diameter 14 inch (Annual Report 2024-25). The study reveals that gas connection yearly complain is around 6466 numbers.

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URBAN INFRASTRUCTURE SECURITY FOCUSING

EARTHQUAKE: BANGLADESH PERSPECTIVE

CHAPTER 1

INTRODUCTION

1.1 Context Information

IT IS MENY PROSTRATE IN THEIR HOMES BEFORE THE MORNING' (7:91). '...CAME AT THEIR BUILDING FROM THE FOUNDATIONS, SO THE ROOF FEL UPON THEM FROM ABOVE THEM' (16:26). 'WHEN THE EARTH IS SHAKEN WITH ITS (FINAL) EARTHQUAKE. AND THE EARTH DISCHARGES ITS BURDEN' (99:1-2)

The point within the earth where earthquake waves originate is called the focus, from where the vibrations spread in all directions. They reach the surface first at the point immediately above the focus and this point is called the epicenter.

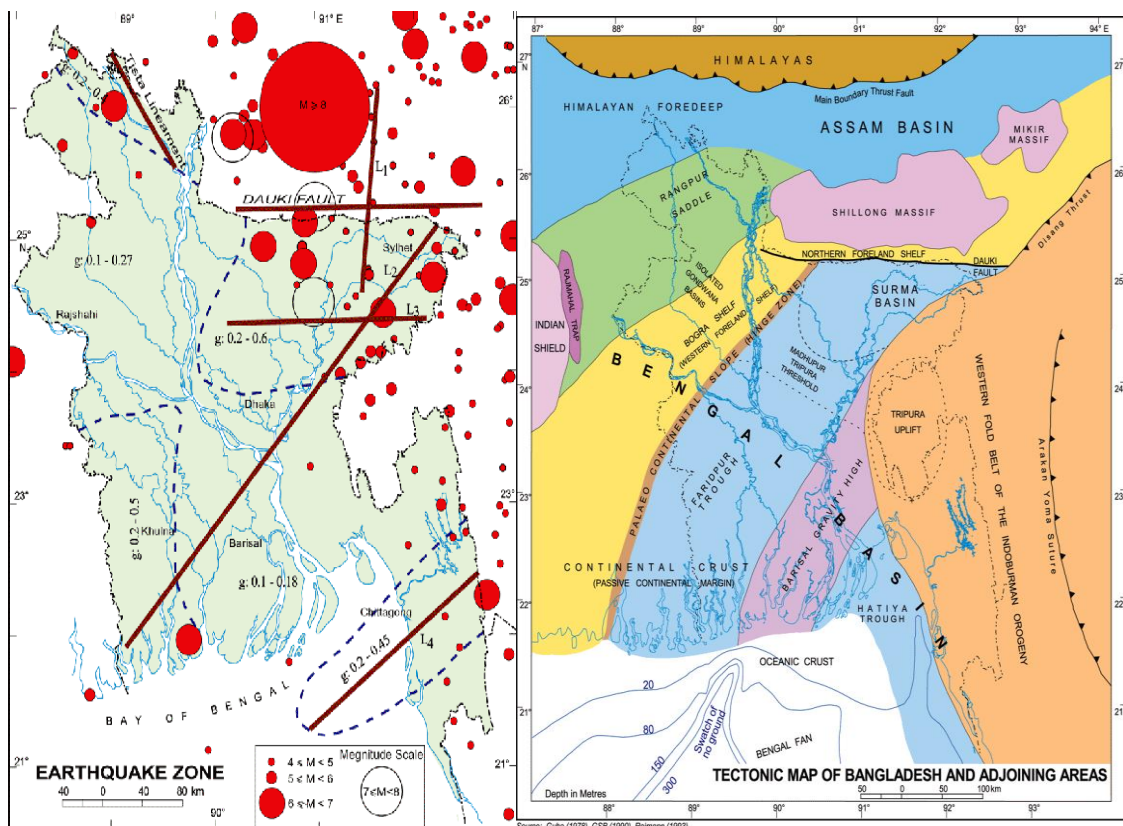


Figure 1.1 Seismo-tectonic lineaments capable of producing damaging earthquake (after Banglapedia, 2004)

Bangladesh is surrounded by regions of high seismicity, which include the Himalayan Arc and Shilong Plateau in the north, the Burmese Arc, Arakan Yoma anticlinorium in the east and complex Naga-Disand-Haflong thrust zone in the northeast shown in figure 1.1. Bangladesh has four seismic zones as shown in figure 1.2. Those are low($Z=0.12$), moderate($Z=0.20$), severe($Z=0.28$) and very severe($Z=0.36$). Part of the country extended from Sylhet to Chittagong is in the high seismic zone whereas Dhaka is in the moderate seismic zone.

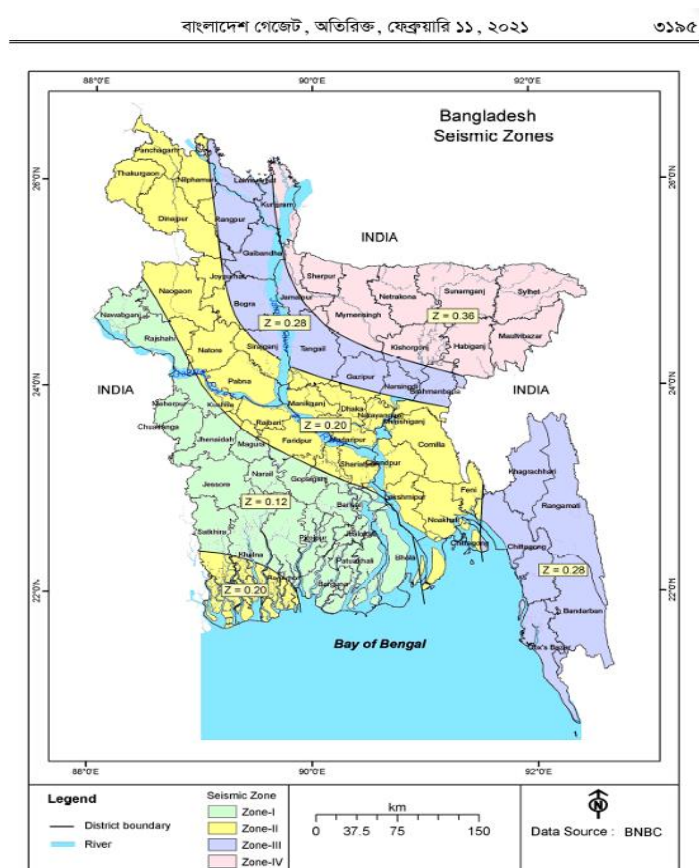


Figure 1.2 Seismic Zoning Map of Bangladesh (BNBC, 2020)

1.1.1 Urban Infrastructure.

Urban areas are very developed, meaning there is a density of human structures, such as houses, commercial buildings, roads, bridges, and railways. Urban infrastructure mainly include airport, building, bridges, electricity, gas supply line, roads, water supply line etc. Bangladesh faces immense urban infrastructure security risks due to its position near highly active tectonic plates, including the Dauki and Madhupur fault lines. Dhaka stands as one of the world's most seismically vulnerable megacities, where a magnitude

6.9 earthquake along the nearby Madhupur Fault could instantly collapse more than 40% to 64% of the capital's total infrastructure. The combination of non-expert design analysis, poorly enforced building codes, soft alluvial soils, and highly volatile utility networks turns Dhaka into a severe urban infrastructure security crisis.

1.1.2 Gas Infrastructure.

Gas distribution systems are one of six broad categories of infrastructure grouped under the heading 'lifelines' (O'Rourke, 1998). Together with water distribution, electric power, liquid fuels, telecommunications, transportation and wastewater facilities they provide the basic services and resources upon which modern communities have come to rely, particularly in the urban context. Disruption of these lifelines through earthquake damage can therefore have a devastating impact, threatening life in the short term and a region's economic and social stability in the long term. Bangladesh gas supply line is yet to be found earthquake resistant.

The direct effects of earthquakes are surface faulting and ground shaking. Secondary or "collateral" effects include liquefaction, landslides, densification and tsunamis. Earthquake effects on buried pipelines are best understood by considering the displacements induced in the surrounding soil. Damage may be caused by transient ground deformation (GDt), or permanent ground deformation (GDp), or a combination of the two. O'Rourke (1998) defines the distinction between these two effects.

Dhaka has rigid, high-pressure underground gas pipeline systems without automated seismic shut-off valves means that structural collapses will instantly trigger city-wide secondary firestorms.

1.1.3 Fire Following Earthquake- Gas Leakage.

Fire following earthquakes due to leakage of gas has resulted in a considerable loss of property during the history. In San Francisco earthquake (1906) the estimated loss due to a widespread fire was about 250 million dollar of its time. The case of Tokyo earthquake of 1923 was much worse where the fire-induced loss shared 77% of the total. In recent times, Loma Prieta (1989), Northridge (1994) and Kobe (1995) earthquakes are new experiences in which ignition of fire was generally because of damages in old gas networks. It is therefore justifiable to concentrate on the performance of city gas networks, among other lifelines, when assessing vulnerability of urban areas to fire following an earthquake.

1.2 Residential Gas Supply System in Saudi Arabia. Residential gas systems in Saudi Arabia differ significantly from many Western nations, as there is no nationwide utility grid supplying piped natural gas directly to individual homes. Instead, residential properties rely entirely on Liquefied Petroleum Gas, a mixture of propane and butane.

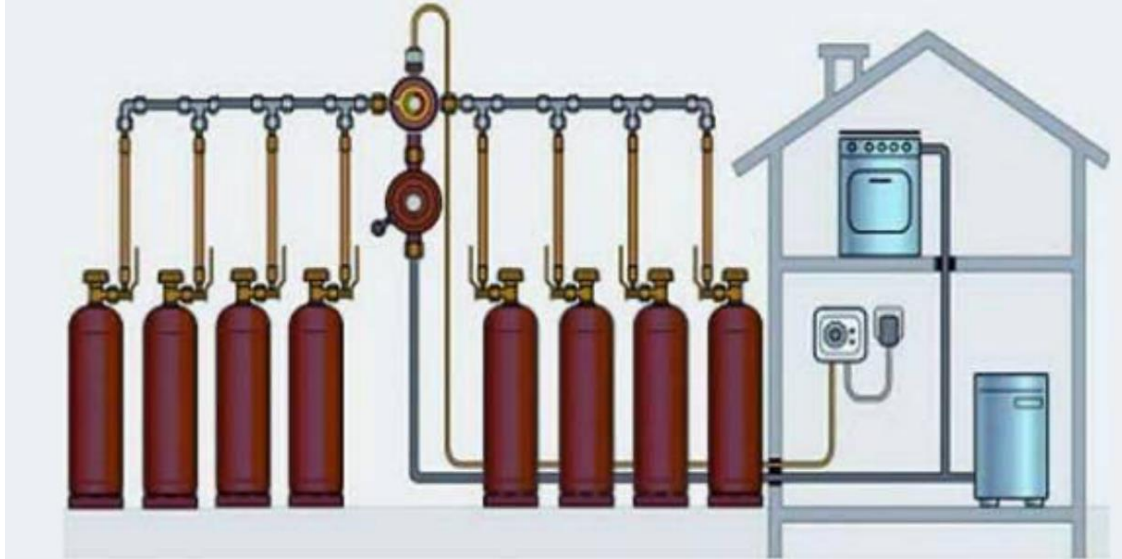


Figure 1.3: Residential Gas System in Saudi Arabia
(Source: Colonel Badr, Saudi Course Member)

1.3 Residential Gas Supply System in Kuwait. There is no national underground municipal natural gas grid in Kuwait, connected directly to homes for residential cooking or heating. Instead, the household gas supply system relies entirely on Liquefied Petroleum Gas, managed primarily through two main distribution methods; Portable Cylinders and Centralized Piped System

Residential Gas Supply System in Kuwait

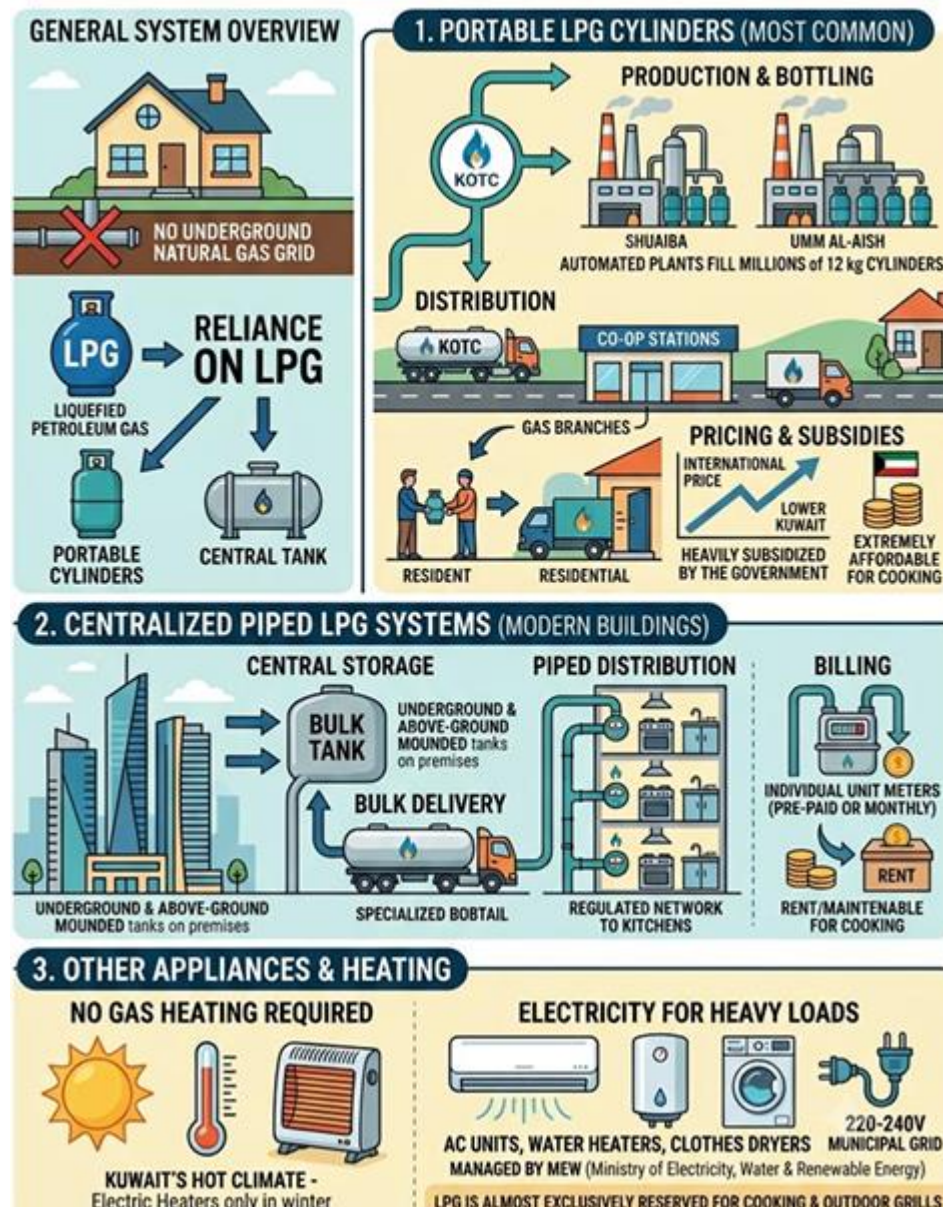


Figure 1.4: Gas Supply System in Kuwait

(Source: Defense Attache, Kuwait)

1.4 Residential Gas Supply System in Japan. Japan has one of the safest household gas systems in the world because it is designed specifically for earthquakes. There are mainly two types of gas used in **Japanese households**: "city gas" and "propane gas (Liquefied Petroleum Gas)". The city gas is highly earthquake resistant with **polyethylene pipelines** and 'Supreme' real time seismic monitoring system (Source: <https://sj.jst.go.jp/stories/2026/s0217-01p.html>, Science Technology Innovation Japan). Gas is supplied to homes in two ways including earthquake protection system.

1.4.1 City Gas

- i) Delivered through underground pipelines.
- ii) Common in Tokyo, Osaka, Yokohama, Nagoya, etc.

1.4.2 Liquefied Petroleum Gas

- i) Stored in cylinders/tanks outside the house.
- ii) Common in rural areas and smaller cities.

1.4.3 Earthquake Protection System (Gas). Every home has a smart gas meter that continuously monitors gas flow and shaking. If it detects:

- i) Earthquake shaking around Japan Meteorological Agency Seismic Intensity 5 or higher.
- ii) Large gas leaks.
- iii) Abnormally high gas flow.
- iv) Gas appliances left on for too long.

The meter automatically shuts off the gas supply.

1.4.4 Gas Phenomenon during an earthquake?

- i) Earthquake occurs.
- ii) Gas meter senses strong shaking.
- iii) Internal valve closes automatically.
- iv) Gas supply to the house stops.
- v) This prevents fires and explosions from broken pipes or damaged appliances.

1.4.5 Gas connected to earthquake monitoring-Procedure

Japan's earthquake network operated by the Japan Meteorological Agency detects earthquakes nationwide. Gas companies also have their own monitoring centers(Large Earthquake).

- i) Gas companies can remotely stop gas supply to entire districts.
- ii) Pipeline pressure and damage sensors are checked.
- iii) Engineers inspect pipelines before restoring service.

1.4.6 Gas stop after a small earthquake

Many foreign residents in Japan are surprised when their stove suddenly stops working after a quake. Often there is nothing wrong. The meter simply activated its safety

function. The resident may press reset button when found no damage or no smell, after safety check is for 3 minutes, stove is turned into active mode.

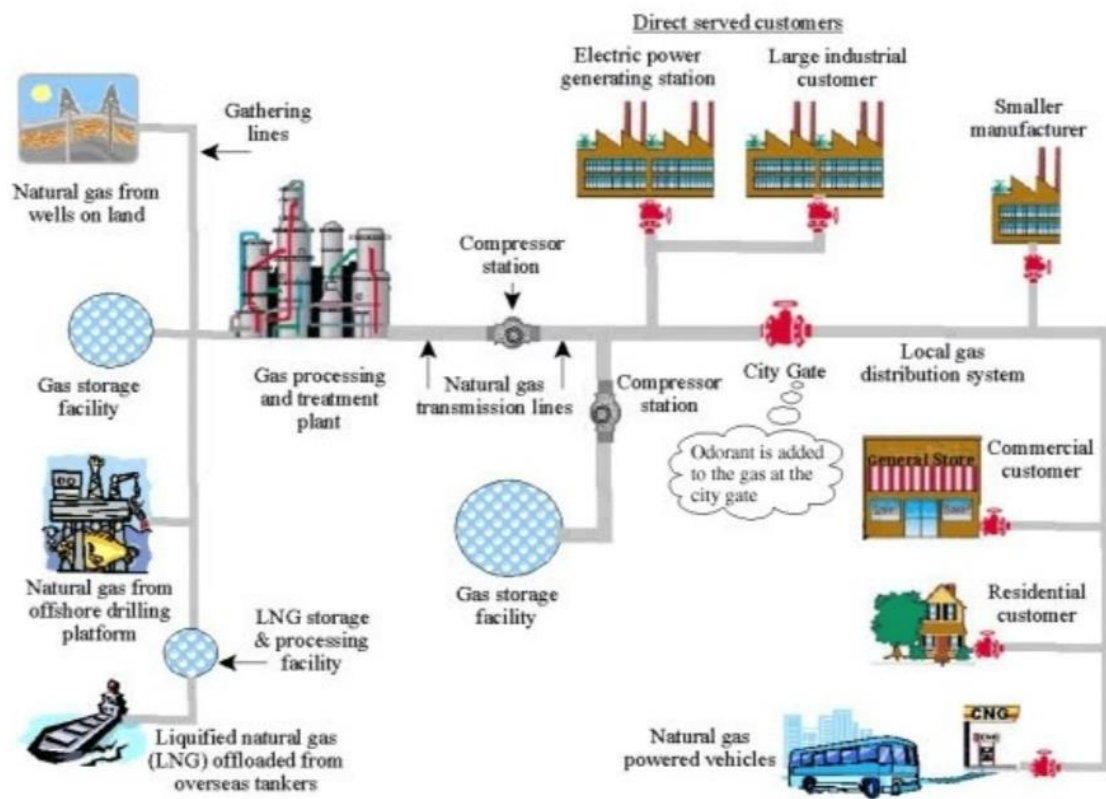


Figure 1.4 Residential Gas Supply System in Japan

(Source: Japan Philips Manager,2026)

1.4.7 Japan Gas Features.

The features are as follows:



Figure 1.3: Types and Features of Gas in Japan (Source: https://www.gtn.co.jp/magazine/en_us/article210/)

1.4.8 The Difference between City Gas and Propane Gas (Japan). The differences are as follows:

Table 1.1: Basic Difference Between City Gas and Propane Gas (Japan)

Item	City Gas	Propane Gas (Lp Gas)
Raw Material	Natural gas(mainly methane)	Liquefied petroleum gas (mainly propane and butane)
Supply Method	Supplied through underground gas pipelines	Delivered to each household via gas cylinders
Weight	Lighter than air	Heavier than air
Heating Value	Approximately 11,000 kcal/m3	Approximately 24,000 kcal/m3

1.5 Residential Gas Supply System in United States of America

1.5.1 A residential natural gas system is designed to safely deliver fuel from the local gas utility to the appliances that use it inside a home. Natural gas is transported through underground pipelines at relatively high pressure to neighborhoods, where it is then distributed to individual homes through smaller service lines. Before the gas enters the home, it passes through a gas meter, which measures the amount of gas consumed for billing purposes. A pressure regulator is typically installed to reduce the gas pressure to a safe level that is suitable for household appliances.

1.5.2 Once inside the home, the natural gas flows through a network of rigid black steel pipe, corrugated stainless steel tubing, or other approved piping materials. These pipes branch off to supply individual appliances such as furnaces, water heaters, stoves, ovens, clothes dryers, fireplaces, and standby generators. Each appliance is equipped with its own shutoff valve, allowing it to be isolated for maintenance or replacement without interrupting the gas supply to the rest of the house.

1.5.3 Overall, a residential gas system provides an efficient, dependable, and cost-effective source of energy for many household needs. By combining pressure regulation, secure piping, controlled combustion, modern safety devices, and proper venting, these systems deliver fuel safely throughout the home while supporting essential appliances that provide heating, hot water, cooking, and backup electrical power.

1.6 Dhaka liquefaction hazard map. A study through Rajdhani Unnayan Kotripokkho and Bangladesh University of Engineering and Technology revealed that 55% to 60% of the land within Rajdhani Unnayan Kotripokkho jurisdiction falls under high-risk "red and magenta zones" on the city's liquefaction hazard map.

1.7 Urban Resilience Project. Dhaka has approximately 2.145 million structures. Rajdhani Unnayan Kotripokkho Urban Resilience Project data confirms that up to 95% of buildings constructed in past decades bypassed strict regulatory safety protocols or lacked approved structural designs.

1.7.1 Rajdhani Unnayan Kotripokkho has identified hundreds of structurally unstable buildings suffering from visible cracking or tilting. Out of a targeted structural audit, 42 major municipal buildings (including key academic institutions and wings of the Bangladesh Medical University/Post graduate Hospital) were ordered for immediate demolition, but yet to be completed.

1.7.2 Dhaka City faces severe structural, geological, and operational gaps regarding earthquake. Recent seismic events in the November 2025 and June 2026 moderate tremors have served as signal that the capital is dangerously exposed to a major catastrophe. An analysis of recent data from Rajuk, Bangladesh University of Engineering and Technology and urban planners reveals few critical gaps in Dhaka's earthquake resilience:

1.8 Widespread Building Code Violations

- i) Lack of Compliance: Up to 95% of buildings in Dhaka were constructed without approved designs, and roughly 90% of older structures fail to comply with the Bangladesh National Building Code.
- ii) Total buildings in Dhaka are 21,45,000. At risk of collapse are Over 8,00,000 (40%) High-rise buildings are over 75,000 which are most built without meeting minimum standards. Between 2006 and 2016, around 95,000 buildings were constructed in the capital of which only 4,147 had Rajuk approval, meaning 95.36% were unauthorised or built without approved designs (Dhaka Tribune, 23 Nov 2025).
- iii) Many high-rises utilize unreinforced "soft-story" ground floors (open spaces dedicated to parking), become highly susceptible to pancaking during structural shaking.

1.9 Hazardous Soil Conditions and Wetland Destruction

- i) **Liquefaction Risk.** Rapid, uncontrolled expansion has seen wetlands and water reservoirs filled with loose earth to build housing, particularly in eastern and southwestern Dhaka.
- ii) **Vulnerable Zones.** Joint studies by Rajdhani Unnayan Kotripokkho and Bangladesh University of Engineering and Technology could find that 55% to 60% of Dhaka falls into a high liquefaction zone. In these areas, shaking turns the filled soil behave like liquid, causing buildings to tilt or sink rapidly.

1.10 Rationale of the Paper

Urban infrastructure security focusing earthquake includes safety of total structure. The safety journey of an infrastructure starts with test, design, construction, post construction and monitoring till last. Urban infrastructure security focusing on earthquakes includes the structural retrofitting of buildings, soil-structure interaction and liquefaction mitigation, and the development of resilient "smart" utility networks. It also relies on strict land-use planning, disaster contingency plans, and structural innovations like base isolation.

1.10.1 The Problem. Bangladesh sits near major active fault zones like the Madhupur Fault and the Dauki Fault, thereby cities like Dhaka is highly vulnerable (Khan et al., 2001).

1.10.2 The Knowledge Gap. Current urban planning often ignores building codes (Dynamic Time History Analysis), having a high density of non-engineered buildings. Existing literature lacks proactive strategy encompassing Bangladesh National Building Code core issue including gas network vulnerabilities.

1.10.3 The Solution. This paper illustrates earthquake resistant structural system including base isolation, Code core issue including structural/soil/gas vulnerabilities and provides an updated, data-driven proactive security strategy to minimize casualties and economic collapse.

1.11 Statement of the Problem

1.11.1 Lack of Dynamic Time History Analysis. Dynamic Time History analysis (Earthquake resistant building structure) to be done for regular buildings with height greater than 40m in Dhaka city. Dynamic Time History Analysis is the most accurate method to simulate how real structures respond to specific earthquake ground motions over time, a lack of its application keeps a real structural risk. (Hasan,2025).

1.11.2 Lack of Soil Liquefaction Analysis. Rapid, unplanned urbanization in Bangladesh has forced structural expansion onto highly vulnerable geological landscapes, including reclaimed wetlands and soft alluvial floodplains filled with uncompacted dredged materials. While superstructure design increasingly attempts compliance with seismic codes, there is a critical lack of comprehensive, site-specific Soil Liquefaction Analysis incorporated into foundation engineering. Standard penetration and structural design processes routinely overlook the hazard of moisture-saturated, loose silty sands transitioning into a liquid state under cyclic earthquake loading. Without integrating automated liquefaction potential mapping and depth-dependent factor-of-safety assessments into pre-construction clearance, major urban

centers face severe ground failure risks. Consequently, even modern infrastructure remains highly susceptible to catastrophic tilting, differential settlement, and sudden foundation collapse during an active seismic event.

1.11.3 Critical Lifeline Infrastructures are Treated Separately. Most existing research and policies focus primarily on structural safety of buildings, while critical lifeline infrastructures (gas, electricity, water, and plumbing) are often given proper importance (Research Question/Survey/Data). This initiates vulnerable loopholes within the infrastructure system that may lead to secondary disruption in the event of an Earthquake. For example:

- i) Gas pipeline ruptures may trigger fires and explosions
- ii) Electrical system failures can hinder emergency response
- iii) Water supply disruption can affect firefighting and sanitation

1.11.4 Identification of Research Problem. The core research problem is:

Proactive measures regarding integrated earthquake infrastructure security including gas, electricity, water, building systems and liquefaction risk focusing earthquake.

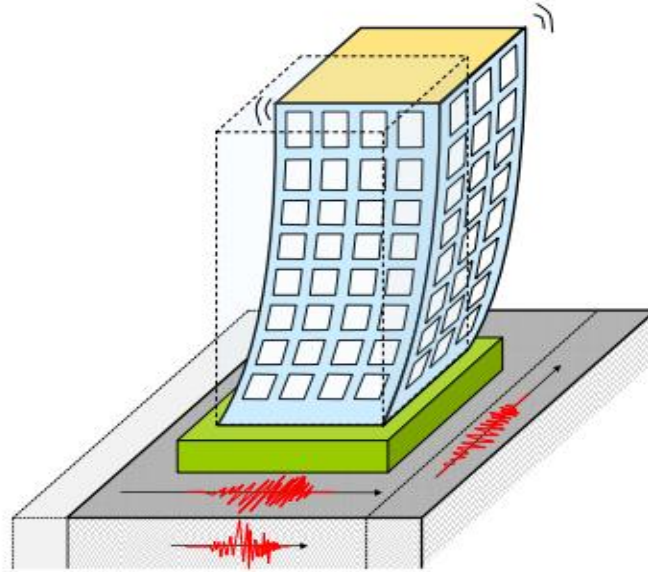


Figure 1.4. Earthquake Behavior of Building (Source: www.iitk.ac.in/nicee/IITK-GSDMA)

1.12 Research Objectives and /or Research Questions. The objective of the research paper is as follows:

1.12.1 Broad Objective.

- i) Broad objective is to establish a comprehensive proactive “**Earthquake Risk Reduction Strategy**” that is linked to an easy implementation at all levels so that “**Urban Infrastructure Security**” preparedness is considered early in Dhaka City.

1.12.2 Specific Research Objectives

- i) To examine tasks that support earthquake risk reduction strategy such as Vulnerability Assessment of Critical and Essential Facilities, Risk Zone, Lifelines (Gas) in Dhaka city,
- ii) To assess what are the damages may occur during earthquake and study the gaps, we have in Dhaka city regarding earthquake.
- iii) To evaluate Early warning systems including forecasting, dissemination of warnings, preparedness measures and reaction capacities
- iv) To strengthening essential support services such as Emergency Operation Centre/Technical Support Centre.
- v) To deliver distant training/education program in implementing ‘Earthquake Risk Reduction Strategy (Design, Vetting, Construction, Supervision and Monitoring)’ at all levels.

1.12.3 Research Question. The research is designed to seek the answer of subsequent questions.

- i) What is the most appropriate proactive measure to address the Infrastructure Security Issue focusing earthquake?
- ii) What is the present state of Dhaka Infrastructure?
- iii) What is the present state of Titas gas line?
- iv) What is the present state of Dhaka soil?
- v) What are Earthquake education platform we have in our country?
- vi) What is the gas system in few other countries?
- vii) How to address vulnerable structures including gas line in Bangladesh?

1.13 Operationalization of Key Terms/Concepts

1.13.1 Urban Infrastructure. This refers to the interconnected physical and organizational structures required to operate an urban area in Bangladesh.

- i) **Critical Building Stock.** Number of hospitals, fire stations, schools, and government headquarters.
- ii) **Utility Networks.** Kilometers of gas pipelines, electricity grids, and water supply lines.
- iii) **Transportation Networks.** Total length of primary roads, flyovers, metro rail tracks, and bridges.
- iv) **Telecommunications.** Density of mobile towers and fiber-optic backbone hubs.

1.13.2 Infrastructure Security (Resilience). This concept measures the capacity of urban systems to resist, absorb, recover from, and adapt to hazardous events.

- i) **Structural Integrity:** Percentage of buildings compliant with the Bangladesh National Building Code.
- ii) **Redundancy:** Availability of backup power systems, alternative evacuation routes, and secondary water sources.
- iii) **Rapidity:** Estimated time (in hours or days) required to restore basic utility services after a disruption.

1.13.3 Earthquake Focus (Hazard & Vulnerability). This defines the specific physical threat and the susceptibility of the urban environment to seismic activity.

- i) **Seismic Hazard Zone:** Location classification according to the BNBC seismic zoning map (Zone 1 to Zone 4).
- ii) **Soil Conditions:** Liquefaction potential index of urban soils, especially in filled and in marshlands.
- iii) **Structural Vulnerability:** Proportion of soft-story buildings, non-engineered brick structures, and buildings with visible structural distress.
- iv) **Spatial Density:** Building footprint ratio and average width of inner-city lanes (for rescue vehicle access).

1.13.4 Bangladesh Perspective. This contextualizes the study within the unique socio-economic, institutional, and geographic realities of Bangladesh.

- i) **Regulatory Compliance:** Percentage of building plans approved vs. actual construction alignment through Rajuk.
- ii) **Institutional Capacity:** Budget allocation, rescue equipment inventory, and manpower of the Fire Service and Civil Defence.
- iii) **Socio-Economic Vulnerability:** Population density per square kilometer in high-risk areas (e.g., Old Dhaka) and percentage of low-income settlements.
- iv) **Community Preparedness:** Frequency of local earthquake drills and public awareness levels regarding emergency protocols.

1.13.5 Earthquake Risk Reduction Strategy. An earthquake risk reduction strategy aims to minimize casualties and property damage. It relies on four primary pillars: Structural Mitigation (building reinforcement), Land-Use Planning, Preparedness, and Early Warning Systems (Unesco).

1.14 Outline of Chapters

1.14.1 The chapter plans are as follows:

- i) Chapter 1: Introduction
- ii) Chapter 2: Literature Review
- iii) Chapter 3: Research Methodology
- iv) Chapter 4: Earthquake Risk Reduction Strategy
- v) Chapter 5: Earthquake Threat and Possible Scenarios in Dhaka
- vi) Chapter 6: Analysis of Data and Results
- vii) Chapter 7: Discussion
- viii) Chapter 8: Bangladesh Earthquake Engineering University
- ix) Chapter 9: Conclusion and Recommendation

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

2.1.1 Bangladesh National Building Code and Infrastructure Security-Literature

Mandated through the Ministry of Housing and Public Works (2021), the Bangladesh National Building Code (BNBC) 2020 serves as the definitive legal and regulatory framework for strengthening **urban infrastructure security** against prevailing seismic hazards across Bangladesh. Directed at protecting public welfare, life, and property, the code establishes rigorous minimum standards for structural design, materials quality, and emergency egress, which are vital for mitigating structural collapses and facilitating rapid evacuations during an earthquake. To ensure strict systemic compliance, the document establishes the Bangladesh Building Regulatory Authority and empowers local jurisdictional Building Officials such as Rajdhani Unnayan Kartripakkha with the statutory power to execute site inspections, mandate structural tests, and issue immediate stop-work orders on non-compliant, unsafe developments. Infrastructure security is further fortified through a mandatory four-stage permit system—comprising Land Use, Large/Specialized Projects, Building, and Occupancy Certificates—which prevents unsafe occupancies and unauthorized modifications. Critically, for structural and earthquake-resilient design, the code completely eliminates informal construction oversight by dictating that sub-soil investigations, geotechnical reports, and structural design blueprints must be certified exclusively by licensed, professional civil and structural engineers with defined clinical design experience. These certified technical specialists are held directly accountable under penalty of professional blacklisting or legal prosecution for any structural lapses or false certifications. Consequently, the BNBC 2020 shifts the engineering paradigm from reactive hazard response **to proactive regulatory compliance**, positioning rigorous **code enforcement as a foundational safeguard for the nation's vulnerable metropolitan infrastructure**.

2.1.2 Earthquake Risk Reduction Strategy Related Literature. Moullier (2018) identify Japan's regulatory system as a "compelling success," highlighting its effectiveness in reducing seismic risk through robust institutional mechanisms. Key

features include: (i) mandatory third-party structural review ensuring independent compliance verification; (ii) multistage inspections linked to occupancy approval; (iii) performance-based codes enabling flexible yet outcome-oriented design (Midorikawa et al., 1999); and (iv) strong professional liability that aligns practitioner incentives with regulatory compliance (Moullier et al., 2018). Pedro, Meijer and Visscher (2009) classify European Union building control systems into three types-public, mixed and dual-with mixed systems predominating, typically combining public regulatory checks with private inspection roles (Pedro et al., 2009).

2.1.3 Dhaka Infrastructure Security Risk Related Literature. Eleous and Al Noman (2025) investigate the widespread structural and systemic vulnerabilities within the Dhaka Metropolitan Area, highlighting how systemic deviations from building regulations directly compromise urban infrastructure security and heighten seismic disaster risks. The study reveals that a staggering 90% of neighborhood structures defy their authorized designs, while approximately 89% to 94.76% of constructions across key residential sectors fail foundational structural pre-requirements, dangerously undermining the city's resilience to the region's active seismic threats. Furthermore, the authors identify that 74.19% of local buildings actively commit setback violations to maximize financial margins. This pervasive encroachment results in exceptionally narrow roadways that inherently block the passage of emergency rescue and response vehicles, crippling the city's post-earthquake evacuation capabilities. Systemic enforcement gaps exacerbate this vulnerability, as 46% of developers bypass occupancy evaluations by obtaining illegal utility connections, and RAJUK's oversight wing suffers a severe 42% manpower shortage. Ultimately, the literature demonstrates that Dhaka's expanding infrastructure remains highly insecure, necessitating strict legal accountability for deed holders and an interconnected "Shared Responsibility and Liability" framework among utilities to enforce the structural baselines mandated by the Bangladesh National Building Code.

2.1.4 Proactive structural and administrative mitigation measures Related Literature. According to Nowroz (2003), the systemic vulnerabilities and extensive disaster risks associated with earthquakes in Bangladesh require an urgent transition from reactive relief operations to proactive structural and administrative mitigation measures. The study establishes that the country's high-density urban infrastructure is

exceptionally vulnerable to catastrophic seismic shocks due to historical seismic gaps along active regional fault lines and the rapid proliferation of non-engineered, high-rise buildings. To systematically minimize potential loss of life and massive financial damage, Nowroz (2003) argues for a multi-tiered disaster management strategy. This framework mandates the rigorous updating and legal enforcement of the Bangladesh National Building Code to secure all ongoing constructions, the implementation of localized seismic monitoring networks to capture micro-tremors, and the creation of highly trained, multidisciplinary emergency response teams comprising engineers, planners, and first responders. Furthermore, the author emphasizes that widespread community-level awareness campaigns broadcast across public media are essential to reduce mass panic and ensure safe evacuation protocols, positioning integrated structural resilience and public preparedness as the foundational pillars of national infrastructure security.

2.1.5 Soil Condition and Liquefaction Risk-Related Literature.

- i) Ansary and Rashid's pioneering liquefaction susceptibility mapping, as synthesized by Anisuzzaman (2021), serves as a critical diagnostic tool for urban infrastructure security by delineating highly vulnerable, fluidization-prone zones across Dhaka's reclaimed lowlands. By identifying these high-risk areas, the framework allows structural engineers and municipal planners to enforce targeted foundation reinforcements and soil stabilization protocols, preventing catastrophic building collapses and utility pipeline failures during a major seismic event (Anisuzzaman 2021).
- ii) The geotechnical characteristics of Dhaka significantly contribute to its earthquake vulnerability. The city is predominantly built on alluvial soil and reclaimed land, which are highly susceptible to liquefaction (Rahman et al., 2020).

2.1.6 Interdependency of Urban Infrastructure Systems-Related Literature

i) Ferooz (2017) in *Seismic vulnerability assessment of transportation networks and lifeline infrastructure in urban centers* demonstrates that earthquakes cause severe urban disruption by simultaneously damaging transportation networks and burying lifeline infrastructure, leading to structural failures and isolated sectors. The study highlights that the rupture of critical lifelines, such as gas and water lines, triggers secondary hazards that exacerbate the immediate, catastrophic damage to roadways and emergency response capabilities.

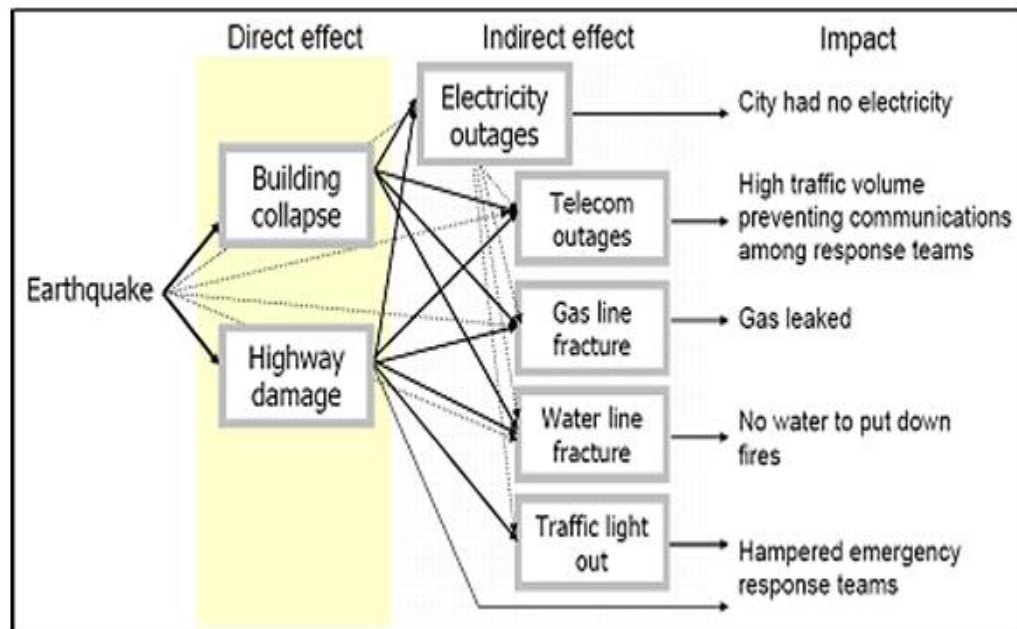


Figure 2.1: Impact of Earthquakes

ii) Xu (2021) in *Cascading failures and interdependency modeling in urban utility networks during extreme hazards*, mentioned that modern metropolitan areas exhibit extreme cascading vulnerabilities during disasters due to the tightly coupled interdependencies of urban utilities. A major outage in the electricity distribution system directly triggers systemic operational failures across both water pumping stations and natural gas regulation systems, effectively crippling a city's vital lifeline infrastructure. Within the context of seismic events, this co-

dependent failure mechanism severely undercuts urban infrastructure security. When an earthquake cuts local power lines, the immediate shutdown of water treatment plants and distribution networks hinders urban firefighting efforts, while unmonitored gas supply system failures elevate the risk of secondary, widespread pipeline explosions. Xu (2021) argues that evaluating infrastructure security from a single-sector perspective is no longer sufficient; instead, municipal authorities must deploy integrated, cross-sector simulation tools and decentralized emergency backup grids. By implementing these measures, cities can prevent localized electrical grid failures from cascading into complete, multi-system lifelines shutdowns, thereby safeguarding public safety during critical post-earthquake response windows.

iii) A key theme emerging from the literature is the interdependency of urban infrastructure systems. Buildings, gas, electricity, and water systems are interconnected, and failure in one sector can trigger cascading effects in others (Alam et al., 2018).

2.1.7 Seismic Risk Profile of Dhaka City-Related Literature

i) Alam et al. (2018) in *Urbanization, megacity growth, and the evolution of seismic vulnerability in Dhaka, Bangladesh*, assert that Dhaka's seismic vulnerability is significantly heightened by rapid urbanization into a megacity that has outpaced the development of resilient infrastructure. The research emphasizes that addressing these risks requires urgent, high-utility, and **structural improvements to critical networks** to protect the dense population during disasters.

ii) A large proportion of structures are **constructed using unreinforced masonry and non-engineered reinforced concrete**, both of which exhibit poor seismic performance (Alam et al., 2018).

iv) The prevalence of **soft-story configurations** further exacerbates structural instability during seismic events. Additionally, **weak enforcement of the Bangladesh National Building Code** has resulted in widespread non-compliance with seismic design provisions (Rahman et al., 2018).

v) Bangladesh being located on the seismically active Indian plate is exposed to significant seismic risk. Earthquake Disaster Risk Index of Stanford University ranks Dhaka among the 20 cities most vulnerable to earthquake in the world (World Bank,2013).

2.1.8 Gas Infrastructure and Fire Hazard Risks-Related Literature

i) Gas infrastructure represents a major source of secondary hazards during earthquakes. Pipeline ruptures can lead to gas leakage, significantly increasing the risk of fire and explosion (Shrestha, 2015).

2.1.9 Electricity Infrastructure Vulnerability-Related Literature

i) Electricity infrastructure is highly susceptible to disruption during earthquakes due to its dependence on both structural stability and network continuity. Damage to transmission and distribution systems can result in widespread power outages and increase the likelihood of electrical fires (Alam et al., 2018).

2.1.10 Water Supply and Plumbing Systems-Related Literature

i) Water supply and sanitation systems are essential for post-disaster survival and recovery. However, Dhaka's water infrastructure is characterized by aging pipelines, leakage, and inadequate maintenance (Islam et al., 2023).

2.1.11 Preparedness and Institutional Challenges-Related Literature

i) Despite the recognized risks, earthquake preparedness in Dhaka remains limited. Kabir et al. (2023) highlight that a significant portion of the population lacks awareness and training in earthquake response.

ii) Institutional challenges include weak coordination among key agencies such as Rajuk, Wasa, and utility service providers. The absence of a unified and integrated disaster management framework further hinders effective risk mitigation and response.

2.1.12 Energy Infrastructure Security-Related Literature. Due to the lack of a comprehensive seismic zonation map and site-specific probabilistic seismic hazard maps, the human casualty, and property losses, specifically, energy infrastructure is going to be disastrous (Sakawat , Masumur and Rabiul,2020)

2.1.13 Earthquake risk reduction-Related Literature. The task of reducing vulnerability of structures and buildings is should be the key to earthquake risk reduction (Raihanul and Nazrul, 2016).

2.1.14 Pre-Disaster preparedness. Emerging studies suggest IoT systems can monitor drainage and infrastructure in real time (Islam et al., 2024).

From an analytical perspective, the literature consistently emphasizes that structural failure is the principal cause of casualties, indicating that improving building resilience should be the foremost priority in earthquake risk reduction strategies.

2.1.15 Urban Planning Strategies and Policy Framework to Make Dhaka City Resilient against Earthquake (Reference: SATREPS-TSUIB Project- 2022) Guidebook on Towards Seismic Resilience in Dhaka City: An Urban Planning Perspective. Tokyo & Dhaka: The University of Tokyo & Housing and Building Research Institute)

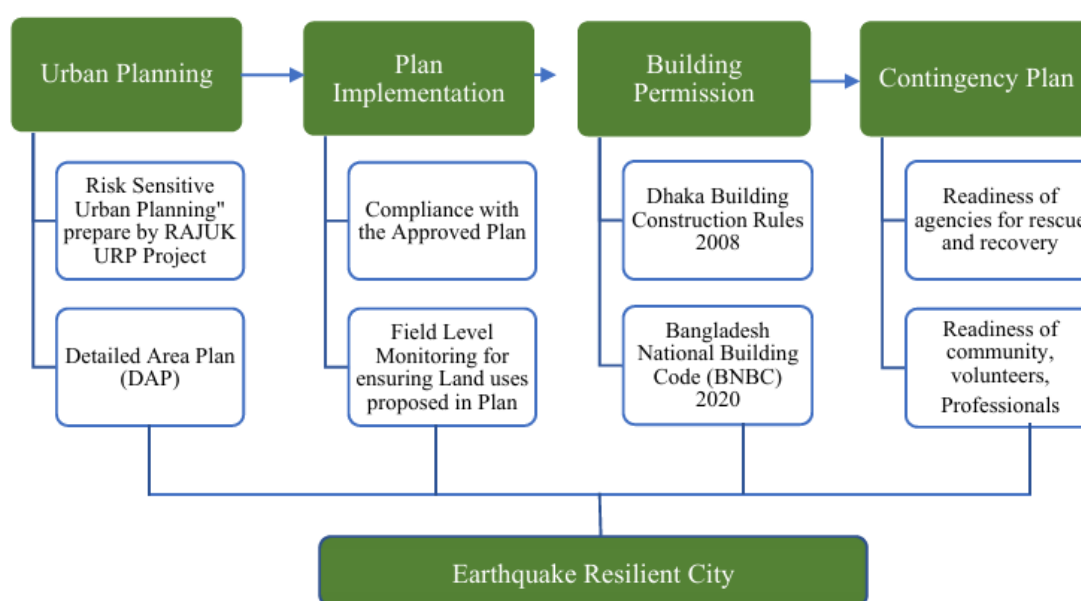


Figure 2.2: Flow Chart of Promoting Dhaka an Earthquake Resilient City

(Source: SATREPS-TSUIB Project-2022)

2.1.16 Urban Resilience Project Related Literature

- i. Under the Urban Resilience Project, Rajdhani Unnayan Karttripakkha conducted an earthquake risk assessment of Dhaka's building stock. Findings indicated that a 6.9-magnitude earthquake along the Madhupur Fault could damage or collapse 40.3–64.8% of the city's 2.145 million buildings (The Business Standard, 2025b). A minor tremor in 2025 damaged at least 300

buildings, including 25 partial collapses and numerous structural cracks (The Business Standard, 2025b). Only 3,252 buildings have undergone formal earthquake-safety assessments, leaving over 99.8% unverified (The Business Standard, 2025). Research on urban seismic vulnerability demonstrates that unplanned, high-density settlements on soft soil amplify earthquake damage, consistent with the situation in Dhaka (Erdik, Sesetyan, & Dogangun, 2019; Shaw & Goda, 2004).

ii. Urban Resilience Project (Rajuk), through Packages S-4 and S-5, implemented a comprehensive, multi-tiered structural assessment (RVA, PEA, DEA) of 3,252 critical buildings in Dhaka, concluding on June 30, 2022, with actionable retrofitting/reconstruction plans for 187 high-risk structures. Parallel to this, the project established a Risk-Sensitive Land Use Planning (Rslup) framework (Package S-5) incorporating Strategic Environmental Assessment (SEA) to guide future sustainable development (Bangladesh Urban Resilience Project, Technical Note 4-11 June 2024).

iii. The Rajdhani Unnayan Kartripakkha is enhancing Dhaka's urban resilience through the Urban Resilience Project, which focuses on institutionalizing building code enforcement, establishing an **Electronic Construction Permitting system**, and strengthening professional accreditation. Key initiatives include the development of an Urban Resilience Unit, the implementation of the 2020 Bangladesh National Building Code, and the training of structural engineers to improve construction safety (Bangladesh Urban Resilience Project, Technical Note 4-11 June 2024).

2.1.17 Bangladesh Seismic Mega Plan- Related Literature.

i) A series of recent earthquakes felt across Dhaka and other parts of the country has prompted the government to begin policy-level discussions on developing a long-term, integrated national earthquake preparedness strategy (Business Times, Dated 14 June 2026).

ii) Officials at the Ministry of Disaster Management and Relief said the proposed framework, being informally referred to as a “mega plan,” would draw

on international best practices from earthquake-prone nations and focus on strengthening Bangladesh's resilience against a potentially devastating seismic event (Business Times, Dated 14 June 2026)..

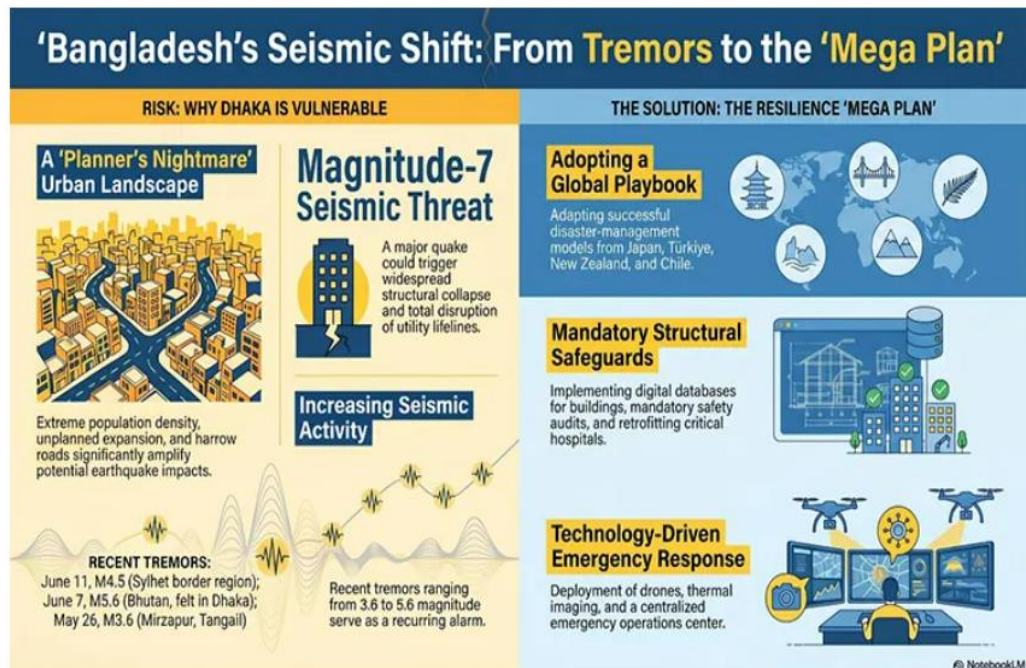


Figure 2.3: Mega Plan

(Source: Business Times, 14 June 2026)

2.1.18 Titas Gas Related Literature

Lifeline earthquake engineering is relatively a new one especially in a country like Bangladesh. Usually within city pipe diameter is small and less dangerous as gas pressure is not so high. But the main distribution line beyond city having larger diameter with high pressure gas is more vulnerable to earthquake. The Titas franchise area map is shown below:



Figure 2.4: Titas Franchise Area

(Source: Titas Gas Official Website)

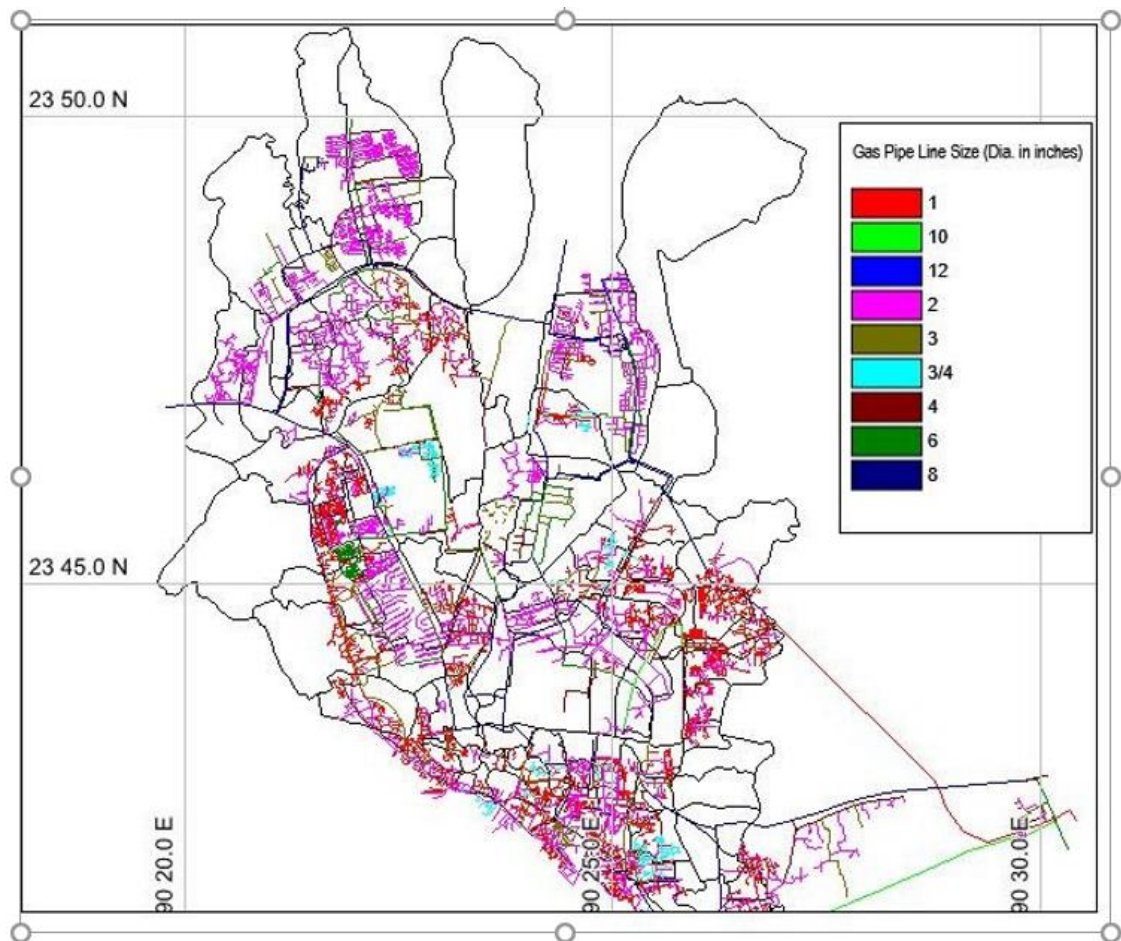


Figure 2.5: Digitized Layout of Gas Pipeline network of Dhaka city showing diameter of pipe

(Source: Sayeed,2013)

Table 2.1 Diameter wise Gas Pipeline length across Dhaka City

Titas Gas Transmission & Distribution PLC
105, Kazi Nazrul Islam Avenue
Karwan Bazar C/A
Dhaka-1215

Diameter wise Gas Pipeline Length across Dhaka City

Dia of Pipeline (Inch)	50 PSIG lines	140 PSIG lines
	Length of Pipeline(m)	Length of Pipeline(m)
0.5	2,274	-
0.75	107,295	-
1	626,365	-
2	953,361	535
3	180,998	1,130
4	275,002	5,615
6	76,147	2,679
8	155,636	25,273
10	30,005	10,686
12	13,674	63,394
16	-	89,669
Total Length	2,418,414	198,981

2.2 Gaps in Past Research and Scope of further Research

2.2.1 Gaps in Past Research. Despite extensive research, several critical gaps remain:

- i) Lack of Integrated Infrastructure Framework. Most studies focus on individual sectors (e.g., buildings, Soil liquefaction or gas systems), with limited emphasis on multi-sector integrated infrastructure security.
- ii) Limited Interdependency Modeling. There is insufficient research on cascading failure mechanisms among gas, electricity, water, building systems and liquefaction risk).
- iii) Proactive mechanism in integrated infrastructure security research is absent.

2.2.2 Scope of further Research

- i) Proactive measures regarding integrated infrastructure security including gas, electricity, water, building systems and liquefaction risk focusing earthquake.

2.2.3 Methodological Bridging and Strategic Paradigm Shift

- i) Ultimately, contemporary literature is limited by viewing urban seismic risks in isolation, failing to account for the cascading operational vulnerabilities of tightly coupled lifelines. It implements an integrated, exploratory, and analytical research design calibrated to the Dhaka Metropolitan Area. By triangulating qualitative and quantitative primary data, gathered through Key Informant Interviews (KIIs), field observations, and specialized questionnaire surveys, the study captures real-world institutional and systemic complexities. These field datasets are directly integrated with structural engineering tools, including site-specific Soil Liquefaction Analysis and Dynamic Time History Analysis for structures exceeding 40 meters. Consequently, this multi-tiered methodology provides the empirical foundation necessary to construct a proactive, data-driven Earthquake Risk Reduction Strategy capable of safeguarding Bangladesh's critical urban infrastructure networks.

2.3 Selected Areas of Research within the Identified Gaps. The selected areas of identified gaps are:

- i) Proactive Strategy regarding integrated infrastructure security focusing earthquake including Building, Electricity, Gas, Water etc since Dhaka has huge building deviation, uncompacted lowland liquefaction,

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Methodology. The research topic ‘Urban Infrastructure security focusing Earthquake: Bangladesh Perspective’ where in research methodology an integrated research approach was adopted. The technological know-how, Data Collection, Interview, expert opinion, strategic views were gathered. The technological know-how (Base Isolation Technique) through Public Works Department including Fire Service and Civil Defense were gathered. ‘Visual Rating method for Potential Seismic Vulnerability Assessment’ of Existing Reinforced Concrete Buildings in Bangladesh was collected through Housing and Building Research Institute Bangladesh. Through this method primary data on building was collected. The data on Dhaka Urban Resilient Project through Rajdhani Unnayan Katripokkho were collected. The Earthquake Risk and Vulnerability map through Disaster Management Bureau were collected. Primary field data including Key Informant Interviews, questionnaires, and field observations is triangulated directly with empirical datasets and technical archives retrieved through key national authorities. These organizations include Professors, Bangladesh University of Engineering and Technology including Military Institute of Science and Technology, Rajdhani Unnayan Kartripakkha, the Housing and Building Research Institute, the Public Works Department and the Dhaka North City Corporation, Titas Gas, Power Grid Bangladesh, Water Supply and Sewerage Authority including Consultants . This multi-tiered approach establishes a scientifically grounded baseline to evaluate cascading failure risks across the capital's utility networks. Ultimately, the design links engineering data directly with state-level strategic command planning. This structure provides a reliable foundation to formulate a comprehensive national Earthquake Risk Reduction Strategy.

3.1.1 Study Area Selection. Dhaka is selected as the primary case study.

3.2 Technology Knowhow Capturing Technique

3.2.1 Urban Infrastructure Security focusing Earthquake encompasses modern technology. In light of that modern technology technique could be captured. The first Base Isolation technique used in construction of Fire Service and Civil Defense build

in Mirpur. This building addresses the soil, resilient gas line. Necessary technical documents could be through assistance of Armed Forces Division, National Defense College, Public Work Development including Fire Service and Civil Defense.

3.2.2 “Development of Low-Cost Methods for Monitoring and Early Warning for Geohazards” Technology. The information collected regarding the Development of low-cost methods for monitoring and early warning for earthquake which is programmed and designed through Bangladesh University of Engineering and Technology team.



Figure 3.1 Station Network Map

(Source: <https://blcasn.tech/stations/map>)

3.3 Data Collection

3.3.1 Primary Data Collection. Visual Rating analysis method adopted.

3.3.1.1 Survey Questionnaire. Question and Answers through experts and officials were collected.

3.3.1.2 Key Informant Interview and in Depth Interviews. These organizations include Professors, Bangladesh University of Engineering and

Technology including Military Institute of Science and Technology, Rajdhani Unnayan Kartripakkha, the Housing and Building Research Institute, the Public Works Department and the Dhaka North City Corporation, Titas Gas, Power Grid Bangladesh, Water Supply and Sewerage Authority including Consultants.

3.3.1.3 Field Visit. Field visit to Bangladesh first Base Isolation Technology Construction at Fire Service and Civil Defense Headquarter was conducted. This technology uses earthquake **resilient** building, gas and other services are taken care of.

3.3.1.4 Types of Interviews.

i) Formal Interviews. Formal interviews were taken of Director General, Disaster management Bureau, Chief Engineer, Public Works Department, Additional Chief Engineer, Public Works Department, Director General, House Building and Research Institute.

ii) Informal Interviews. Informal interviews were taken of Head, Civil Engineering Department, Military Institute of science and Technology, Titas gas officials, Power Grid Bangladesh Officials, Wasa Officials.

iii) Conversation Through Mobile. Conversation with Joint Secretary, Ministry of Disaster management and Relief, Director General and Staff officer of Armed Forces Division was held.

3.3.2 Secondary Data Collection.

- i) Government Organizations were the main source of secondary data.
- ii) Existing Literatures

3.4 Structural Model Analysis. Normal Structure and Base Isolation Structure model was analyzed through Etabs Software. The aim was to see time difference when real time earthquake input data is given.

3.5 Data Analysis

Data and Map collected through Government Organizations were analyzed.

3.6 Sampling Plan for the Research

The topic itself is technical. Their target people were professionals, Engineers, Government Officials.

3.7 Research Ethics and Limitations. The research work was conducted with maximum effort on research ethics. Necessary permission was acquired before collecting data. Necessary citing of the source was tried to be mentioned as humanly possible if otherwise not mistaken. Intentional non citation of any source was not done.

CHAPTER-4

EARTHQUAKE RISK REDUCTION STRATEGY

4.1 Earthquake Risk Reduction Planning Assumptions (Source: National Earthquake Contingency Plan)

- i) Earthquakes are impact type events and provide no warning preventing any pre-event response activities.
- ii) Earthquakes within Dhaka city is going to cause large numbers of deaths and injuries and extensive damage and destruction of buildings, emergency facilities and infrastructure.
- iii) There is likelihood of secondary effects following an earthquake or aftershocks which may include tsunami, fire, flood, liquefactions, subsidence, damming of rivers, landslides, and dam failure, release of hazardous and toxic chemicals.
- iv) Strong aftershocks are going to be continued for several days resulting in further building collapse.
- v) Large number of persons (hundreds of thousands) are going to be in needs of shelter, welfare, relief assistance, medical care etc.
- vi) Access is going to be severely restricted due to debris, landslides, collapsed bridges etc.

4.2 Intended Users of the Earthquake Risk Reduction Plan

4.2.1 First Responder Institutions/Organization (Source: National Earthquake Contingency Plan). Following agencies are identified as First Responder Institution/Organizations:

- i) Armed Forces Division
- ii) Fire Service and Civil Defense
- iii) Director General of Health Services
- iv) Disaster Management Bureau
- v) Department of Relief and Rehabilitation
- v) City Corporations

4.2.2 Second Level Responder Institutions (Source: National Earthquake Contingency Plan).

- i) Utility and Life Line Institutions (water supply, electricity, gas supply, telecommunications, waste disposal etc).
- ii) Transportation systems (road, air, rail, and sea).

- iii) Para military forces and Police

4.2.3 Third Responder Organizations

- i) Ministries
- ii) Departments
- iii) City authorities
- iv) Non-Government Organisation
- v) Community Based Organizations
- vi) Civil Society Organizations
- vii) Academia etc

4.3 Earthquake Risk Reduction Plan Implementation Strategies. The following strategy to be adopted in earthquake risk reduction plan implementation:

- i) A system to be set up for regular reporting and updates on the readiness of first responder institutions.
- ii) Plan and Development of national capabilities to translate earthquake risk reduction into preparedness and Response Plans.
- iii) Propose a mechanism to build the capacity of government and nongovernment partners by providing training and education on earthquake risk reduction.
- iv) Transfer knowledge and state of the art of technologies necessary to support institutional operations and implement operations plans.
- v) The plan requires that the National Emergency Operations Center is to be established on priority basis to have service functions for a 24/7 schedule with duty officers in place with clearly identified notifications protocols.

4.4 Concept of Operations

4.4.1 General Operations. It has been identified that the North Eastern and South Eastern regions of Bangladesh are vulnerable to earthquake (Bangladesh Disaster Risk Management Profile 2006).

4.4.2 Operations Principles.

- i) Pre- Earthquake Phae-Disaster Management Bureau in Lead Role.** The Disaster Management Bureau is going to play a lead role in the pre-earthquake phase in coordinating and ensuring preparedness of first responder organizations for effective management of response activities in partnership with other stakeholder agencies.
- ii) During/Post Earthquake Phase – Armed Forces Division.** During Earthquake the Armed Force Division is going to take the lead role in responding to the emergency situation. The main function of Search and Rescue of victims to be handled by Armed Forces Division with support from Fire Services and Civil Defence. Para military forces,

Police, Community first responders to assist the professional search and rescue personnel employed by Armed Forces Division including Fire Services and Civil Defence.

4.4.3 Capacity Assessment

- i) The shortfalls in existing capacities for responding to the needs during the disaster situation to be tied to a scenario-based assessment and can be analysed only after vulnerability of existing structures including gas, water and electricity.

4.5 Urban Services Development/Restoration

4.5.1 Objective

- i) Identification of critical public facilities vulnerabilities and strengthening the same to a higher safety level.
- ii) Spatial Planning and land use control for earthquake vulnerability reduction
- iii) **Building Control and ensure compliance to Building code practices**

4.5.2 Spatial Planning. When a spatial planner receives a model showing that a magnitude 6.9 earthquake on the Madhupur Fault is going to collapse structures on Dhaka's soft, liquefied soil, they use spatial planning to fix the issue by:

- i) **Zoning Laws.** Banning the construction of high-rises on highly liquefiable soils.
- ii) **Infrastructure Routing:** Moving critical command centres, hospitals, and emergency supply roads into safer, seismically stable zones.
- iii) **Open Space Preservation:** Setting aside large public parks or playgrounds that can double as safe emergency evacuation camps during a disaster.

4.6 Establishment of a National Emergency Operation Centre

4.6.1 Objectives. The objectives of the National Emergency Operation Centre is going to be as shown below:

- i) Ensuring readiness among various agencies for anticipated Earthquake.
- ii) Provide Technical Assistance to institutions for plan development review and revisions.
- iii) Monitoring of reporting of the readiness by various first responder institutions

4.6.2 Organization Structure

- i) Planning Desk
- ii) Operation Desk
- iii) Logistics Desk
- iv) Administration and Logistics

4.6.3 Roles and Responsibilities of Various Desks. National Emergency Operation Centre is going to work 24x7. The National Emergency Operation Centre is going to work during peace and disaster time with all capacities.

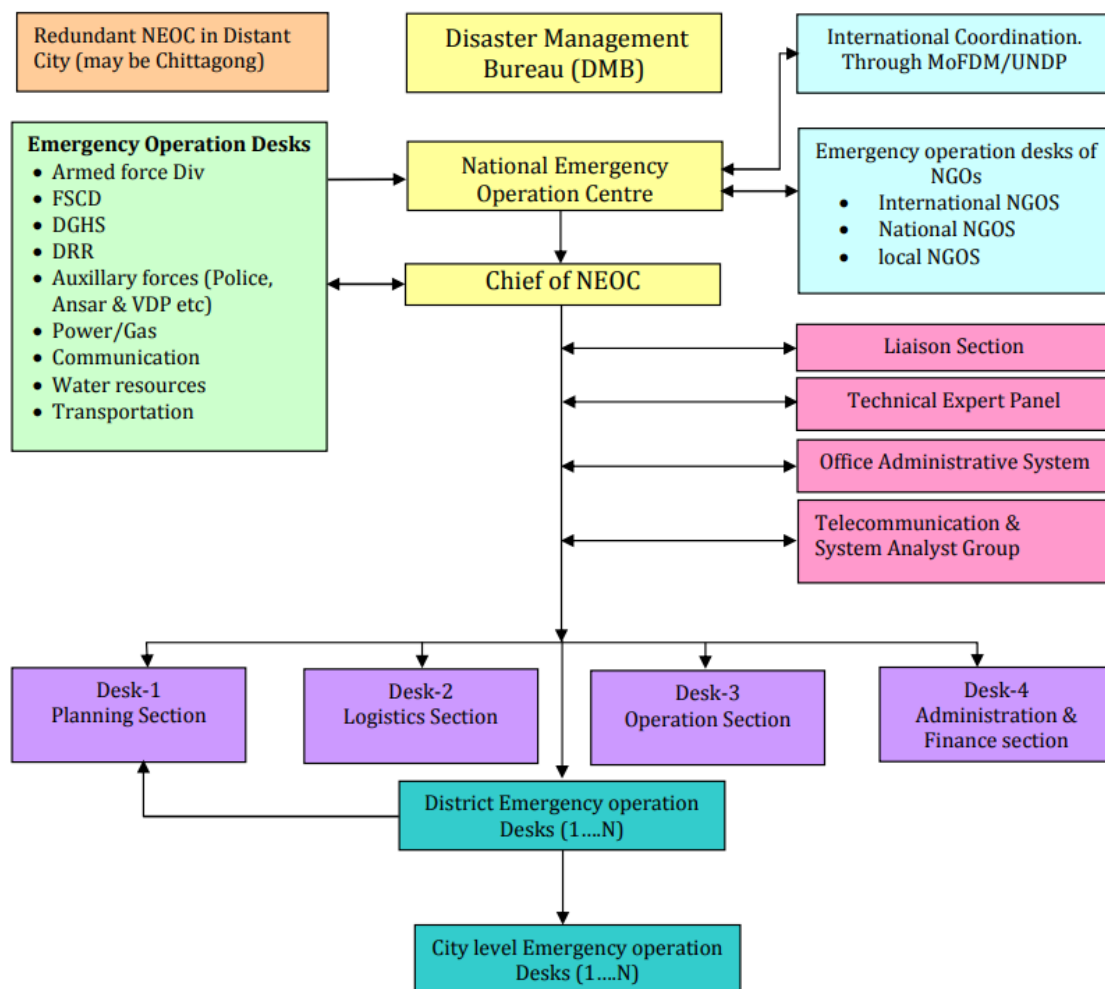


Figure 3.1 Organizational Structure for National Emergency Operation Centre
(Source: National Earthquake Contingency Plan)

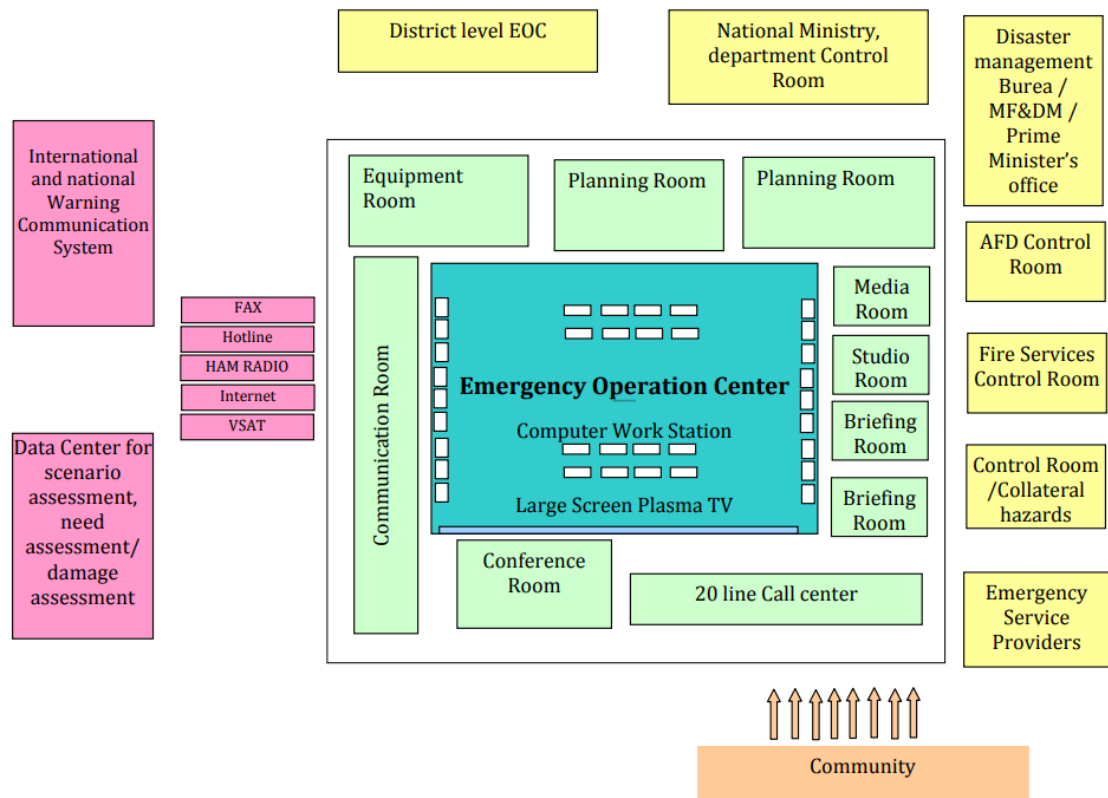


Figure 3.2 Emergency Operation Centre
(Source: National Earthquake Contingency Plan)

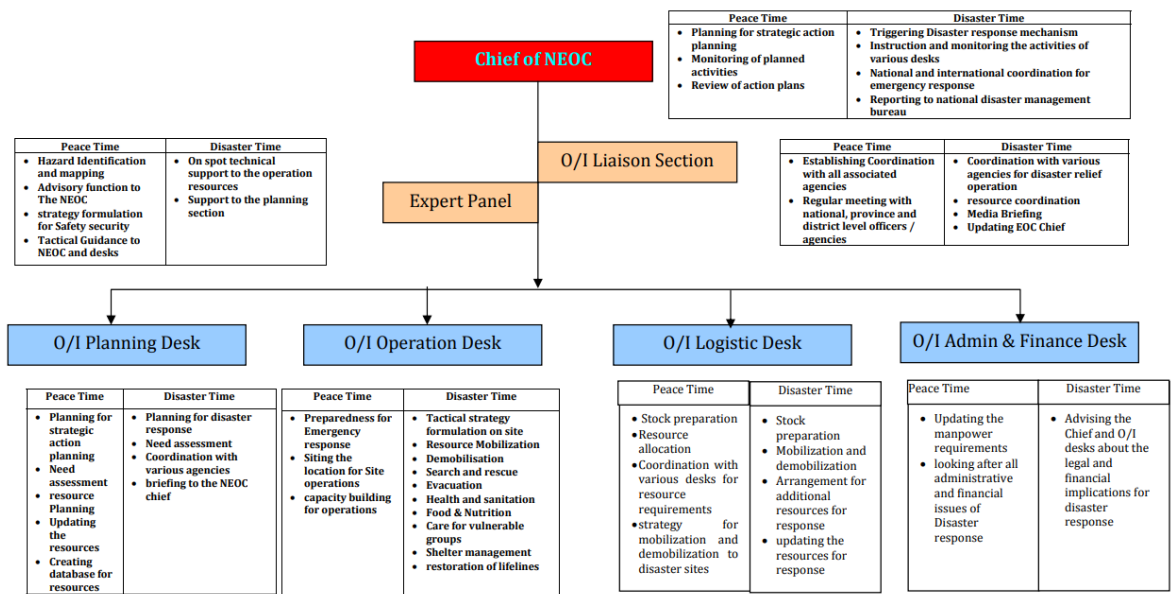


Figure 3.3 Roles and Responsibilities of National Emergency Operation Centre
(Source: National Earthquake Contingency Plan)

4.7 Establishment of Reporting System for Reporting the Readiness of Different Agencies. During the plan development stage each and every first responder agency should agree to provide a bi-annual report to the National Emergency Operation Centre about the preparedness level. Each agency to minimum provides information on:

- i) Whether the staff within the agency is familiar with of the Earthquake risk reduction strategy.
- ii) Orientation and Training for Earthquake Risk Reduction Strategy.
- iii) Understanding of each relevant officer.
- iv) The vulnerability assessment of buildings.
- v) Earthquake Precautionary measures and procedures.
- vi) No of Building which were strengthened to have standard of safety.
- vi) The vulnerability assessment of gas, electricity and water supply line.

4.8 Virtual Earthquake Engineering University. The virtual Earthquake Engineering University is being launched which is named as ‘Bangladesh Earthquake Engineering University’. The details are in Chapter 8.

4.9 Gas Sector Efficiency Improvement and Carbon Abatement Project. Installation of **Smart Prepaid Gas meter** for Titas Gas is under process where 11 lakh smart prepaid gas meter to be installed. The project estimated cost is 3542.71 crore taka. The duration is January 2024 to December 2028. The area of the project is Dhaka Distribution Division (North), Gazipur, Mymensingh. The Project being approved in Ecneec in last 31 Oct 2023.

CHAPTER 5

EARTHQUAKE THREAT AND POSSIBLE SCENARIOS IN DHAKA

5.1 Elements at Risk. In the current study three different scenarios have been developed to identify the possible damage to infrastructures, buildings, transportation and number of casualties. The scenarios are least, moderate and worst case as assumed based on different magnitude of earthquake.

5.1.2 Buildings Damage in Dhaka City Corporation Area- Earthquake of 5.7 Magnitude on 21/11/2025 at 10.38 minute. Origin was in Narsingdi Zilla Madhabdi area 10 km beneath the ground

1	Zilla	Information	Building Damaged	Source
01	Dhaka	Dead 04, Injured 59	10	Disaster management Bureau Dt 22/11/2025
Bangladesh	Total	Dead 10, Injured 461	50	

During an earthquake at 7.5 Mw originated from Madhupur fault, about 53,166 buildings will be moderately damaged. This is about 15.00 % of the total number of buildings in the city. About 72,316 buildings are going to be totally damaged during this time. If the magnitude of the earthquake is 8.0 Mw, about 34,326 buildings are going to be at least moderately damaged. During an earthquake originated from under the city at 6.0 Mw about 78,323 buildings are going to be completely damaged. An earthquake at 8.5 originated from plate boundary fault-2 is going to completely destroy 238,164 buildings.

5.1.3 Essential facilities Damage

i) Essential Facilities Damage in Dhaka City Corporation Area. There are going to be severe damage to essential facilities like Hospital, schools, police stations after different level of earthquakes. An earthquake at 7.5 Mw in Dhaka City, originated from Madhupur fault are going to moderately damage about 241 hospital or clinics. At the same time 10 hospitals are going to be completely damaged. About 90 schools are going

to be totally damaged at the same time. 30 police stations and 4 fire service stations are going to be moderately damaged during this earthquake.

5.2 Availability of Hospital Facilities after Earthquake in Dhaka City Corporation Area

In Dhaka there are about 59,849 hospital beds available for use. On the day after an earthquake of 7.5 Mw, it is estimated that only 24,242 hospital beds (41%) will be available for use by patients already in the hospital and those injured by the earthquake. After one week, 54% of the beds will be back in service. By 30 days, 72% will be operational. After an earthquake at 8.0 Mw only about 37,625 hospital beds (63%) are available for use by patients already in the hospital and those

5.3 Utility System Economic Loss (Millions of Dollars)

System	Component	Scenario 1			Scenario 2			Scenario 3			Scenario 4		
		Inventory Value	Economics Loss	Loss Ratio (%)	Inventory Value	Economics Loss	Loss Ratio (%)	Inventory Value	Economics Loss	Loss Ratio (%)	Inventory Value	Economics Loss	Loss Ratio (%)
Potable Water	Pipelines	13.10	1.61	12.32	13.10	0.80	6.13	13.10	1.74	13.29	13.09	0.80	6.13
	Facilities	140.20	20.58	14.68	140.20	10.18	7.26	140.20	36.32	25.91	140.19	10.18	7.26
	Subtotal	153.28	22.19		153.28	10.99		153.28	38.06		153.28	10.99	
Waste Water	Pipelines	2.90	3.18	111.13	2.90	1.98	69.15	2.90	3.92	137.22	2.86	1.98	69.15
	Facilities	60.60	5.33	8.79	60.6	4.13	6.81	60.60	10.02	16.53	60.62	4.13	6.81
	Subtotal	63.48	8.51		63.48	6.10		63.48	13.94		63.48	6.10	
Natural Gas	Pipelines	7.70	0.43	5.66	7.70	0.20	2.58	7.70	0.45	5.82	7.66	0.20	2.58
	Facilities	7.00	1.28	18.35	7.00	0.49	6.94	7.00	1.94	27.72	7.00	0.49	6.94
	Subtotal	14.66	1.72		14.66	0.68		14.66	2.39		14.66	0.68	
Electrical Power	Facilities	75.80	15.6	20.59	75.80	5.40	7.13	75.80	21.77	28.74	75.76	5.40	7.13
	Subtotal	75.76	15.6		75.76	5.40		75.76	21.77		75.76	5.40	
Communication	Facilities	81.00	11.24	13.87	81.00	5.20	6.42	81.00	19.79	24.44	81.00	5.20	6.42
	Subtotal	81.00	11.24		81.00	5.20		81.00	19.79		81.00	5.20	
Total		388.18	59.25		388.18	28.37		388.18	95.96		388.18	28.37	

Table 5.1 Utility System Economic Loss

(Source: Hazus calculation based on database, engineering geology and seismic hazard)

5.4 Earthquake Map of 50 Year Return Period- Dhaka Division. Earthquake hazard map of 50 year return period(Dhaka Division) is shown below:

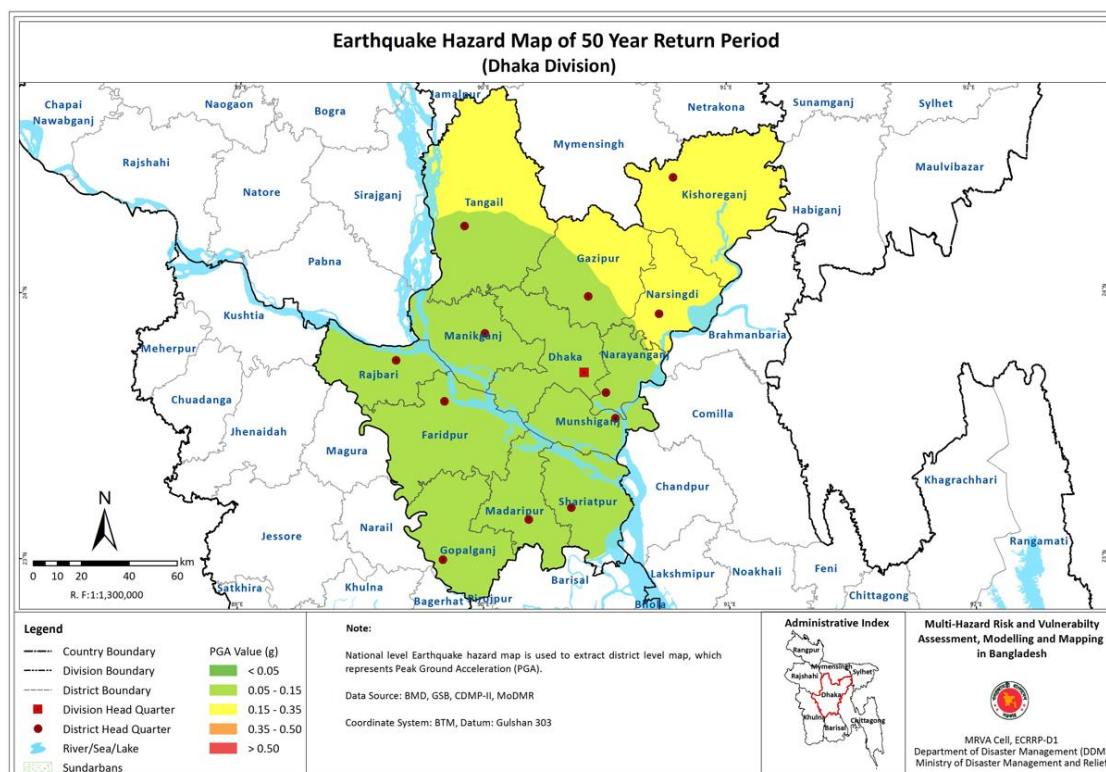


Figure 4.1: Dhaka Division Earthquake Map of 50 Year Return Period
(Source: Disaster Management Bureau)

5.5 Exposure Category of Population to Earthquake- Dhaka Division. Exposure Category of Population to Earthquake (Dhaka Division) is shown below:

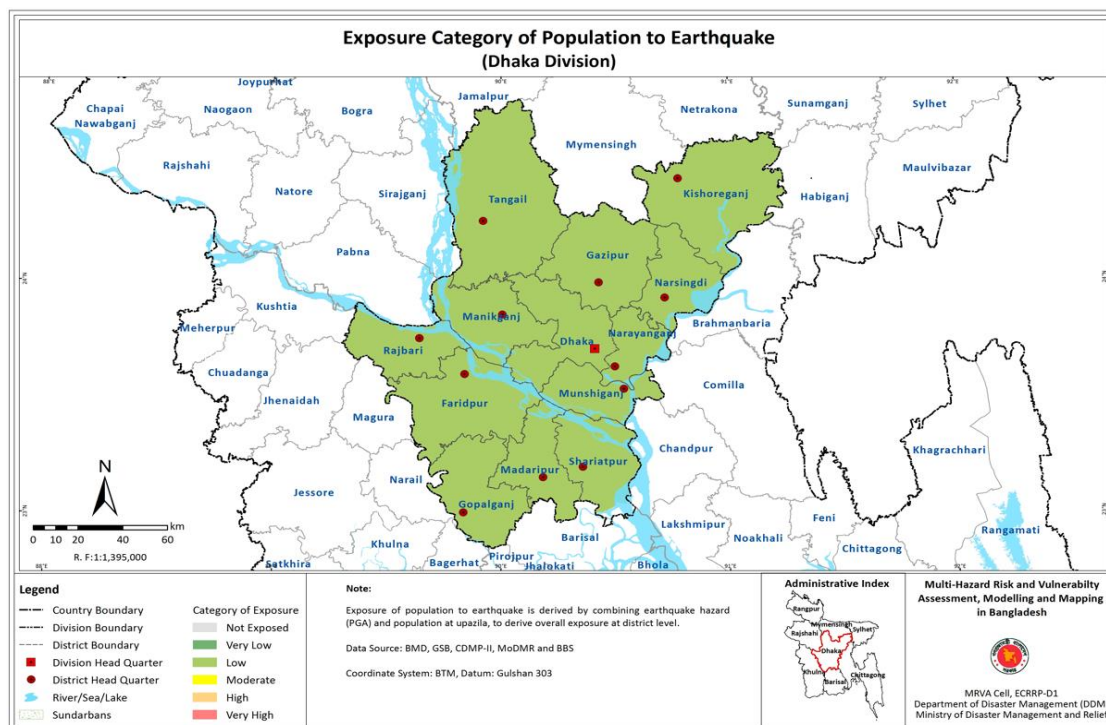


Figure 4.2: Exposure Category of Population to Earthquake- Dhaka Division
(Source: Disaster Management Bureau)

5.6 Risk Levels of Housing (Pucka and Semi Pucka) to Earthquake. Risk Levels of Housing (Pucka and Semi Pucka) to Earthquake (Dhaka Division)

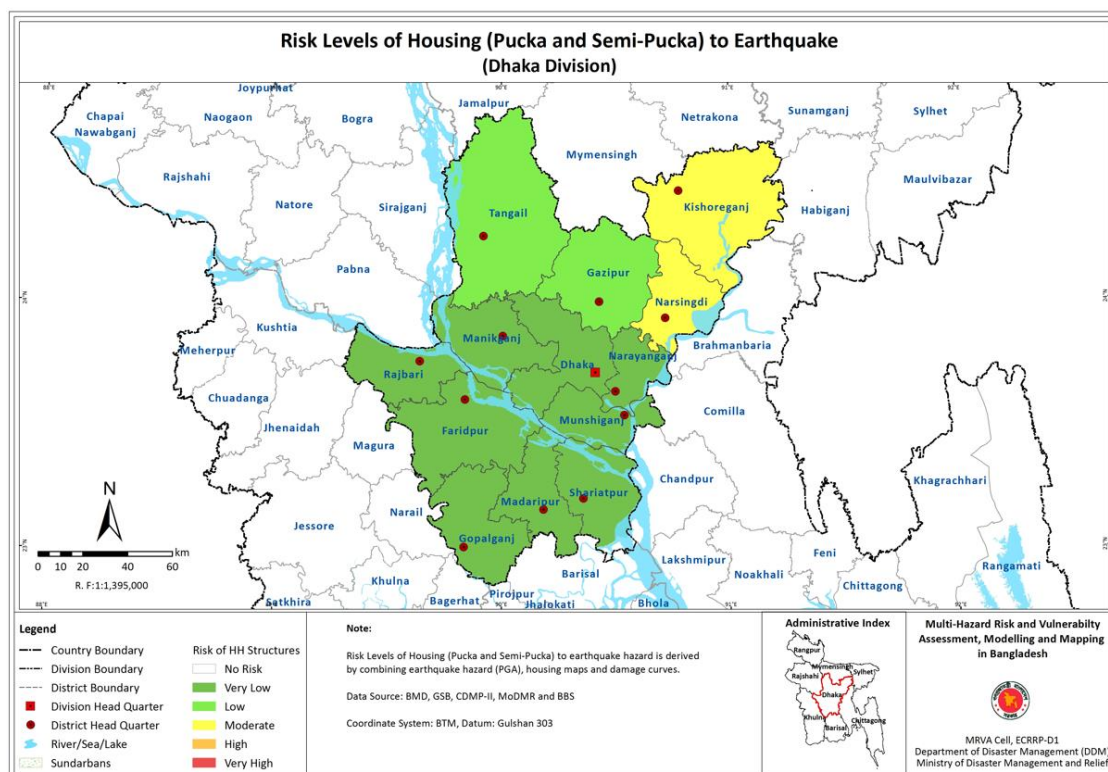


Figure 5.3: Risk Levels of Housing (Pucka and Semi Pucka) to Earthquake (Source: Disaster Management Bureau)

CHAPTER 6

ANALYSIS OF DATA AND RESULTS

6.1 Earthquake Resistant Building/Infrastructure (Area Wise- Dhaka): Can Withstand or Damage/Vulnerability Assessment and Prioritized Investment plan for Critical Assets in Dhaka

Funded through Government of Bangladesh and the World Bank in "Bangladesh Urban Resilience Project", the consultancy firm Joint Venture of Nky (Turkey) – Protek-Yapi (Turkey) – Sheltech (Bangladesh) was appointed for consultancy on "Vulnerability Assessment and Prioritized Investment Plan for Critical Assets in Dhaka." The consultancy firm estimated earthquake risk through Rapid Visual Assessment of 3252 number buildings, Preliminary Engineering Assessment of 579 buildings.

Table 6.1

The Distribution of Different Building Types in Dhaka Metropolitan Region

Structure Type	Count	Percentage (%)
RC	513,507	24%
Semi-RC/URM	877,620	41%
Others	735,660	34%
Under Construction	20,432	1%
Sum	2,147,219	100%

Source: Urban Resilience Project, Tech Note

The rapid visual assessment of 3252 buildings which includes schools & colleges, hospitals, police stations, other buildings also which are around 0.6% of total buildings in Dhaka Metropolitan Region.

Table 6.2 Rapid Visual Assessment

General Type	Number of buildings
Schools and Colleges	2,705
Hospitals	207
Police Stations	36
Others	304
Total	3,252

Source: Urban Resilience Project, Tech Note

6.1.1 Preliminary Engineering Assessment Result

579 buildings were surveyed and assessed in the Preliminary Engineering Assessment Stage. Therefore, preliminary engineering assessment done are 17.8% of total rapid visual assessment (3352 buildings) done.

Table 6.3 Preliminary Engineering Assessment

General Type	Number of buildings
Schools and Colleges	568
Hospitals	5
Fire Stations	3
Others	3
Total	579

Analysis of outcomes at this stage are demonstrated in Histogram. The age distribution of the buildings in terms of the selected benchmark years is shown in Figure 1. More than 51 percent of the buildings were constructed before 1996.

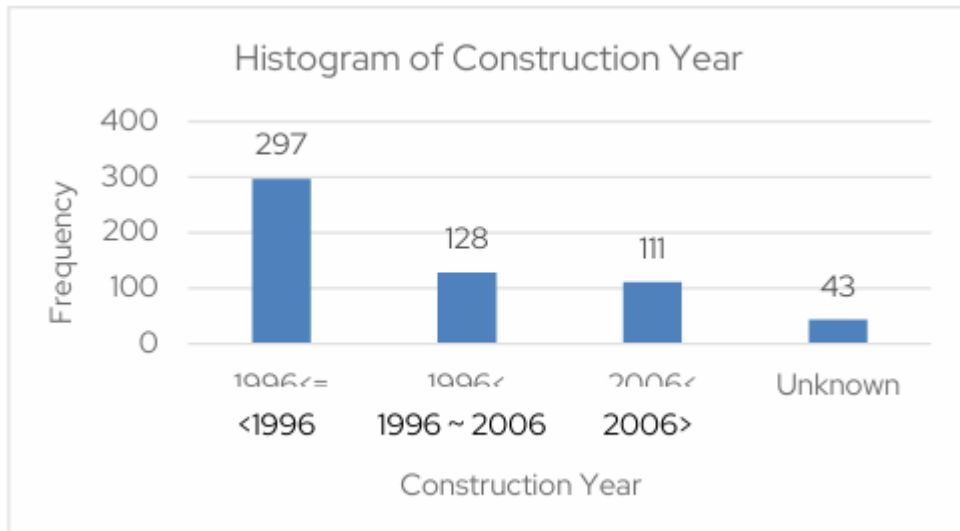


Figure 6.1 Consultancy Report of Rajuk 2022

The buildings surveyed revealed that most of the buildings are concrete moment resisting frame. About 380 buildings have reinforced concrete frame structural system, 98 buildings have unreinforced masonry structural system and 101 buildings have Dual system.

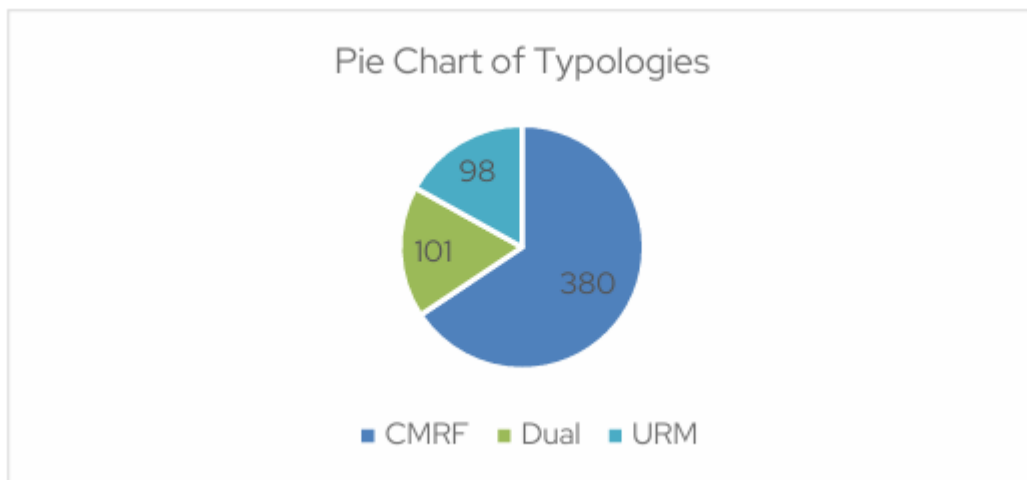


Figure 6.2: Distribution of Buildings for Types of Structural System

Figure 3 depicts the concrete strength of Concrete Moment Resisting Frame buildings. As shown, more than 79% of concrete strength of Concrete Moment Resisting Frame buildings are less than 25 MPa. In addition, less than 5% of buildings have concrete strength more than 30 MPa.

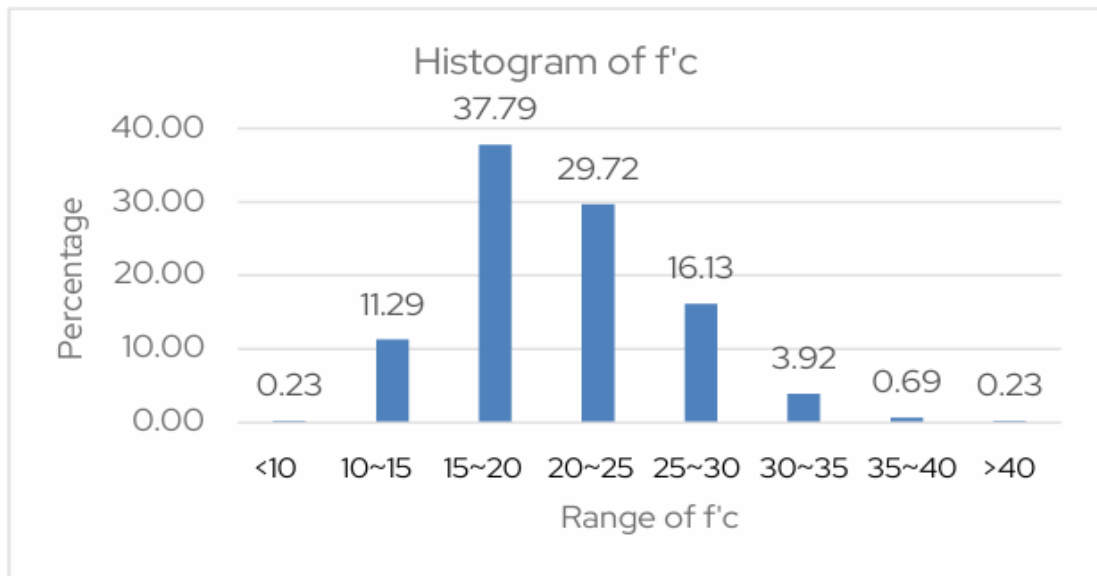


Figure 6.3: Concrete Strengths (Mpa) of Concrete Moment Resisting Frame Buildings

(Source: Vulnerability Assessment and Prioritized Investment Plan for Critical Assets, Rajuk Final Consultancy Report, 15 June 2022)

The concrete strength based on the construction year has been depicted in the following figure:

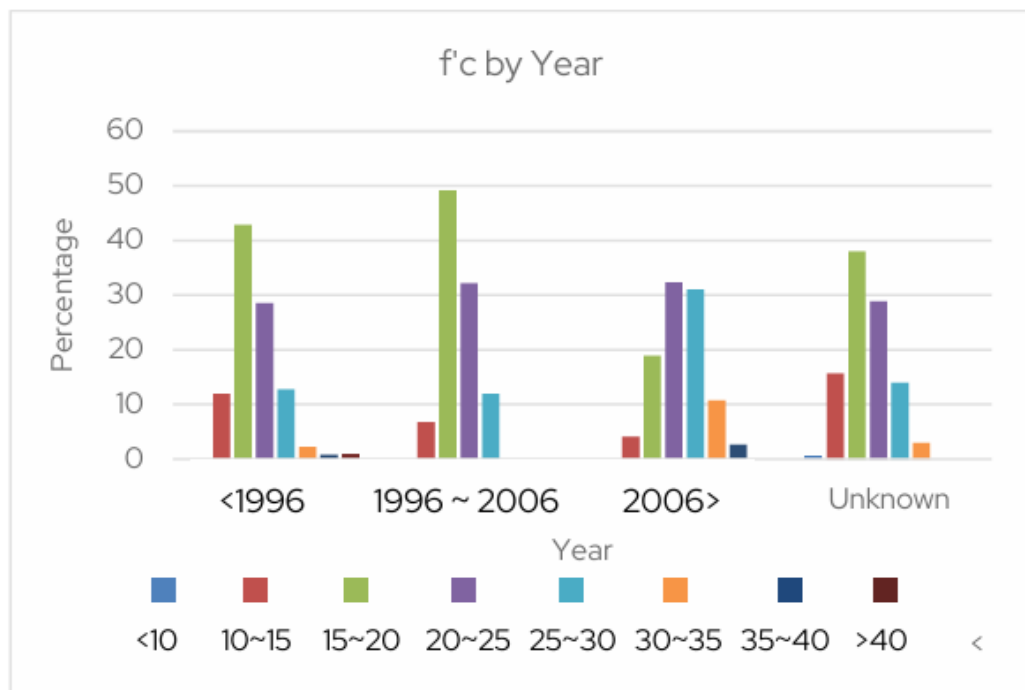


Figure 6.4: Distribution of Concrete Strengths (MPa) based on the Construction Year

As it can be seen the concrete strength is significantly increased after 2006. The distribution of buildings based on the function is demonstrated in figure 5 below:

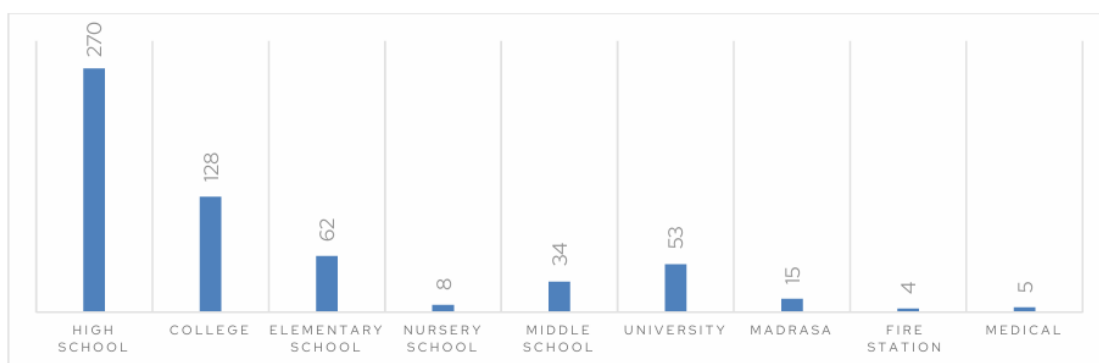


Figure 6.5: Distribution of Buildings based on the Function

The Histogram of most important deficiencies of Concrete Moment Resisting Frame buildings are illustrated in the following figure:

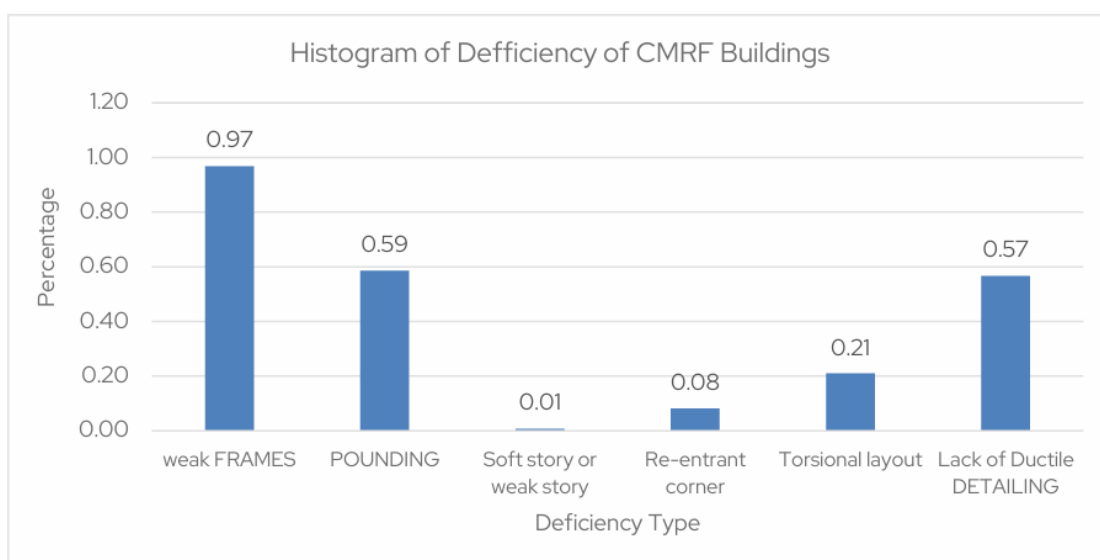


Figure 6.6: Histogram of Deficiencies of Concrete Moment Resisting Frame Buildings

It can be seen, most of buildings have the problems of weak frames (368 out of 380) and lack of ductile detailing (217 of 380). In addition, more than 59% of buildings have the risk of poundings since the separation joints were not sufficient. Also, more than 21% buildings have torsional layout.

6.1.2 Detail Engineering Assessment Results

Detail Engineering Assessment of 187 buildings (School/Colleges/Universities/Hospitals) were done which are of Government/Semi governments/Autonomous organisations which are within Dhaka Metropolitan Development Plan.

Table 6.4 Urban Resilient Project Summary (Category Wise)

General Type	Number of Buildings
Schools	94
Collages	78
Universities	11
Hospitals	4
Total (need to be retrofitted)	187
Total (need to demolished)	42

According to the recommendation of consultancy firm 187 risky buildings were recommended for retrofitting/strengthening and 42 very risky buildings were recommended to be demolished. The risky building list is appended below(Source: Rajuk):

Table 6.5 Urban Resilient Project Summary (Department Wise)

Sl. No.	Organization	Strengthening	To be Demolished
1	Bsmmu	–	3
2	Institute of leather Engineering and Technology	3	–
3	Jogonnath University	–	4
4	Department of Health Engineering	4	–
5	Madrasa Board	6	1
6	Directorate of Primary Education	10	1
7	Jahangirnagar University	10	3
8	Education Engineering Department	154	30
9	Total	187	42

The age distribution of the 187 risky buildings in terms of the selected benchmark years is presented in Figure below. 64 buildings were constructed before 1993, 66 buildings were constructed between 1993 till 2006 and about 57 buildings were constructed after 2006.

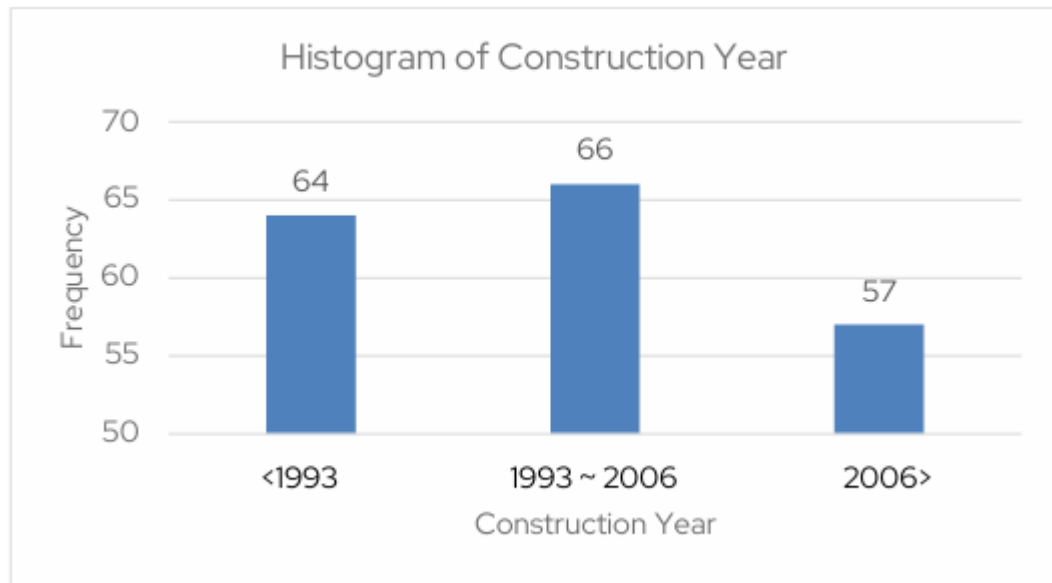


Figure 6.7: Distribution of Buildings according to Year of Construction

The building statistics in terms of number of stories are displayed in Figure 8. The majority buildings are of less than 6 stories (more than 87% buildings). 25 buildings out of 187 buildings, have more than 6 stories. This indicates most of the buildings are low-to mid-rise.

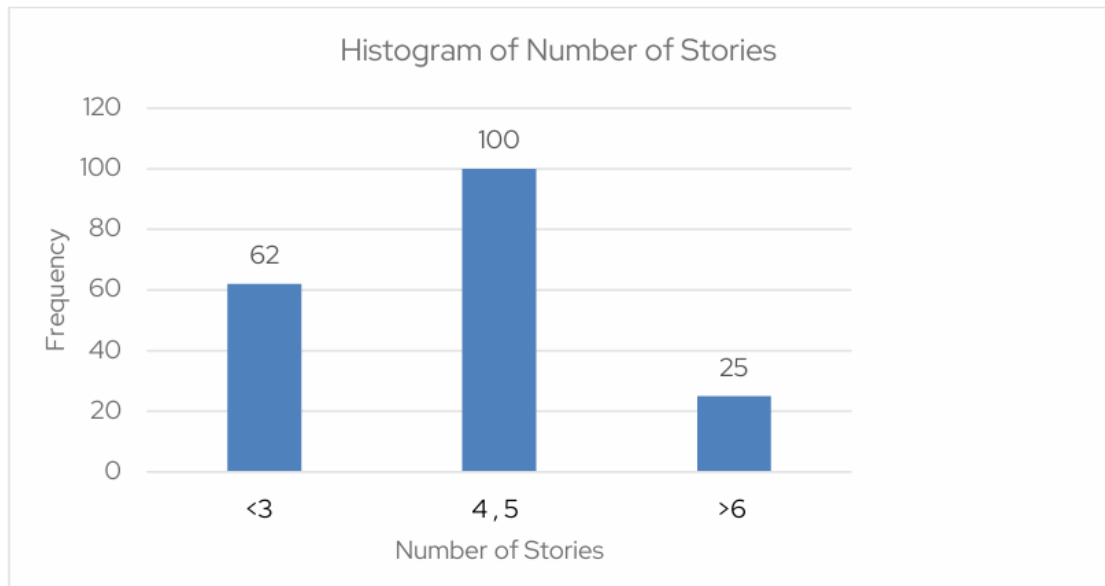


Figure 6.8: Number of Floor Distribution of the Buildings

The buildings surveyed revealed that most of the buildings are concrete moment resisting frame. Among 187 buildings 162 buildings have reinforced concrete frame structural system, 2 buildings have unreinforced masonry structural system and 23

buildings have Mixed system (Concrete moment resisting frame + unreinforced masonry)

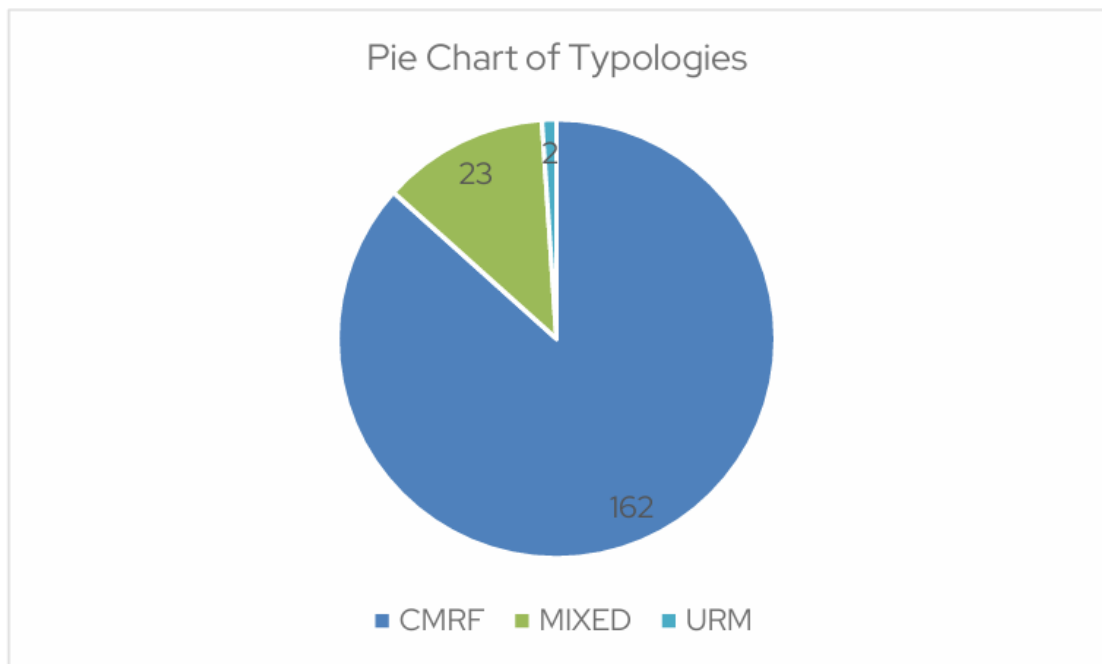


Figure 6.9: Distribution of Buildings for Types of Structural System

In the Detail Evaluation Analysis stage buildings are assessed for two performance levels. In this regard, for educational facilities, the vulnerability assessment of the structures is carried out for Damage Control and life safety performance levels, against an earthquake with 475 years and 2475 years return, respectively. For the health care facilities, the vulnerability assessment of the structures is carried out for immediate occupancy and Damage Control performance levels, against an earthquake with 475 years and 2475 years return, respectively. **Page 66**

The data reveals that out of 3252 buildings (187+42=)229 buildings are risky. That means in Dhaka city around 7.04% buildings are risky and $(42/3252)=1.29\%$ buildings are to be destroyed.

6.1.3 Seismic Hazard Analysis

A comprehensive seismic hazard/vulnerability analysis was performed for Dhaka Metropolitan Region under the Urban Resilience Project. An investigation into the structural geology regime indicates that the interaction between plates had formed a compressional regime in the region. After understanding the structural geology regime affecting Dhaka Metropolitan Region, the detail analysis was done within 300-km radius of Dhaka Metropolitan Region and potential seismic sources (faults) were identified. According to the scenario-based Seismic Hazard Analysis, the worst-case scenario in the study area is related to the Madhapur fault (a segment of Jamuna fault with a maximum magnitude of 6.9).

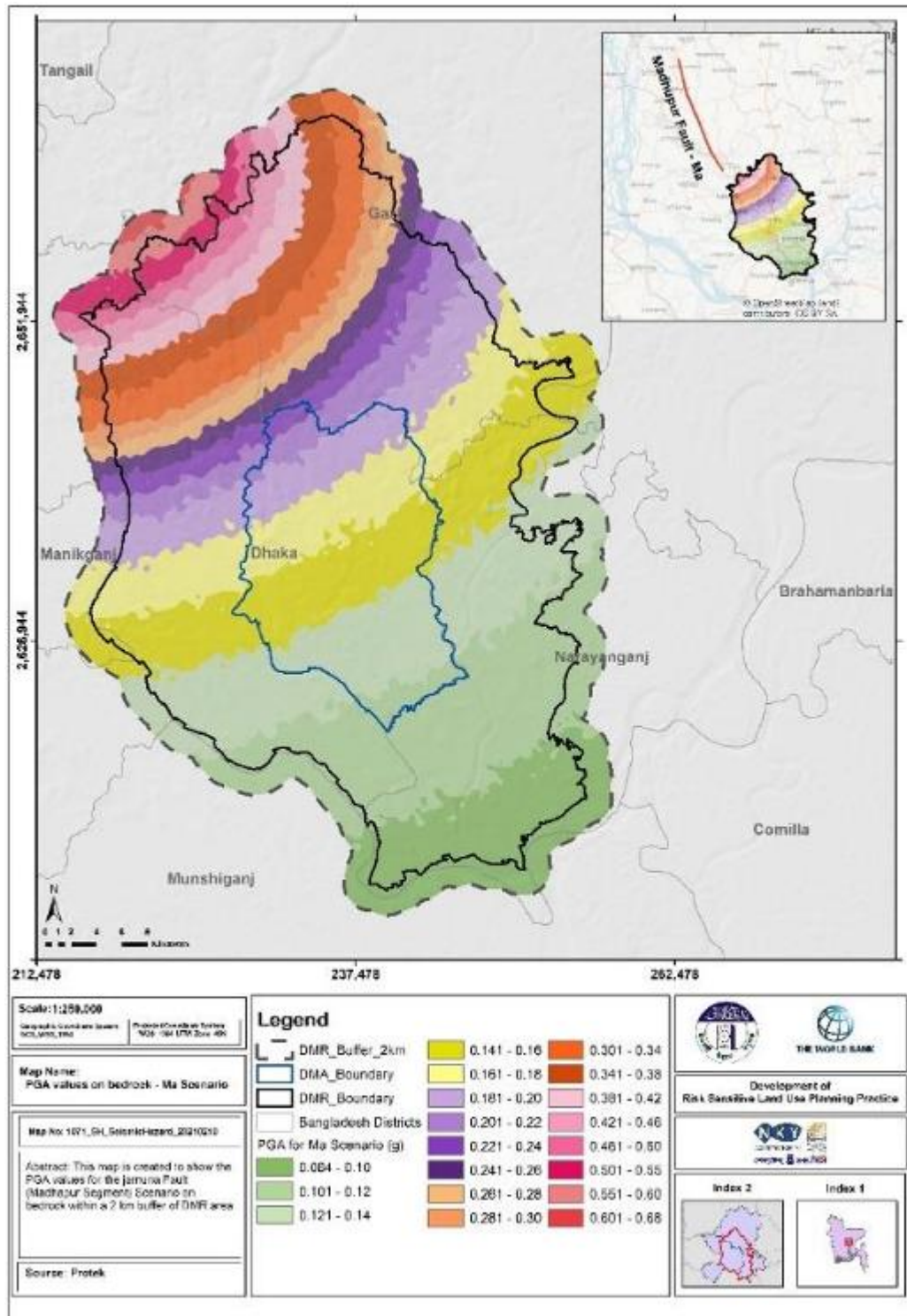


Figure 6.10: Scenario-based Peak Ground Acceleration map on the bedrock for Madhupur fault

Using the amplification factor obtained by applying seismic site response analysis as a reliable method, Peak Ground Acceleration on the surface was calculated for 10000 points in Dhaka Metropolitan Region .

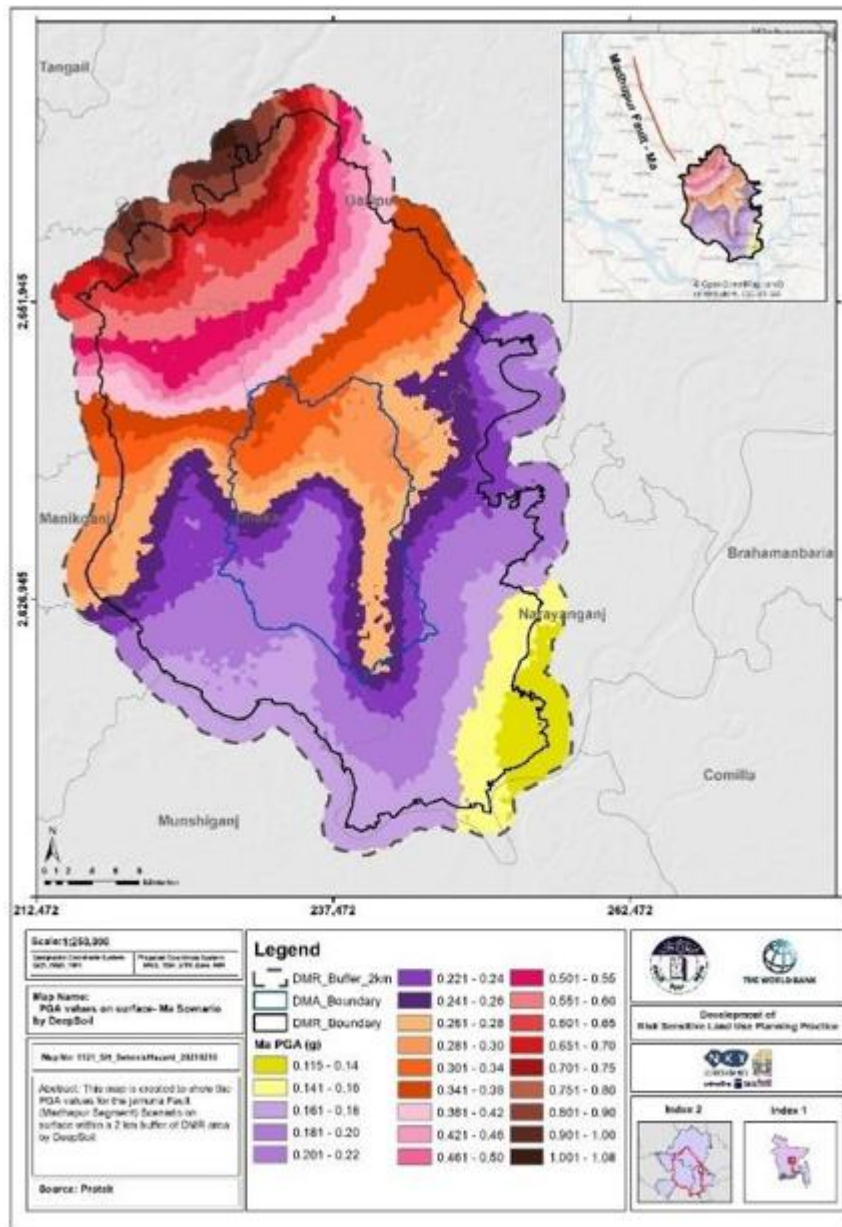


Figure 6.11: Peak Ground Acceleration zonation map on the surface using seismic site response analysis for Madhupur Fault

According to the scenario-based Seismic Hazard A, the worst-case scenario in the study area is related to the Madhupur fault (a segment of the Jamuna fault) with a maximum magnitude of 6.9. The Peak Ground Acceleration due to the Madhupur fault changes between 0.08 and 0.68 on bedrock.

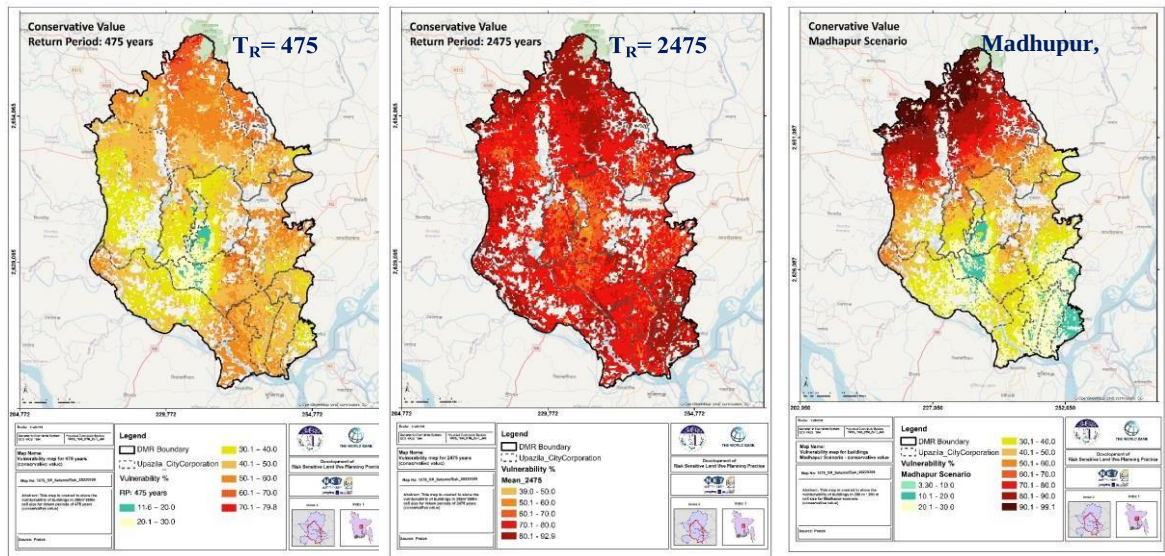


Figure 6.12 Vulnerability Map

The summary of estimated Peak Ground Acceleration on the bedrock and surface for various scenarios are presented:

Table 6.6 Peak Ground Acceleration on the bedrock and surface for various scenarios

Seismic Hazard Level/ Case		Scenario Magnitude	PGA (g) on seismic bedrock	PGA (g) on surface
PSHA	50 years		0.032-0.047	0.05-0.10
	75 years		0.044-0.066	0.07-0.14
	100 years		0.056-0.082	0.088-0.18
	200 years		0.089-0.130	0.14-0.29
	475 years		0.141-0.192	0.18-0.41
	2475 years		0.325-0.455	0.37-0.63
	10000 years		0.613-0.800	0.67-1.21
Scena	Chittagong Coastal Fault (segment 1)	7.2	0.058-0.100	0.074-0.186
	Chittagong Coastal Fault (segment 2)	6.8	0.035-0.061	0.045-0.117
	Dauki Fault (central segment)	7.6	0.044-0.069	0.06-0.14
	Dauki Fault (eastern segment)	7.1	0.030-0.042	0.04-0.08
	Dauki Fault (easternmost segment)	6.6	0.019-0.025	0.025-0.05
	Dauki Fault (western segment)	7.4	0.041-0.065	0.055-0.13

	Jamuna Fault	6.5	0.038-0.084	0.05-0.177
	Madhapur Fault	6.9	0.084-0.680	0.115-0.9
	Kaladan Fault 1 (first segment)	7.1	0.027-0.035	0.034-0.07
	Kaladan Fault 2 (second segment)	7.3	0.030-0.039	0.038-0.08
	Paoma2 Fault	6.9	0.025-0.040	0.033-0.078
	Paoma3 Fault	7.0	0.048-0.110	0.06-0.19
	Sylhet Fault	7.1	0.065-0.240	0.09-0.32

6.1.2 Seismic Vulnerability Analysis

The activities under seismic vulnerability analysis involve developing exposure models for both buildings and infrastructures. The proposed taxonomy for buildings is based on building type (concrete, masonry, and Katcha), building height (high-rise, mid-rise, and low-rise), and building quality (high, medium, and low); and consists of ten different categories.

The transportation system, gas system, potable and wastewater, and electrical power network were investigated as infrastructures and lifelines. In this regard, two main components were considered for transportation systems containing roads and bridges. Two classes were suggested for roads: urban roads and major roads. Also, major bridges, urban bridges, and pedestrians were defined as three main categories of bridges.

The gas system consists of pipelines which all have steel material. The electrical network contains substations (low voltage and medium voltage), electrical poles and powerlines (unanchored standard components), and generation plants (medium and small size).

Total 2,146,626 numbers buildings were analysed. The number and percentage of damaged buildings according to various seismic scenarios (the return period of 475 and 2475 years, Madhupur fault, and Sylhet lineament) are presented in Table 4–2. In this table, the thresholds of 40%, 35%, and 30% were considered for reinforced concrete, masonry, and Katcha building to determine whether a building is damaged. Also, the distribution of the building vulnerability map ratio for the expected earthquakes hazard with the return period of 475 and 2475 years and for the Madhapur fault with M6.9 is shown in Figure 4–5.

Table 6.7: Number and Percentage of damaged buildings for various scenarios

Taxonomy	Total No. of Buildings	Damaged Buildings	TR= 475 years	TR= 2475 years	Madhupur Fault	Sylhet Lineament
KA_LR_LQ	735,416	Number	578,472-735,416	735,416	395,703-673,877	40,935-299,587
		%	79-100 %	100%	54-92 %	6-41 %
MA_LR_LQ	893,108	Number	149,683-699,375	893,108	401,740-616,130	0-15,086
		%	17-78 %	100%	45- 69 %	0-2 %
MA_MR_LQ	763	Number	323-763	763	511-726	0-50
		%	42-100 %	100%	67-95 %	0-7 %
RC_LR_LQ	162,445	Number	605-44,623	162,445	47,735-69,756	0-19
		%	0.37-27 %	100%	29-43 %	0-0.01%
RC_LR_MQ	205,709	Number	0	114,762-204,958	9,903-13,143	0
		%	0%	56-99.6 %	5-6 %	0%
RC_MR_LQ	44,122	Number	115-8,778	44,122	7,310-14,457	0
		%	0.26-20 %	100%	17-33%	0%
RC_MR_MQ	95,675	Number	0	82,482-95,675	1,193-2,093	0
		%	0%	86- 100%	1-2 %	0%
RC_HR_LQ	3,226	Number	53-1,263	3,226	482-1,314	0
		%	2-39 %	100%	15-41 %	0%
RC_HR_MQ	5,668	Number	0-18	5,657-5,668	42-189	0
		%	0-0.32 %	99.8-100%	1-3 %	0%
RC_HR_HQ	494	Number	0	1-195	0	0
		%	0%	0.2-39 %	0%	0%
All Building	2,146,626	Number	729,251-1,490,236	2,041,982-2,145,576	864,619-1,391,685	40,935-314,742
		%	33.97-69.42%	95.13-99.95%	40.28-64.83%	1.91-14.66%

6.2 Emergency Gas Control and Management Statistics- Titas Gas

Natural gas is the primary energy source of Bangladesh. Approximately 2,300 million cubic feet of gas are distributed daily to more than 2.3 million consumers. Natural gas plays a vital role in electricity generation, fertilizer production, industry, commerce, residential use etc.

Natural gas is a finite resource, ensuring the efficient use and proper management is the responsibility of the Government. Therefore, the Gas Distribution Rules, 2004 were updated, resulting in the promulgation of the Gas Distribution Rules, 2014.

6.2.1 Gas Complain and Repair (July 2025 to May 2026). Gas Complain Summary shown below:

Table 6.7 Gas complain Summary

Month	Leakage Complain	Total Registered Complain	Leakage Complain Addressed/ Repaired	Remarks
July 25 to June 2026	3694	6466	6466	

(Source: Titas Gas)

6.2.2 Gas vigilance Team. Gas Vigilant operation summary shown below:

Table 6.8 Gas Vigilant operation summary

Month	Vigilant Operation	Domestic Stove Numbers	Commercial Connection Numbers	Industrial Connection Numbers	Spot Numbers	Evicted Pipe Length (feet)
Dec 2025	361	10596	53	61	859	21883
Jan 2026	289	8443	30	46	595	11319
Feb 2026	182	3431	16	24	349	7349
March 2026	143	4709	21	31	260	16702
April 2026	432	12789	42	44	1127	15188
May 2026	289	16801	40	46	788	17483

(Source: Titas Gas)

6.3 Structural Model Analysis. A four storied structural model focusing earthquake is analyzed. The story response is shown below:

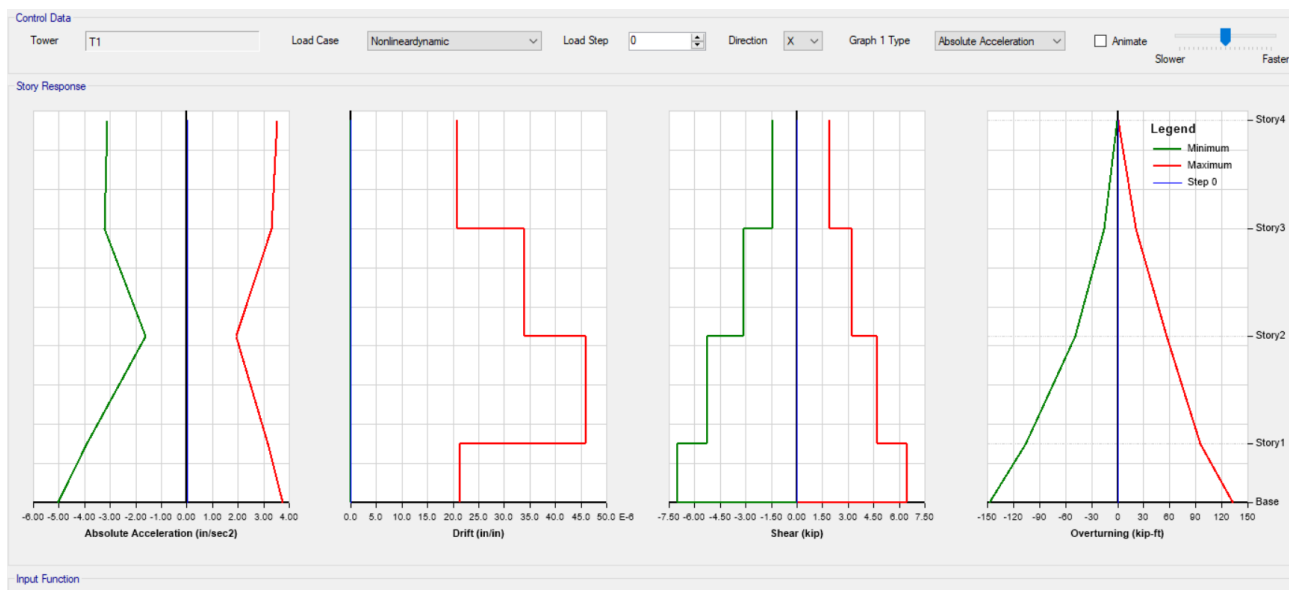


Figure 6.13: Story Response

CHAPTER 7

DISCUSSION

7.1 Bangladesh sits over a complex network of active fault lines, notably the Madhupur Fault in Central Bangladesh and the Dauki Fault system in Sylhet. Between 1869 and 1930, the country experienced five major earthquakes that exceeded a magnitude of 7.0 on the Richter scale.

The most destructive historical event was the Great Indian Earthquake of 1897. This 8.7 magnitude tremor caused catastrophic losses in Dhaka, Sylhet, Mymensingh and Rangpur. More recently a 5.7 magnitude earthquake on November 21, 2025, with its epicenter in Narsingdi, killed 10 people, injured over 461, 50 buildings were destroyed across the country.

Seismologists state that these recurring minor to moderate tremors are clear indicators of stress accumulation along major underground faults.

7.2 High Risk Areas and the Threat to Dhaka. According to the official seismic zoning map of Bangladesh, the country is split into distinct zones of risk:

7.2.1 High Risk (Zone I): The northern and northeastern regions, including Sylhet, Rangpur and Dinajpur are highly unstable due to their proximity to the active Dauki Fault and the Himalayan front.

7.2.2 High Density Risk (Zone II): The central region, which includes the Dhaka metropolitan area, falls under a major risk zone due to the nearby Madhupur Fault.

7.3 What is going to Happen in Dhaka.

A comprehensive survey conducted by the capital's development authority, Rajdhani Unnayan Katripakha in 2003, reveals an alarming scenario. If a 6.9 magnitude earthquake strikes Madhupur Fault near Tangail, approximately 865,000 buildings in

Dhaka could collapse. If this occurs during the daytime, it is projected to cause 210,000 deaths and 229,000 injuries.

The total economic damage is estimated at Usd 25 billion(Tk 2.62 trillion), with an additional Usd 44 billion required for rebuilding. Even a distant 7.1 magnitude quake on the Dauki Fault in Sylhet would collapse over 40,000 buildings within Dhaka city due to the fragile state of its soil and structures.

7.4 Major Concerns-Dhaka at Risk

7.4.1 Housing Regulation and Maintenance. The primary factor driving Bangladesh's high vulnerability is the widespread disregard for safety regulations. The institute for Planning and Development notes that urban centers face extreme risks because building codes, master plans and land use zoning are routinely ignored.

7.4.2 Illegal Expansion and Weak Ground. Over 80% of buildings in the capital lack proper clearance from Rajuk and 60% of approved buildings were constructed by violating their original structural designs. Many are built illegally on filled up wetlands and marshy tracts that suffer from severe soil liquefaction during a tremor.

7.4.3 Soft-Storey Structures. Thousands of buildings use the ground floor entirely for car parking without installing reinforced concrete shear walls. These "soft stories" easily pancake under horizontal seismic forces.

7.4.4 Maintenance and Fire Risk. Most high rise buildings lack standard fire safety measures, escape routes, or regular structural maintenance, creating a secondary hazard environment where gas and electrical line explosions would follow any tremor.

7.5 Expert Opinions on Structures and Prevention

7.5.1 Structural Condition Assessment

The late Professor Jamilur Reza Choudhury, a prominent construction and earthquake expert, in his lifetime repeatedly warned that a large percentage of buildings in Dhaka flouted the building construction rules, placing millions of lives at risk.

Professor Mehedi Ahmed Ansary (2019) from the Department of Civil Engineering at the Bangladesh of Engineering and Technology emphasized that nearly 90 percent of older structures in the capital completely bypassed modern seismic design codes. He points out that the unchecked boom of high-density housing has left no space between buildings, which is going to cause a domino effect of collapses and block narrow roads, making rescue operations impossible.

7.5.2 Expert Advice on Prevention. To avoid a mass casualty catastrophe, urban planners and structural engineers recommend immediate, legally binding steps:

- i) **Independent Regulatory Authority.** Professor Ansary advises that the government must establish a dedicated urban planning and building safety authority that operates independently of bureaucratic and real estate developer influence.
- ii) **Mandatory Retrofitting and Enforcement.** Old, highly vulnerable buildings must be reinforced using modern engineering practices or demolished. The Bangladesh National Building Code must be strictly enforced for all new construction.
- iii) **Adapting Seismic Isolation Technology.** Adapting Seismic Isolation Technology. Experts suggest transitioning toward advanced construction methods like “base isolation” a system using rubber lead bearings under foundations to absorb seismic shocks. Important public structures, select metro rail stations and new power generation centers in Bangladesh have already begun incorporating this technology.

iv) **Community Readiness.** The government must fast track the creation of dedicated urban emergency shelters and cultivate public awareness through regular drills. Currently, the ministry of Disaster Management and Relief has identified 445 safe shelters across Dhaka and to create a database of 100,000 local volunteers to assist in post disaster rescue.

7.6 Infrastructure/Soil Condition in Dhaka if Earthquake Occurs

National Earthquake Hazards Reduction Program (Nehrp) has developed a site classification system representing the soil types and classes. It is an internationally well-established site classification to date.

Table 7.1: Nehrp Site Classification

Soil Type	Brief Description	Average shear wave velocity at 30m (m/s)
A	Hard Rock	$V_s > 1500$
B	Rock	$760 < V_s \leq 1500$
C	Hard soil, stiff/very stiff soil, mostly gravels	$360 < V_s \leq 760$
D	Sand, silts, stiff or very stiff clay, some gravels	$180 < V_s \leq 360$
E	Soft clay	$V_s < 180$
F	Need specific calculations	--

All the 87 borehole locations were analyzed, as shown above. The results are as follows:

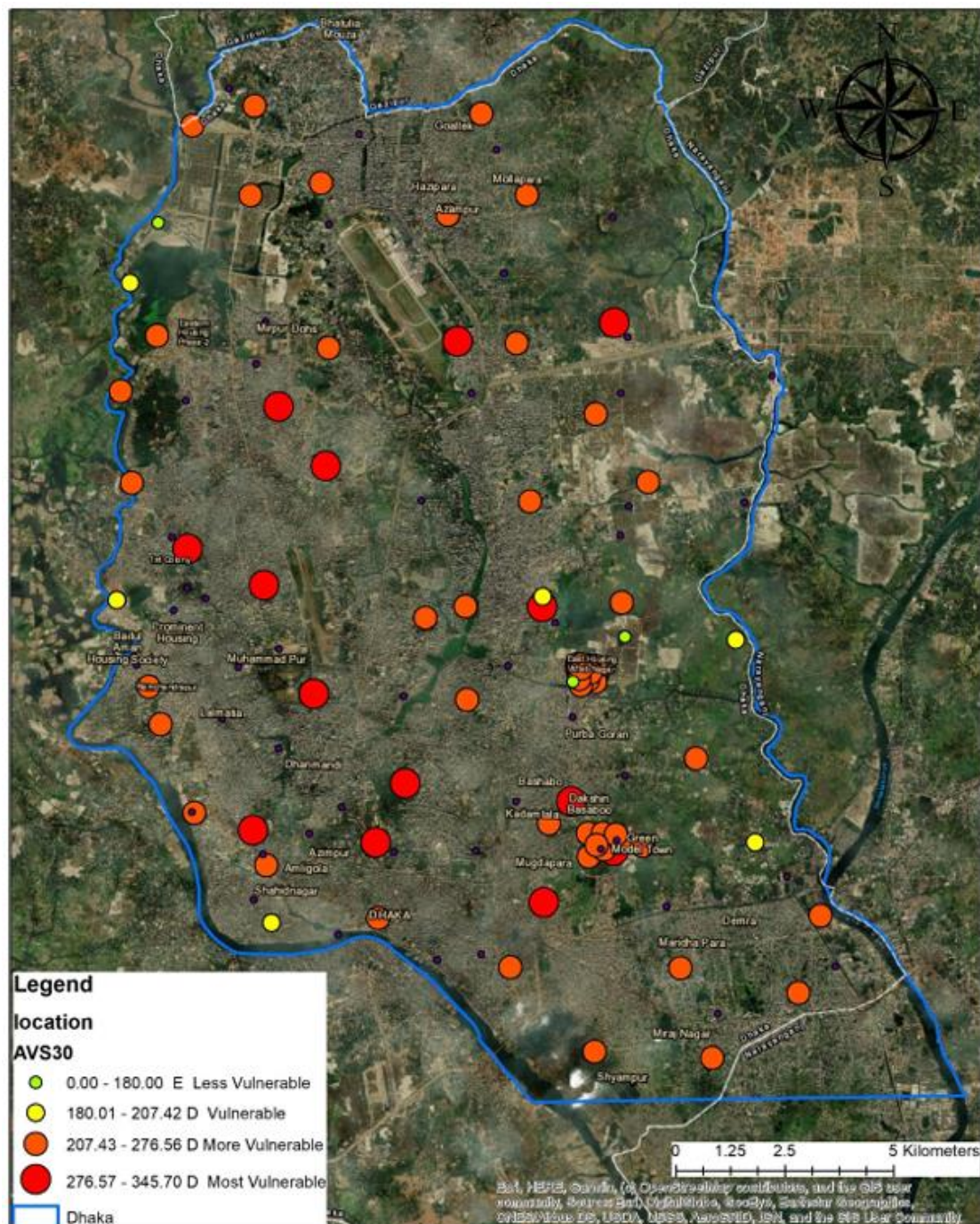
Table 7.2: Soil Condition in Dhaka City

No	Borehole	Average Shear Wave Velocity	Amplification Factor	Soil Type	Soil Class
1	DK_BM2948	131.34	1.92	Soft Soil	E
2	DK_BM2897	160.47	2.87	Soft Soil	E
3	DK_BM2914	167.42	3.2	Soft Soil	E
4	DK_BM1866	181.73	3.35	Stiff Soil	D
5	DK_BM2969	188.72	3.08	Stiff Soil	D
6	DK_BM2971	189.17	2.38	Stiff Soil	D
7	DK_BM2976	195.27	2.93	Stiff Soil	D
8	DK_BM1863	195.64	2.65	Stiff Soil	D
9	DK_BM2952	206.3	2.63	Stiff Soil	D
10	DK_BM3276	207.72	3.07	Stiff Soil	D
11	DK_BM3282	208.03	3.53	Stiff Soil	D
12	DK_BM2905	208.45	2.82	Stiff Soil	D
13	DK_BM2917	208.45	2.82	Stiff Soil	D
14	DK_BM2931	210.55	2.67	Stiff Soil	D
15	DK_BM2994	211.47	2.56	Stiff Soil	D
16	DK_BM3293	212.33	3.41	Stiff Soil	D
17	DK_BM3284	215.28	3.48	Stiff Soil	D
18	DK_BM2890	215.69	2.66	Stiff Soil	D
19	DK_BM3266	219.31	3.45	Stiff Soil	D
20	DK_BM2978	219.56	2.49	Stiff Soil	D
21	DK_BM2993	219.62	2.44	Stiff Soil	D
22	DK_BM2921	220.47	2.41	Stiff Soil	D
23	DK_BM2956	222.23	2.42	Stiff Soil	D
24	DK_BM3272	224.82	2.81	Stiff Soil	D
25	DK_BM2981	226.76	2.31	Stiff Soil	D
26	DK_BM3277	227.53	2.72	Stiff Soil	D
27	DK_BM2962	228.3	2.38	Stiff Soil	D
28	DK_BM3290	229.94	3.2	Stiff Soil	D
29	DK_BM3271	230.17	2.87	Stiff Soil	D
30	DK_BM2922	231.79	2.36	Stiff Soil	D
31	DK_BM2911	232.03	2.33	Stiff Soil	D
32	DK_BM3278	233.42	2.88	Stiff Soil	D
33	DK_BM3279	233.67	2.95	Stiff Soil	D

No	Borehole	Average Shear Wave Velocity	Amplification Factor	Soil Type	Soil Class
34	DK_BM2913	234.01	2.44	Stiff Soil	D
35	DK_BM3270	234.98	2.96	Stiff Soil	D
36	DK_BM3288	236.49	3.07	Stiff Soil	D
37	DK_BM3297	236.75	2.63	Stiff Soil	D
38	DK_BM2972	238.1	2.28	Stiff Soil	D
39	DK_BM3296	239.97	3.01	Stiff Soil	D
40	DK_BM3280	241.3	3.25	Stiff Soil	D
41	DK_BM3274	241.57	2.89	Stiff Soil	D
42	DK_BM2979	243.28	2.23	Stiff Soil	D
43	DK_BM2974	244.3	1.94	Stiff Soil	D
44	DK_BM3275	244.44	2.81	Stiff Soil	D
45	DK_BM3295	246.26	2.98	Stiff Soil	D
46	DK_BM2891	246.74	2.19	Stiff Soil	D
47	DK_BM3291	246.87	2.92	Stiff Soil	D
48	DK_BM2906	247.21	2.23	Stiff Soil	D
49	DK_BM3292	247.27	2.95	Stiff Soil	D
50	DK_BM3294	247.66	2.81	Stiff Soil	D
51	DK_BM2918	249.25	2.2	Stiff Soil	D
52	DK_BM3312	250.77	2.73	Stiff Soil	D
53	DK_BM3283	251.68	2.62	Stiff Soil	D
54	DK_BM2959	256.06	2.26	Stiff Soil	D
55	DK_BM2849	256.27	2.2	Stiff Soil	D
56	DK_BM3285	258.19	2.84	Stiff Soil	D
57	DK_BM2950	258.42	2.48	Stiff Soil	D
58	DK_BM2975	261.93	2.12	Stiff Soil	D
59	DK_BM2942	264.57	2.03	Stiff Soil	D
60	DK_BM2899	264.92	2.1	Stiff Soil	D
61	DK_BM3273	265.07	2.65	Stiff Soil	D
62	DK_BM2904	265.39	2.09	Stiff Soil	D
63	DK_BM3267	267.16	2.08	Stiff Soil	D
64	DK_BM2910	267.27	2.12	Stiff Soil	D
65	DK_BM2951	267.57	1.99	Stiff Soil	D
66	DK_BM1865	268.47	1.92	Stiff Soil	D
67	DK_BM3281	268.61	2.75	Stiff Soil	D

No	Borehole	Average Shear Wave Velocity	Amplification Factor	Soil Type	Soil Class
68	DK_BM2943	272.26	2.09	Stiff Soil	D
69	DK_BM3286	274.38	2.6	Stiff Soil	D
70	DK_BM2947	279.61	1.97	Stiff Soil	D
71	DK_BM3287	279.65	2.58	Stiff Soil	D
72	DK_BM3303	279.86	2.55	Stiff Soil	D
73	DK_BM2903	283.55	2.9	Stiff Soil	D
74	DK_BM2919	283.73	1.98	Stiff Soil	D
75	DK_BM3269	283.93	2.12	Stiff Soil	D
76	DK_BM3268	290.19	1.94	Stiff Soil	D
77	DK_BM2963	303.74	1.8	Stiff Soil	D
78	DK_BM2977	304.32	1.8	Stiff Soil	D
79	DK_BM2900	304.69	1.58	Stiff Soil	D
80	DK_BM2902	310.41	1.78	Stiff Soil	D
81	DK_BM2973	321.33	1.41	Stiff Soil	D
82	DK_BM2961	323.2	2.13	Stiff Soil	D
83	DK_BM2912	327.02	1.9	Stiff Soil	D
84	DK_BM2958	332.19	1.69	Stiff Soil	D
85	DK_BM3309	339.72	2.18	Stiff Soil	D
86	DK_BM3309	339.72	2.18	Stiff Soil	D
87	DK_BM3311	345.7	2.03	Stiff Soil	D

The peripheral areas of Dhaka once were water bodies, wetlands and flood plains land. Due to rapid growth and urbanization, the khals, canals, and water bodies were changed into urban areas. It was found that most of the area of Dhaka city can be classified as D according to Nehrp soil classification (Table: 7.2). Only three locations were found as E where the average shear wave velocity was below 180 m/s.



(Source: A Japan-Bangladesh Joint Research Project)

The peripheral areas of Dhaka are more vulnerable in terms of shear wave velocity. It happened because of landfilling, which was previously low land and wetland, and developed without proper maintenance of rules and regulations, making the areas more vulnerable to seismic hazards and disasters

Figure 7.1, for which the amplification factors are interpolated to grids in Dhaka city area, shows that some of the peripheral areas of Dhaka are more vulnerable in terms of

amplification factors, making the areas more vulnerable to seismic hazards and disasters. It shows that some parts of the north-western region of Dhaka are most vulnerable; in the late 80's it was primarily wetlands. It is also applicable for the eastern part and south-eastern part of Dhaka. The amplification factor also shows that Dhaka's areas are not evenly distributed in terms of soil amplification.

The central part of this city consists of upper, middle, and lower Madhupur terraces. The peripheral areas have consisted of high floodplains and low floodplains with wetland, marshland, and water bodies. Over the year, to manage the enormous population pressure, Dhaka started expanding towards the peripheral areas. Land development turned into a necessity for the inhabitants as well as for the government. Unplanned development without monitoring made this city seismically more vulnerable. After analysing 87 borehole data, it was found that the soil type of Dhaka is 'D' according to the Nehrp soil classification system. On the other hand, the amplification factor of this city is also considered very high. Especially the eastern part, north-western part and some portions of the western part are considerably high amplifiable areas. More in-depth studies in the geotechnical area are required for more accurate and reliable results from academicians and professionals.

The stratified areas showing some of the areas are vulnerable, and some are less vulnerable, but the average value indicating the amplification situation is not safe enough for generalization. There needed the most crucial intervention for making the planning rules and regulations for Dhaka.

7.7 Vulnerability assessment of critical and essential infrastructures in Dhaka (Findings of the Urban Resilience Project) :

- i) Rapid Visual Assessment (RVA) done for total number of buildings=3252
- ii) Preliminary Engineering Assessment (PEA) done for total number of buildings=579
- iii) Detail Engineering Assessment (DEA) done for total number of buildings=187
- iv) Total number of buildings (Schools, Colleges and Hospitals) should be retrofitted= 187 and

- v) Total number of buildings (Schools, Colleges and Hospitals) should be demolished= 42

7.7.1 The detail engineering assessments of the structures which were carried out are to be retrofitted. This implies that structures which are detail analyzed are weak in strength below the standard.

7.7.2 Total Building vulnerability due to Madhupur fault, M 6.9:

- 864,619-1,391,685 (40%-64%) of the 2.15 million buildings
-

7.7.3 The Urban Resilience Program is a ground-breaking project that:

- a. Assessed the vulnerability of essential infrastructure, public facilities, and lifelines to better guide future investments in retrofitting.
- b. Equipped national and local disaster risk management agencies with relevant response resources, including emergency operation centers, communication systems, and related training.
- c. Improved construction practices by integrating disaster risk into development planning and zone processing.

7.8 The primary reasons identified for building vulnerability

7.8.1 Soft-Storey Defect. Many multi-storey residential and commercial buildings feature open ground floors used for parking or retail. These ground floors lack shear-resisting walls, making them highly flexible and prone to collapsing under seismic pressure.

7.8.2 Plan and Vertical Irregularities. Buildings often feature asymmetric mass distribution, geometric irregularities, and uneven layouts. This produces hazardous torsional twisting and uneven stress concentrations during earthquakes.

7.8.3 Substandard Construction Materials. The widespread use of poor-quality concrete mixtures, low-grade rebar, and weak mortar significantly compromises a building's baseline structural load-bearing strength.

7.8.4 Failure of Non-Structural Elements. Modern facades, heavy glass cladding, false ceilings, and poorly anchored partition walls are highly unstable. They pose severe risks of collapsing outward into narrow streets.

7.8.5 Construction on Unstable Filled Land. Massive migration has forced rapid urban expansion onto filled-up canals, low-lying wetlands, and floodplains. This soft soil significantly exacerbates the risk of liquefaction, soil settlement, and foundation tilting during earthquakes.

7.8.6 Aging Infrastructure. A vast percentage of urban housing consists of buildings that are 30 to 40 years old. These structures have deteriorated over time and lack the foundational engineering required to withstand modern seismic activity.

7.8.7 Violation of Approved Designs. Developers frequently deviate from officially approved architectural blueprints to maximize floor space. This adds uncalculated weight that overburdens the structural foundations.

7.8.7 Low Adherence to Building Codes. While the Bangladesh National Building Code mandates earthquake and fire-resistant standards, enforcement mechanisms historically remain opaque and lack direct accountability.

7.8.8 Inadequate Quality Control. Construction processes lack standardized on-site verification, leading to common operational issues like poor concrete curing, inadequate compaction, and weak joints.

7.8.9 Extreme Building Proximity. Due to a lack of setback space, buildings are constructed directly adjacent to one another. This increases the likelihood of a "domino effect" structural collapse and prevents safe emergency evacuation.

7.9 Urban Planning Strategies and Policy Framework to Make Dhaka City Resilient against Earthquake

7.9.1 Visioning Seismic Resilient City. Ultimate vision of this endeavour is to gradually develop efficient and effective urban planning strategies to make cities safer and more resilient against earthquake. Pursuing of this vision is not an isolated work, it needs multi-stakeholders' efforts. Following interventions are extremely necessary to make Dhaka city resilient:

- i) Seismic hazard responsive land use planning and enforcement of that plan;
- ii) Construction of seismic responsive buildings in compliance with building construction rules and code;
- iii) Evaluation of the building distribution by seismic performance in the city;
- iv) Prioritization of public and private building for retrofit;
- v) Knowledge sharing with community about community resilience and retrofitting;
- vi) Strong integrated effective disaster risk governance (Legal, Institutional, and Execution);
- vii) Resilience oriented investment of resources to make city safer and resilient.

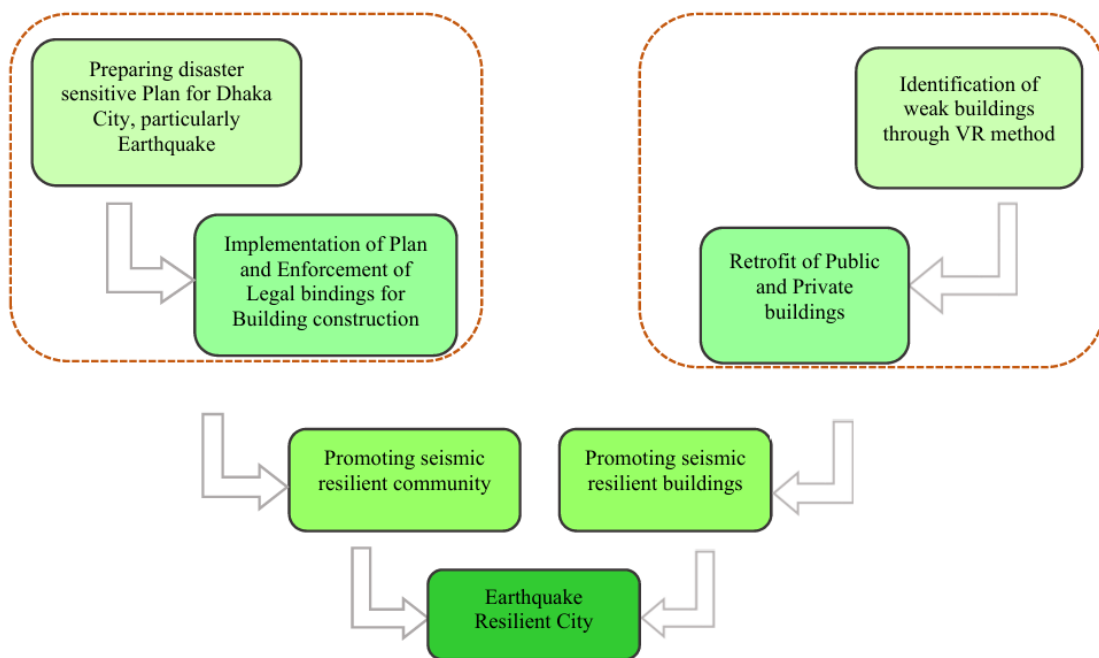


Figure 7.2: Vision of building Earthquake Resilient City

A long-term endeavour that city authorities and the citizen must pursue with combined efforts keeping a vision ahead for earthquake resistant city. The schematic diagram (Figure 7.2) shows us the process of developing an earthquake resilient city in comprehensive manner.

7.10 The Fields Where to Focus for Building Seismic Resilient City.

Development of efficient and effective urban planning strategies to make cities safer and more resilient against disasters

7.10.1 Earthquake Responsive Urban Planning City development authority need to consider all the shocks and threats related to earthquake in the city including seismic fault lines, geology, land formation and soil condition of the city. Plan must be prepared on the considering the finding of the above-mentioned investigation to reduce the seismic vulnerability of Dhaka city. Geological matters in urban planning to be taken into consideration.

7.10.2 Enforcement of Formulated Plan and Policy. The formulated policy is enforced to be complied.

7.10.3 Building distribution of seismic performance. As a part of improving seismic resilience of the city, chapter 3 of this guidebook guides through a process developing a method for identifying the buildings with low seismic performance using Japanese Standard. The method of estimating seismic performance helped in addressing ‘building condition’ in understanding the city’s overall seismic condition.

7.10.4 Prioritization of Building for Retrofit. For improving seismic resiliency of Dhaka city, it is required to improve seismic performance of the buildings through seismic retrofitting. Random selection of buildings for seismic retrofitting might not help in achieving efficiency. Buildings should be prioritized for retrofitting to maximize efficiency as a part of improving seismic resiliency.

7.10.5 Compliance of Building Rules and Code. Building rules and code should be complied.

7.10.6 Revision of Contingency Plan. Earthquake Contingency Plan was prepared in 2009 under the Comprehensive Disaster Management Programme. It was solely prepared for the unified Dhaka city corporation to withstand the adversity of an earthquake. In this document it clearly mentioned that twenty-seven stakeholder organizations are directly related to execute the plan. However, most of the stakeholder organizations are unaware of this plan. Disowning such a plan merely indicate the lack of interest among the organizations.

7.10.7 Measures to be Taken to Improve City’s Resilience. Here is the summary of policy interventions need to be taken for building a seismic resilient Dhaka city.

Serial No	Existing Situation	Necessary Policy Intervention	
1	Building Permission Process	To reduce building vulnerability, new section should be added in the Building construction rules that structural design has to approved through Rajuk	Dynamic Time History analysis including earthquake resistant electricity,gas, water line to be attached

7.11 Earthquake Resistant Lifeline Utilities (Water, Gas, and Power). A city cannot survive an earthquake if its core lifelines are disrupted:

7.11.1 Flexible Pipeline Joints. Rather than rigid cast-iron pipes, modern water and gas networks use segmented, high-density polyethylene lines. These pipes pull, push, and bend at the joints without bursting during severe ground displacement.

7.11.2 Automated Shutoff Valves. Natural gas grids are equipped with smart seismic sensors. The moment a critical threshold of ground velocity is reached, the system automatically shuts down regional gas valves to prevent catastrophic urban firestorms.

7.12 The primary reasons identified for building vulnerability include:

7.8.1 Structural and Engineering Flaws

i) **Soft-Storey Defect.** Many multi-storey residential and commercial buildings feature open ground floors used for parking or retail. These ground floors lack shear-resisting walls, making them highly flexible and prone to collapsing under seismic pressure.

CHAPTER 8

ONLINE/VIRTUAL EARTHQUAKE ENGINEERING UNIVERSITY

8.1 Vision

Vision to build a successful Online/Virtual Earthquake University must combine flexible digital learning with practical, real-world simulation tools.

8.2 Operational Modalities

The key operational modalities which are required, this vision being in reality:

8.2.1 Academic & Curriculum Structure

The academic and curriculum structure is as follows:

- i) **Degree Pathways.** Targeted in Offering fully accredited Bachelor's, Master's, and Ph.D. programs in Seismology, Earthquake Engineering, and Disaster Management.
- ii) **Micro-Credentials.** Short, targeted certificate, courses for working professionals like architects, city planners, and first responders.
- iii) **Global Faculty Network.** Hosting live and recorded lectures from top global scientists, pooling research from seismically active regions like Japan, Turkey and the United States of America.

8.2.2 Technology & Learning Methods

- i) **Virtual Reality Labs.** Using virtual reality headsets so remote students can safely simulate structural stress tests, building collapses, and emergency rescues.
- ii) **Cloud-Based Modeling.** Providing students access to high-powered software (like SAP2000 or OpenSees) via the cloud to design earthquake-resistant structures.

8.2.3 Hybrid & Field Partnerships

- i) **Local Practical Hubs.** Partnership with local engineering universities worldwide so online students can complete mandatory, hands-on lab work locally.

8.2.4 Global Impact & Access

- i) <https://github.com/mohammademadulhaque/Bangladesh-Earthquake-Engineering-University>
- ii) An Open-Source, Global Digital Portal for Seismic Design, Structural Dynamics and Geotechnical Education.
- iii) This open-access digital portal is dedicated in providing high-quality, structured and free educational resources for students, researchers, and structural engineers worldwide to design a safer, resilient built environment.

8.2.1 Academic Departments & Curriculum

- i) Department of Structural Dynamics
- ii) Department of Seismology & Earth Sciences
- iii) Department of Seismic Design & Geotechnical Engineering
- iv) Virtual Simulation Laboratory
- v) Global Research & Data Center
- vi) How to Contribute to Virtual Earthquake Engineering University
- viii) Readme.md (Uploaded Files)

CHAPTER 9

CONCLUSIONS AND RECOMMENDATIONS

9.1 Conclusions

Dynamic Time History Analysis (Nonlinear Dynamic Analysis) is generally not carried out in Bangladesh, during the design and vetting process of either regular or irregular buildings. For buildings requiring dynamic analysis, Linear Dynamic Analysis (Response Spectrum Analysis) is commonly performed in accordance with the provisions of BNBC 2020. In particular, irregular buildings are typically evaluated using the linear response spectrum method as specified by the code. Nonlinear Dynamic Analysis is primarily limited to specialized research studies and performance-based engineering assessments and is not a routine design requirement in current engineering practice.

Dynamic time history analysis (Non-linear dynamic analysis) is mainly performance-based design suitable for research purpose, Detailed Engineering Assessment of existing buildings and special type projects. Bangladesh National Building Code-2020 allows both linear and non-linear dynamic analysis for design of structures. This type of analysis is time consuming and requires high amount of computing resources.

Presently Linear dynamic analysis i.e., Response Spectrum Analysis is carried out for design and review of regular buildings with height greater than 40 meters in zone 2,3,4. (Dhaka city falls in zone-2).

Linear dynamic analysis i.e. Response Spectrum Analysis is carried out for design and review of irregular buildings with height greater than 12 meters in zone 2,3,4. (Dhaka city falls in zone-2).

The factor of safety against earthquake-induced soil liquefaction is commonly evaluated during geotechnical investigations for sites susceptible to liquefaction. The factor of safety is generally determined as the ratio of the soil's cyclic resistance to the cyclic stress induced by earthquake loading. A factor of safety greater than 1.0 indicates that the soil possesses adequate resistance against liquefaction under the considered seismic loading. Where necessary, appropriate ground improvement measures are recommended to achieve acceptable safety levels.

Most of the soil test companies in Bangladesh do not provide the liquefaction data in soil test. In practice we calculate susceptibility to liquefaction as per code guideline for specific project as per requirement. Pile is designed considering earthquake induced lateral force and bending moment. Pile size and rebar requirement increases due to increased lateral force. Pile embedment length is increased as well.

Although provisions for the seismic design and restraint of non-structural components, including gas piping, HVAC systems, and air-conditioning lines, are provided in ASCE 7-16 (Chapter 13: Seismic Design Requirements for Nonstructural Components), such checks are generally not carried out as part of routine building design and vetting practice in Bangladesh. Current engineering practice primarily focuses on the seismic performance of structural components, while detailed seismic qualification and restraint design of utility systems and mechanical services are not commonly implemented except in specialized or critical facilities.

Urban Resilience project surveyed 3252 important buildings composed of 5.3 million square meters, using the Rapid Visual Assessment methodology. Then 578 of the buildings prioritized in the Rapid Visual Assessment stage, composed of 1 million square meters went through Preliminary Engineering Analysis. Finally, 187 of the buildings, composed of more than 0.4 million square meters went through in-depth Detail Engineering Analysis. Out of 187 buildings 82 buildings were earmarked to be demolished.

9.2 Recommendations

9.2.1 Bangladesh's number one priority action is the strict enforcement of the Bangladesh National Building Code and the rapid, systematic retrofitting of vulnerable, high-risk existing structures. Experts emphasize that earthquakes do not kill people—collapsing, non-compliant buildings do.

9.2.2 Advanced Monitoring Systems. In 2026, the use of smart sensors and IoT in construction is not just a trend but a necessity. These systems provide real-time data on building integrity, allowing for timely maintenance and upgrades. Consider installing seismic sensors and structural health monitoring systems for proactive safety management.

9.2.3 Regular Safety Drills and Education. Educate occupants about earthquake safety protocols. Conduct regular drills and provide resources on emergency preparedness. Awareness and preparedness are key to minimizing risks during an actual event.

9.2.4 Retrofitting Older Structures. Older buildings may not meet the current seismic standards. Assess such structures and undertake necessary retrofitting using modern techniques like steel bracing, shear walls, and flexible diaphragms.

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ANNEXURE A

Japan's city gas system is highly earthquake-resistant

Yes, Japan's city gas system is highly earthquake-resistant and heavily engineered to prevent secondary disasters like fires or explosions. While a massive earthquake will temporarily interrupt your gas supply, the entire system is built to minimize risk and withstand significant ground movement.

The network achieves its structural resilience through a mix of high-tech prevention, automatic mitigation, and advanced materials:

1. Flexible Polyethylene Pipelines

Older, rigid cast-iron pipes have been extensively replaced across Japan. Major providers like [Tokyo Gas Network](#) use **polyethylene (PE) pipelines** for low-pressure distribution. Polyethylene is highly flexible and durable; it can bend and twist with shifting soil during an earthquake without cracking, preventing underground gas leaks. High- and medium-pressure trunk lines are made of welded steel with specialized earthquake-resistant joints that withstand major ground deformation.

2. The SUPREME Real-Time Monitoring System

Major city gas networks utilize dense real-time seismic monitoring systems, famously known as **SUPREME**. Thousands of seismic sensors are installed across the service grids.

- If an earthquake strikes, these sensors instantly measure the shaking intensity across different blocks.
- If the shaking passes a dangerous threshold (usually lower 6 on the Japanese seismic intensity scale), the district regulators automatically or remotely **shut off the gas supply** to that specific area to eliminate the risk of fire.

3. Intelligent Consumer Meters (*Mai-con Meter*)

Every residential property in Japan is equipped with a microcomputer-controlled gas meter (*Mai-con* meter).

- **Automatic Shutoff:** If the meter detects seismic vibrations of **Magnitude 5 or higher**, it instantly shuts off the gas right outside your home.
- **Leak Detection:** It also shuts off automatically if it detects an unnatural, continuous flow of gas (indicating a broken indoor pipe) or a sudden drop in pressure.
-

What Happens After a Major Earthquake?

Because safety is the absolute priority, restoring city gas takes longer than electricity or water. The gas network is divided into isolated "disaster prevention blocks". After an intense quake, technicians must physically check the integrity of the pipelines block-by-block and ensure there are zero leaks before gas can be safely pumped back into the neighbourhood.

If your gas ever cuts off during a tremor, you can easily **reset your own meter** by pressing a button on the physical unit once the shaking completely stops and you have confirmed there is no smell of gas.

If you are currently planning emergency preparedness, would you like to know **how to locate and manually reset your home's gas meter**, or do you need a list of **essential emergency supplies** recommended for households in Japan?

2.14 SEISMIC FORCE FOR NON-STRUCTURAL COMPONENTS

2.14.1 General

Non-structural components are permanently installed building items and BNBC 2020 gives the provisions for these in BNBC 2020 Part 6, § 2.5.15. To ensure life safety and preserve the functionality of essential facilities, these components should be designed to resist the lateral seismic forces. Seismic design for non-structural components also should be done with great importance.

Examples of architectural components are un-braced parapet walls on roof, interior partition walls, interior and exterior non bearing walls, retaining walls and penthouses. Mechanical/electrical components include water tanks and vessels with support systems; electrical, mechanical and plumbing equipment; and emergency communication equipment.

It is essential to consider the following facts in Bangladesh:

The out-of-plane mode of failure of infill walls has been widely observed in reinforced concrete frame structure after severe earthquakes (*See D-14 in Appendix-D of the Manual*). One of its countermeasures is shown at Sec. B-5 in Appendix-B of the Manual.

2.14.2 Seismic Design Force

Table. 6.2.22 or Table. 6.2.23 in BNBC 2020 Part 6, given a list of amplification factors (α_c) and response reduction factor (R_c). The seismic design force, F_c applied in the horizontal direction can be calculated by Equation 2.35:

$$F_c = \alpha_c a_h W_c I_c (1 + 2z/h) / R_c \quad (2.35)$$

Where,

$$0.75 \alpha_c W_c I_c \leq F_c \leq 1.5 \alpha_c W_c I_c$$

α_c = components amplification factor which varies from 1.0 to 2.5

(See BNBC 2020, Tab. 6.2.22 or Tab. 6.2.23)

a_h = expected horizontal peak ground acceleration (in g) for design = $0.67 \times z \times s$

W_c = weight of component

R_c = component response reduction factor which varies from 1.0 to 12.0

(See BNBC 2020, Tab. 6.2.22 or Tab. 6.2.23)

I_c = component importance factor (See BNBC 2020 Part 6, §2.5.15.1)

z = height above the base of the point of attachment of the component, but z shall not be taken less than D and the value of z/h need not exceed 1.0.

h = roof height of structure above the base

The force F_c shall be independently applied in at least two orthogonal horizontal directions in combination with service loads associated with the component. In addition, the component shall also be designed for a concurrent vertical force of $\pm 0.5 a_h W_c$.

The force F_c is used to design the members and connections that transfer the force into the seismic-resisting systems. (See BNBC 2020 Part 6, § 2.5.15.3)

Where non-seismic loads on non-structural components exceed F_c such loads shall govern the strength design, but seismic detailing requirement and limitation shall apply. Drift, movement and other standards also apply.

2.14.3 A Water Tank as an Example

A water tank laterally braced below the center of mass is assumed to be set on the roof of the structure built in Dhaka. In this case, seismic lateral static force is assumed to act at lateral center of gravity of the water tank and the frame is modeled by pinned support at base on roof.

According to Table 6.2.22 in BNBC 2020 Part 6, a_c and R_c are 2.5 and 3.0, respectively.

The seismic design force F_c is

$$\begin{aligned} F_c &= a_c a_h W_c I_c (1 + 2z/h) / R_c & F_c \min &= 0.75 a_h \times W_c \times I_c \\ &= 2.5 \times 0.15 \times W_c \times 1.0 (1 + 2h_{\text{roof}}/h_{\text{roof}}) / 3 & &= 0.75 \times 0.15 \times W_c \times 1.0 \\ &= 0.38 \times W_c & &= 0.11 \times W_c \end{aligned}$$

Where,

$$a_h = 0.67 \times 0.2 \times 1.15 \approx 0.15$$

$$I_c = 1.0$$

Such a component is popular in Japan, and the seismic co-efficient “ k ” is multiplied by component weight W_c .

In Japan the seismic design force F_c is

$$F_c = 1.0 W_c$$

The ratio 1 to 0.38 is approximately 2.53 times that is between 2.5 and 3.0. This ratio is the same to the earthquake intensity ratio (1/2.5 ~ 1/3.0) assumed when seismic intensity is compared between Bangladesh and Japan. (See Figure 2.8 and Figure 2.9 in Chapter 2 of Part-II of the Manual).

2.15 SEISMIC DESIGN REQUIREMENT FOR NON-BUILDING STRUCTURES

Non-building structures include all self-supporting structures that carry gravity loads and that may be

2.15 SEISMIC DESIGN REQUIREMENT FOR NON-BUILDING STRUCTURES

Non-building structures include all self-supporting structures that carry gravity loads and that may be required to resist the effects of earthquake. Chimney, self-supported overhead water/fluid tank, silo, trussed tower, storage tank, cooling tower, monument etc. are the example of Non-building Structures. Non-building structures supported by the earth or supported by other structures shall be designed and detailed to resist the minimum lateral forces.

New BNBC 2020 (Ref. 2.1) has referred “Chapter 15 : Seismic Design Requirements for Non-Building Structures, Minimum Design Loads for building & Other Structures, ASCE Standard ASCE/SEI 7-05”(Ref. 2.3) for calculation of seismic design forces on non-building structures.

Questionnaire and Answers(Kii)- Mist, Civil Department Head

Section A: Nonlinear Dynamic Analysis

1. Design vetting is done in Engineering Universities. Do we carry out Nonlinear Dynamic Analysis of Regular buildings with height greater than 40 meter (Dhaka City)? Do we carry out Nonlinear Dynamic Analysis of Irregular buildings with height greater than 12 meter (Dhaka City)? Statistics if possible.

Answer: *In Bangladesh, Nonlinear Dynamic Analysis is generally not carried out during the design and vetting process of either regular or irregular buildings. For buildings requiring dynamic analysis, Linear Dynamic Analysis (Response Spectrum Analysis) is commonly performed in accordance with the provisions of BNBC 2020. In particular, irregular buildings are typically evaluated using the linear response spectrum method as specified by the code. Nonlinear Dynamic Analysis is primarily limited to specialized research studies and performance-based engineering assessments and is not a routine design requirement in current engineering practice.*

Section B: Soil Safety Against Earthquake

2. Soil liquefaction may be defined as the state of soil which loses its shear strength and subsequently behaves like a liquid. During soil test do find the factor of safety against earthquake greater than 1. Factor of Safety = Cyclic Strength / Disturbing Strength > 1. Statistics if possible.

Answer: *Yes. For sites susceptible to liquefaction, the factor of safety against earthquake-induced soil liquefaction is commonly evaluated during geotechnical investigations. The factor of safety is generally determined as the ratio of the soil's cyclic resistance to the cyclic stress induced by earthquake loading. A factor of safety greater than 1.0 indicates that the soil possesses adequate resistance against liquefaction under the considered seismic loading. Where necessary, appropriate ground improvement measures are recommended to achieve acceptable safety levels.*

Section C: Earthquake Resistant Gas and Air Condition Line

3. A building is vetted including all utilities. During vetting do you check earthquake resistant gas and air condition line? If checked then which code is followed? Code or Reference if possible.

Answer: *Although provisions for the seismic design and restraint of non-structural components, including gas piping, HVAC systems, and air-conditioning lines, are provided in ASCE 7-16 (Chapter 13: Seismic Design Requirements for Nonstructural Components), such checks are generally not carried out as part of routine building design and vetting practice in Bangladesh. Current engineering practice primarily focuses on the seismic performance of structural components, while detailed seismic qualification and restraint design of utility systems and mechanical services are not commonly implemented except in specialized or critical facilities.*

**Questionnaire Answer-Key Information Interview (Staff Officer-1, Army
Headquarter, Works Directorate)**

**Research Title: Urban Infrastructure Security Focusing Earthquake-Bangladesh
Perspective**

Section A: Nonlinear Dynamic Analysis

1. Design vetting is done in Engineering Universities. Do we carry out Nonlinear Dynamic Analysis of Regular buildings with height greater than 40 meters (Dhaka City)? Do we carry out Nonlinear Dynamic Analysis of Irregular buildings of height greater than 12 meters (Dhaka City)? Statistics if possible.

Answer:

Dynamic analysis can be both linear and non-linear. Nonlinear Time History Analysis (NTHA) is one of the most advanced and reliable methods for evaluating the seismic performance of structures.

Non-linear dynamic analysis (time history analysis) is mainly performance-based design suitable for **research purpose, Detailed Engineering Assessment (DEA) of existing buildings and special type projects**. BNBC-2020 allows both linear and non-linear dynamic analysis for design of structures. This type of analysis is time consuming and requires high amount of computing resources.

Linear dynamic analysis i.e. Response Spectrum Analysis is carried out for design and review of **regular buildings** with height greater than 40 meters in zone 2,3,4. (Dhaka city falls in zone-2).

Linear dynamic analysis i.e. Response Spectrum Analysis is carried out for design and review of **irregular buildings** with height greater than 12 meters in zone 2,3,4. (Dhaka city falls in zone-2).

Section B: Soil Safety Against Earthquake

2. Soil liquefaction may be defined as the state of soil which loses its shear strength and subsequently behaves like a liquid. During soil test do find the factor of safety against earthquake greater than 1. Factor of Safety = Cyclic Strength / Disturbing Strength > 1

Statistics if possible

Answer:

Most of the soil test companies in Bangladesh **do not provide the liquefaction data in soil test**. In practice we calculate susceptibility to liquefaction as per code guideline for specific project as per requirement. Pile is designed considering earthquake induced lateral force and bending moment. Pile size and rebar requirement increases due to increased lateral force. Pile embedment length is increased as well.

Section C: Earthquake Resistant Gas and Air Condition Line

3. A building is vetted including all utilities. During vetting, do you check earthquake resistant gas and air condition lines? If checked then which code is followed? Code or reference if possible

Answer:

Earthquake resistant gas and air condition lines are **not checked in our current practice**.

হিমায়েশ গ্যাস কন্ট্রোল এন্ড ম্যানেজমেন্ট (সেন্ট্রাল) বিভাগের অধীনে কার্যকরীকরণ (২০১২-২০২৬ অর্থবছর)।
ইমার্জেন্সি গ্যাস কন্ট্রোল এন্ড ম্যানেজমেন্ট বিভাগ (সেন্ট্রাল)
১০০, অস্ট্রী নগরকনা ইসলাম রোড, এমএনটি
কম্পিউটার ভিত্তিক সিস্টেম, ঢাকা-১২১৫।

হিমায়েশ গ্যাস কন্ট্রোল এন্ড ম্যানেজমেন্ট (সেন্ট্রাল) বিভাগের অধীনে কার্যকরীকরণ (২০১২-২০২৬ অর্থবছর)।
ইমার্জেন্সি গ্যাস কন্ট্রোল এন্ড ম্যানেজমেন্ট বিভাগ (সেন্ট্রাল)

মাসের নাম	মোট অডি যোগ সংখ্যা	রেকর্ডারভুক্ত অধীনে অডি যোগ সমূহ							মোট অডি যোগ সংখ্যা	সিবেক মোবাইলকৃত অডি যোগ সমূহ						
		অধি ন	লিঙ্ক জ	খার লটি	গ্যাস নাই	চাপ কম	হাউজ লাইন	অন্য না		বিস্তার ও	সার্ভিস স্টেশন	হাইড্রা স্টিক	লক উইং কক	হাউজ স্টেশন	হাউজ লাইন	অন্যান্য
জুলাই-১৫	৫৯১	২৫	৪৪৫	২২	৫৩	৬	১০	৩৩	৫৯১	৫৯৬	১০	৯৬	১৫	৩	১০	৬৮
আগস্ট-১৫	৫২২	২৬	৩৬৭	৪৪	৪৭	৬	৯	৩৩	৫২২	৫৫২	১০	১২৫	১	১	১৬	২৫
সেপ্টেম্বর-১৫	৬৫৩	২০	৩৫২	২৭	১৯৯	১৪	১৬	২৫	৬৫৩	৪৩৬	৯	১৬০	৩	৪	১৫	৫৫
অক্টোবর-১৫	৬৮৩	২৪	৪১৯	২৫	১০০	১৫	১৫	৫৫	৬৮৩	৪৭০	৪	১১৬	১৮	৫	১৭	৪৩
নভেম্বর-১৫	৫৩২	২১	৩২৯	১৩	৯২	১৭	১৭	৪৩	৫৩২	৩৮৬	৬	৫৪	৮	১	১২	৪৪
ডিসেম্বর-১৫	৪৬৯	২৭	২৭৮	২৩	৭৪	১১	১১	৪৪	৪৬৯	৩৫৫	৬	৩৬	২	১	১৭	২৮৭
জানুয়ারী-১৬	৭৯০	২০	৩১৯	১২	৩৪৬	১৯	১৭	৪৭	৭৯০	৩৪১	৪	১১৯	৬	১১	১৩	১০৬
ফেব্রুয়ারী-১৬	৪৯৫	২৮	২৩৪	১৩	১৭০	১৫	১৩	২২	৪৯৫	২৭৫	২	১৩	৫	১১	১৭	৪৮
মার্চ-১৬	৪৫০	৩৪	২৮১	১৩	৭৩	১১	১৭	২১	৪৫০	৩২৮	৪	৫০	২	১	১৪	১৫৮
এপ্রিল-১৬	৫৫৯	৩১	২৪০	৪৪	১৩৮	১৪	১৪	২৬	৫৫৯	২৮৬	৪	৪১	৩	৩	১৭	১৯৪
মে-১৬	৭৭২	৩২	৪৩২	৩২	১৮৬	২০	১৭	৫৩	৭৭২	৪৬৬	৯	৮৩	১	২	১৭	১৯৪
জুন-১৬								০								
মোট	৬৪৬৬	২৮৮	৩৬৯৪	২৬৭	১৫০৮	১৫৭	১৫৭	৩৯৪	৬৪৬৬	৪০০৬	৬৮	৯৭৬	৬৪	৪৫	১৫৭	১০৫২

মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাপ্রায়কৃত গ্যাস ঘ.মি	দৈনিক সাপ্রায়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
Dec-25	মোবাইল কোর্ট	৩৯	৭৫৭৫	৪৯	৩০	৭৬০২৮	#####	১৩০	১৭৮০৭
	ভিজিল্যান্স	৩৭	৪১	০	১৩	৮১১২৫	#####	৮৮	০
	সেলস	২৪	১৬৪৩	৪	১৮	৩৭২৮৩	#####	৭৮	৪০৭৬
	রেভিনিউ	২৬১	১৩৩৭	০	০	২৬৭৪	৪৮১৩২	৫৬৩	০
মোট		৩৬১	১০৫৯৬	৫৩	৬১	১৯৭১১১	#####	৮৫৯	২১৮৮৩

মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাপ্রায়কৃত গ্যাস ঘ.মি	দৈনিক সাপ্রায়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
Jan-26	মোবাইল কোর্ট	৩৪	৭০০৪	২৯	৩৪	৬৮৪১১	#####	১০৫	৯৭৫০
	ভিজিল্যান্স	৩৭	২১	১	৬	৪৫৭২৩	#####	৬৯	০
	সেলস	৭	৭২২	০	৬	৯৪৬৩	#####	১৮	১৫৬৯
	রেভিনিউ	২১১	৬৯৬	০	০	১৩৯২	২৫০৫৬	৪০৩	০
মোট		২৮৯	৮৪৪৩	৩০	৪৬	১২৪৯৮৯	#####	৫৯৫	১১৩১৯

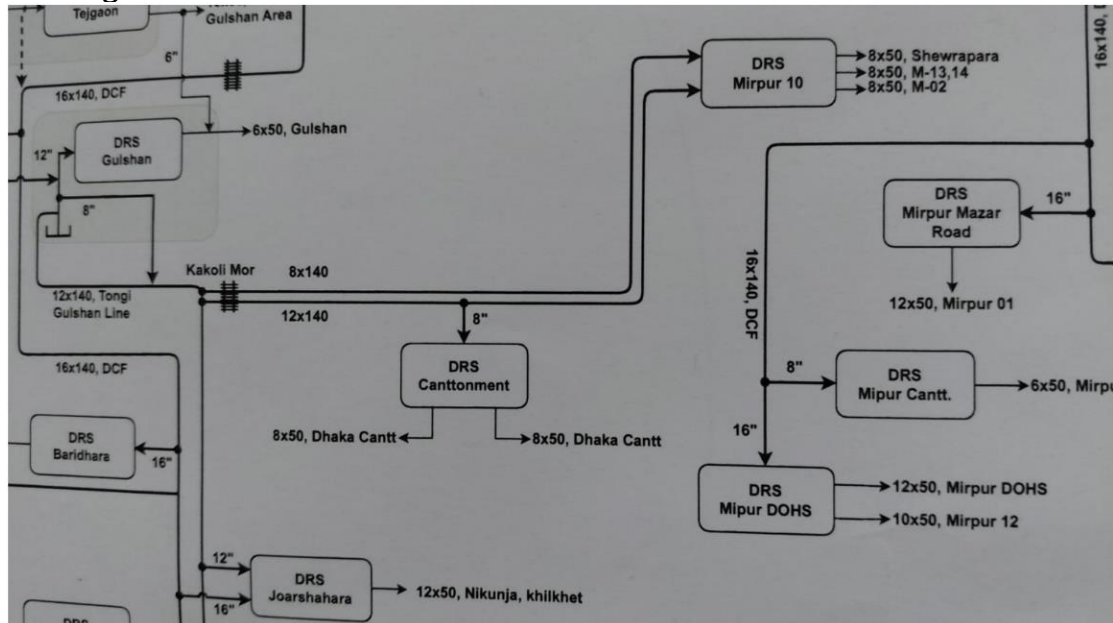
মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাপ্রায়কৃত গ্যাস ঘ.মি	দৈনিক সাপ্রায়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
Feb-26	মোবাইল কোর্ট	২০	২৯৫০	১৫	১৭	৬০৫৯৭	#####	৬০	৬৪৯৯
	ভিজিল্যান্স	৪২	৫	১	৪	১৮৫৮৪	#####	৯৪	০
	সেলস	৪	১২২	০	৩	৫১৩৭	১৮০৫৪৫	৮	৮৫০
	রেভিনিউ	১১৬	৩৫৪	০	০	৭০৮	১২৭৪৪	১৮৭	০
মোট		১৮২	৩৪৩১	১৬	২৪	৮৫০২৬	#####	৩৪৯	৭৩৪৯

মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাশ্রয়কৃত গ্যাস ঘ.মি	দৈনিক সাশ্রয়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
Mar-26	মোবাইল কোর্ট	২৪	৩৭৩২	১৮	২৮	৬৩৮৩১	#####	৭০	১৪৭১৯
	ভিজিল্যান্স	২৪	২৯	৩	৩	৪৩৪৪	#####	৬৫	০
	সেলস	৩	৭৬৯	০	০	১৫৩৮	২৭৬৮৪	৭	১৯৮৩
	রেভিনিউ	৯২	১৭৯	০	০	৩৫৮	৬৪৪৪	১১৮	০
মোট		১৪৩	৪৭০৯	২১	৩১	৭০০৭১	#####	২৬০	১৬৭০২

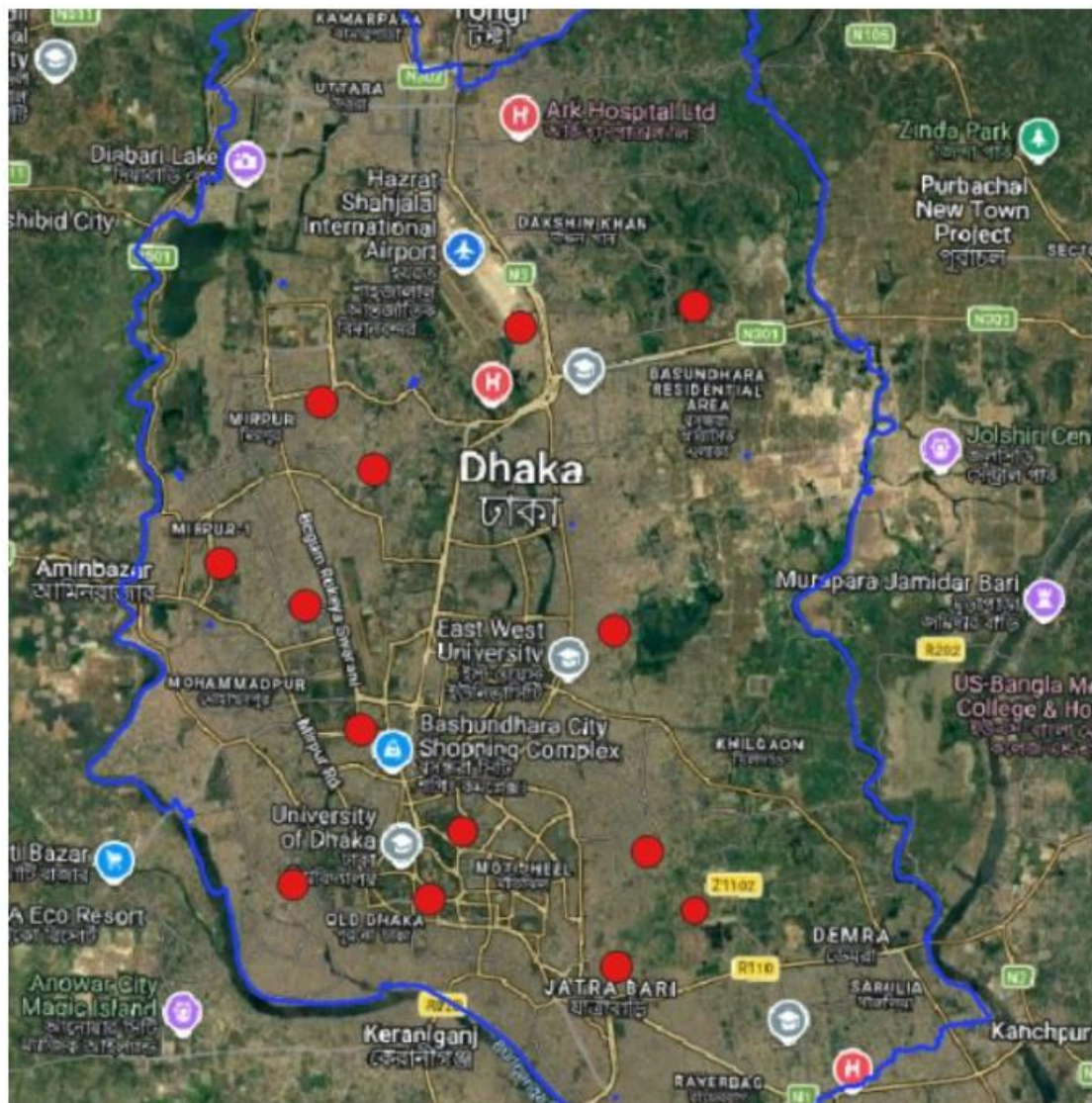
মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাশ্রয়কৃত গ্যাস ঘ.মি	দৈনিক সাশ্রয়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
Apr-26	মোবাইল কোর্ট	৪২	৯৭২২	৩৩	৩৪	৮২৩৬৫	#####	১৩৮	১২৯৬৬
	ভিজিল্যান্স	৪২	১৮১	৪	৭	২১৭২৭	৭৭৭১৪৬	১০৪	৫০
	সেলস	২৭	১৪০৬	৫	৩	৮২০৮	২৪৪৮৭৪	১৩৭	২১৭২
	রেভিনিউ	৩২১	১৪৮০	০	০	২৯৬০	৫৩২৮০	৭৪৮	০
মোট		৪৩২	১২৭৮৯	৪২	৪৪	১১৫২৬০	#####	১১২৭	১৫১৮৮

মাস	ডিপার্টমেন্ট	অভিযান এর সংখ্যা	আবসিক চুলা সংখ্যা	বাণিজ্যিক সংযোগ সংখ্যা	শিল্প সংযোগ সংখ্যা	দৈনিক সাশ্রয়কৃত গ্যাস ঘ.মি	দৈনিক সাশ্রয়কৃত গ্যাসের আনুমানিক মূল্য (টাকা)	স্পট সংখ্যা	উচ্ছেদকৃত পাইপের দৈর্ঘ্য (ফুট)
May-26	মোবাইল কোর্ট	৩৩	১৪২১২	২৯	৩৩	১২৩০১৫	#####	৯৮	১৪৬১৪
	ভিজিল্যান্স	৪০	১৫৮	৬	৫	২৭৩৬১	৯৭৯২৯৪	১০৫	০
	সেলস	২২	১২৯১	৫	৮	১৭৯৭৮	#####	৬০	২৮৬৯
	রেভিনিউ	১৯৪	১১৪০	০	০	২২৮০	৪১০৪০	৫২৫	০
মোট		২৮৯	১৬৮০১	৪০	৪৬	১৭০৬৩৪	#####	৭৮৮	১৭৪৮৩

Existing Titas Gas Line

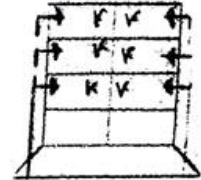
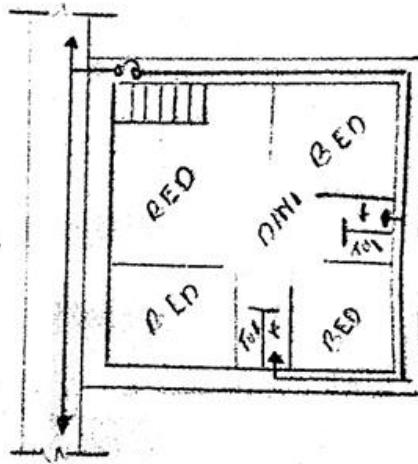


5.3 Vulnerable Soil. The vulnerable soil area shown in red circle



গ্রাহকের নাম : শ্রী: কুমার
 ঠিকানা : ১৫/১০/১১, পূর্ববঙ্গ রাস্তা
১৫৮
 তারিখ : ১১/১১/১১
 গ্রাহক সংকেত : ১১১-১১১১

১। ১ জি, আই পাইপ :
২। $\frac{3}{8}$ জি, আই পাইপ : ৬০'-০"
৩। $\frac{1}{2}$ জি, আই পাইপ : ৬'-০"
৪। পাক ঘরের সংখ্যা : ৬ (৬২) টি
৫। হুনার সংখ্যা : ৬ (৬২) টি
৬। গ্যাসের পরিমাণ : ৬০৬২ ফুট



১। ১ জি, আই পাইপ :	
২। ২ জি, আই পাইপ :	৩০'-০"
৩। ২ জি, আই পাইপ :	৬'-০"
৪। পাক ঘরের সংখ্যা :	১ (১) কক্ষ
৫। চুপার সংখ্যা :	১ (১) কক্ষ
৬। গ্যাসের পরিমাণ :	৬০০০ ঘুট,

[illegible]

M. Ullah
আইকের স্বাক্ষর

20/02/2022

মেসার্স তাজ এন্টারপ্রাইজ
ঠিকানাঃ, মঙ্গলাড়া, ঢাকা।

In the current study three different scenarios have been developed to identify the possible damage to infrastructures, buildings, etc. The scenarios are least, moderate and worst case as assumed based on different magnitude of earthquake.

During an earthquake at 7.5 Mw originated from Madhupur fault, about 53,166 buildings are going to be moderately damaged. This is about 15.00 % of the total number of buildings in the city. About 72,316 buildings are going to be totally damaged during this time. If the magnitude of the earthquake is 8.0 Mw, about 34,326 buildings are going to be at least moderately damaged. During an earthquake originated from under the city at 6.0 Mw about 78,323 buildings will be completely damaged. An earthquake at 8.5 originated from plate boundary fault-2 will completely destroy 238,164 buildings.

Essential Facilities Damage in Dhaka City Corporation Area

There is going to

be severe damage to essential facilities like Hospital, schools, police stations after different level of earthquakes. An earthquake at 7.5 Mw in Dhaka City, originated from Madhupur fault is going

to moderately damage about 241 hospital or clinics. At the same time 10 hospitals will be completely damaged. About 90 schools are going

to be totally damaged at the same time. However 30 police stations and 4 fire service stations are going to be moderately damaged during this earthquake.

Table Expected Damage to Essential Facilities in Dhaka City Corporation Area

<i>Classification</i>	<i>Total</i>	<i>Facilities</i>		
		<i>At Least Moderate Damage >50%</i>	<i>Complete Damage >50%</i>	<i>With Functionality >50% on day 1</i>
Case 1				
Hospitals	600	241	10	224
Schools	2,737	1,173	90	895
EOCs	18	8	2	7
Police Stations	62	30	0	15
Fire Stations	10	4	0	3
Case 2				
Hospitals	600	22	0	431
Schools	2,737	99	2	2,029
EOCs	18	7	0	11
Police Stations	62	1	0	46
Fire Stations	10	0	0	7
Case 3				
Hospitals	600	364	23	128
Schools	2,737	1,567	165	495
EOCs	18	9	0	4
Police Stations	62	39	0	2
Fire Stations	10	6	0	0
Case 4				
Hospitals	600	483	482	114
Schools	2,737	2,292	2,268	406
EOCs	18	14	14	4
Police Stations	62	61	61	1
Fire Stations	10	10	10	0

Source: National Earthquake Contingency Plan

Utility Services Damage in Dhaka City Corporation Area

An earthquake at 7.5 Mw originated from Madhupur fault is to be responsible for moderate damage to potable water at 153 points, waste water at 2, natural gas at 2 points, electrical power at 15,200 and communications at 5 places. During this period there are going to be 79 leaks and 272 breaks in water supply system, 107 leaks & 360 breaks in waste water system and 56 leaks & 191 breaks in gas supply network.

Table Utility Services Damage in Dhaka City Corporation Area

System	Scenario 1					Scenario 2					Scenario 3					Scenario 4				
	Number of Locations					Number of Locations					Number of Locations					Number of Locations				
	Total Number	With at Least Moderate Damage	With Complete Damage	With Functionality >50%		Total Number	With at Least Moderate Damage	With Complete Damage	With Functionality >50%		Total Number	With at Least Moderate Damage	With Complete Damage	With Functionality >50%		Total Number	With at Least Moderate Damage	With Complete Damage	With Functionality >50%	
				After Day 1	After Day 7				After Day 1	After Day 7				After Day 1	After Day 7				After Day 1	After Day 7
Potable Water	748	153	0	0	747	748	0	0	0	748	748	748	0	0	457	748	748	134	0	0
Waste Water	14	2	0	0	12	14	0	0	0	14	14	14	0	0	0	14	14	3	0	0
Natural Gas	7	2	0	0	7	7	0	0	0	7	7	7	0	0	1	7	7	2	0	0
Electrical Power	54,815	15,200	0	0	0	54,815	0	0	0	0	54,815	0	0	0	0	54,815	54,815	0	0	0
Communication	30	5	0	0	29	30	0	0	10	29	30	29	0	0	29	30	29	6	0	3

Pipeline Design Calculation Sheet (Transcribed)

Parameter	Value
Code	ASME B31.8
Pipe Spec	API 5L Grade B
ANSI Class	300
Design Pressure	5.10212 MPa (740 psig)
Outside Diameter	406.4 mm (16 in)
SMYS	244.76398 MPa (35,500 psi)
Joint Factor E	1.0
Temperature Factor T	1.0
Design Factor F	0.80
Corrosion Allowance	1.50 mm
Mill Tolerance	12.5%
Calculated Wall Thickness	5.29463 mm
Required Wall Thickness	7.644 mm
Selected Wall Thickness	7.92 mm (0.31181 in)
Flow Rate	2.83168 MMSCMD (100 MMSCFD)
Flowing Pressure	2.06843 MPa (300 psig)
Temperature	297.33 K (24.18 °C)

Internal Diameter	398.48 mm
Specific Gravity	0.5718
Compressibility Z	0.96234
Gas Constant	8.314 J/mol·K
Calculated Gas Velocity	13.19784 m/s (43.30 ft/s)

Pipeline Design for 1.92 km x 300 Psig Main Distribution Pipeline for CEZL	
CALCULATION OF WALL THICKNESS ALONG WITH PIPELINE GAS VELOCITIES THROUGH STRAIGHT SECTIONS OF PIPELINE IN ACCORDANCE WITH ASME B31.8: 2018	
CLASS :	ANSI 300
CODE :	ASME B31.8
DESIGN PR. :	740 Psig
MATERIAL :	API 5L Gr.B
FORMULA :	$t = \frac{PD}{2SFET}$ $t_{cm} = (t + c_A) \times M_T$

INPUT AREA		
Parameters	Values	[SI Units]
Fluid	Natural Gas	
Design Pressure	5.10212	Mpa
Outside Diameter	406.40000	mm
Pipes Grade		
Specified Minimum Yield Strength	244.76398	Mpa
Longitudinal Joint Factor	E	
	1.00000	ND
Temperature Derating Factor	250_de	
°F Factor	1.00	ND
Class Design Factor	1	
(Accoding to Location)	0.8	ND
Corossion Allowance	1.500124	mm
Mill Tolerance	12.50000%	

OUTPUT AREA		
Calculated Wall Thickness of Pipe	5.29463	mm
Required Wall Thickness of Pipe	7.6441	mm
Selected Wall Thickness of Pipe as per API 5L	7.92	mm

CALCULATION OF GAS VELOCITY IN A PIPELINE					
INPUT AREA					
Symbol	Parameters	Values	[SI Units]	Converted Values	[USCS Units]
Fluid		Natural Gas		Natural Gas	
P_1	SCFH at 14.69594878 Psia @ 60°F	2.83168	MMSCMD	100.00000	MMSCFD
		117986.84947	SCMH	4166666.66667	SCFH
P_f	Pressure of Gas Flowing	2.06843	Mpa	300.00000	Psig
				314.69595	Psia
T_f	Absolute Temperature of Flowing Gas	297.33000	°K	24.18000	°C
				535.52400	°R
OD	Outer Diameter of Pipe	406.40000	mm	16.00000	Inch
ID	Internal Diameter of Pipe	398.48	mm	15.688189	Inch
WT	Selected Pipe Wall Thickness as Per API 5L	7.92	mm	0.311811	Inch
SG	Specific Gravity of Gas	0.57118	ND	0.57118	ND
Z	Gas Compressibility Factor	0.96234	ND	0.96234	ND
R	Universal Gas Constant	8.31400	J/mol·K	10.73160	(ft ³ psia)/(lbmol·°R)
OUTPUT AREA					
V_g	Pipeline Gas Velocity (Calculated)	13.19784	m/s	43.29999	ft/s

প্রকল্প গ্রহণের যৌক্তিকতা

ঢাকা মহানগরের গ্যাস বিতরণ নেটওয়ার্ক স্থাপন এবং মেয়াদ উত্তীর্ণ বিতরণ লাইন পরিবর্তন/প্রতিস্থাপন বিষয়ে মাননীয় প্রধানমন্ত্রীর নির্দেশনার প্রেক্ষিতে এ প্রকল্প গ্রহণের উদ্যোগ নেয়া হয়। ঢাকা ও নারায়ণগঞ্জ শহরের ক্রমবর্ধমান পার্শ্ব ও উর্ধ্বমুখী সম্প্রসারণের ফলে ক্রমান্বয়ে বর্ধিত চাহিদার বিপরীতে ১৯৬৮ সাল হতে নির্মিত গ্যাস বিতরণ নেটওয়ার্কটি অপরাধ হতে পড়েছে। এছাড়াও চাহিদার বিপরীতে গ্যাসের সরবরাহের ঘাটতির কারণে ঢাকা শহরের অভ্যন্তরে নির্মিত প্রায় সকল ডিআরএস এর ইনলেট চাপ ১৫০ পিএসআইজি ও আউটলেট চাপ ৫০ পিএসআইজি এর পরিবর্তে ক্ষেত্র বিশেষে পিক-আওয়ারে ইনলেট চাপ গড়ে ৩০-৬০ পিএসআইজি ও আউটলেট চাপ গড়ে ১০-৩০ পিএসআইজিতে নেমে আসে। ফলে টিজিটিডিপিএলসি এর বিদ্যমান এই পুরনো পাইপ লাইনসমূহের মাধ্যমে গ্রাহকদের প্রয়োজনীয় চাপে গ্যাস সরবরাহ করা সম্ভব হচ্ছে না। কোম্পানির বিতরণ লাইনের আড়াআড়ি/সমান্তরালে বিভিন্ন সেবা প্রদানকারী সংস্থার ইউটিলিটি লাইন স্থাপনকালে এবং বিভিন্ন স্থানে রাস্তা মেরামত, ড্রেন নির্মাণ ও বিভিন্ন উন্নয়ন কাজের ফলে গ্যাস পাইপ লাইনের র‍্যাপিং ক্ষতিগ্রস্ত হয়েছে। এতে গ্যাস পাইপের ক্যাথডিক প্রটেকশন (সিপি) বিনষ্ট হয়ে পাইপসমূহে অতি দ্রুত লিকেজের সৃষ্টি হচ্ছে। অপেক্ষাকৃত কম ব্যাসের (৩/৪"থেকে ২") ব্যাসের লাইন পাইপসমূহের পুরুত্ব কম থাকায় এসমস্ত ব্যাসের পাইপ লাইনসমূহে লিকেজের পরিমাণ আশঙ্কাজনক হারে বৃদ্ধি পেয়েছে। গ্যাস লিকেজের কারণে অগ্নি দুর্ঘটনা ও অন্যান্য কারণে জননিরাপত্তা বিঘ্নিত হওয়ার সাথে সাথে পাইপ লাইনে পানি প্রবেশ করে গ্রাহক প্রান্তে গ্যাস প্রাপ্তিতে মারাত্মক সংকট সৃষ্টি হয়েছে। ঢাকা শহরের পাইপ লাইনে লিকেজ, পানির অস্তিত্ব ও পাইপ লাইন ক্ষতিগ্রস্ত বিষয়ে কোম্পানির জরুরী গ্যাস নিয়ন্ত্রণ বিভাগ প্রতিদিন গড়ে প্রায় ২৫ টি অভিযোগ গ্রহণ করছে।

টিজিটিডিপিএলসি এর বিস্তৃত, পরিবর্তনশীল ও পুরনো গ্যাস বিতরণ নেটওয়ার্কের ব্যবস্থাপনা ঝুঁকি মোকাবেলা এবং যথাযথ পরিকল্পনা প্রণয়ন, উন্নয়ন, নেটওয়ার্ক বিশ্লেষণ এবং রক্ষণাবেক্ষণ ও অপারেশন কার্যক্রম সহজতর করার লক্ষ্যে টিজিটিডিপিএলসি এর ঢাকা শহরের গ্যাস নেটওয়ার্কের সমন্বিত GIS নক্সা প্রস্তুতকরণ ও টিজিটিডিপিএলসি এর সমগ্র নেটওয়ার্কে SCADA System প্রবর্তন করার প্রয়োজনীয়তা রয়েছে। SCADA System স্থাপনের ফলে গ্যাস নেটওয়ার্কে গ্যাস প্রবাহ এবং চাপের তথ্যাদি অনলাইন মনিটরিং ও লোড ব্যালান্সিং (Load Balancing) করা সম্ভব হবে। টিজিটিডিপিএলসি এর বিদ্যমান সিজিএস, টিবিএস, ডিআরএস, ভল্ট স্টেশন, বৃহৎ গ্রাহক (বিদ্যুৎ, সারকারখানা, বান্ধ ক্যাপটিভ) এর সিএমএসসমূহে স্থাপিত মিটারের মাধ্যমে রিয়েলটাইম গ্যাস প্রবাহ ডাটা আহরণ করা সম্ভব হবে।

এসকল বিবেচনায় পুরাতন গ্যাস পাইপ লাইন প্রাতিস্থাপন, সমন্বিত GIS নক্সা প্রস্তুতকরণ ও সমগ্র নেটওয়ার্কে SCADA System প্রবর্তনের লক্ষ্যে ঢাকা ও নারায়নগঞ্জ সিটি কর্পোরেশন এলাকায় বর্তমান ও ভবিষ্যত গ্রাহকদের গ্যাস সরবরাহ বৃদ্ধি, স্বল্পচাপ পরিস্থিতি নিরসন, গ্যাস লিকেজজনিত অপচয় রোধ, গ্যাস লিকেজজনিত দুর্ঘটনা প্রতিরোধ, কার্বন নিঃসরণ হ্রাস সহ জনগণের জীবন ও সম্পদের নিরাপত্তা প্রদান করা এবং টিজিটিডিপিএলসি এর গ্রাহকদের উন্নত সেবা প্রদানের লক্ষ্যে New Development Bank (NDB), জিওবি ও টিজিটিডিপিএলসি এর অর্থায়নে আশোচ্য প্রকল্পটি বাস্তবায়নের উদ্যোগ গ্রহণ করা হয়েছে। এই প্রকল্পটি বাস্তবায়িত হলে, তিতাস-অধিভুক্ত মোট গ্রাহকের প্রায় ৭৩% এই প্রকল্প থেকে উপকৃত হবেন।

তিতাস গ্যাস ট্রান্সমিশন এন্ড ডিস্ট্রিবিউশন পিএলসি
ইমার্জেন্সি গ্যাস কন্ট্রোল এন্ড ম্যানেজমেন্ট বিভাগ (মেট্রো-ঢাকা)
১০৫, কালী নজরুল ইসলাম এভিনিউ
কম্পোন বাজার বারিষাটিক এলাকা, ঢাকা-১২১৫।

ক্র. নং	মাসের নাম	মোট অতিযোগ সংখ্যা	রোজিটারভুক্ত জরুরী অতিযোগ সমূহ						মোট অতিযোগ সংখ্যা	লিকেজ মেরামতকৃত অতিযোগসমূহ							
			আগুন	লিকেজ	থার্মাল	গ্যাস নাই	চাপ কম	হাউজ লাইন		অন্যান্য	বিতরণ ও সার্ভিস লাইন	সার্ভিস টি	রাইজার	লক উইং কক	গ্রামীণ রেগুলেটর	হাউজ লাইন	অন্যান্য
১	জুলাই-২৫	৫৯১	২৫	৪৪৩	২১	৫৩	৬	১০	৩৩	৫৯১	৩৮৬	১০	৯৯	১৫	৩	১০	৬৮
২	আগস্ট-২৫	৫২২	২৬	৩৬৭	৪৪	৪৭	৬	৯	২৩	৫২২	৩৫২	১০	১২৫	১	১	৯	২৪
৩	সেপ্টেম্বর-২৫	৬৫৩	২০	৩৫২	২৭	১৯৯	১৪	১৬	২৫	৬৫৩	৪৩৬	৯	১৬০	৩	৪	১৫	৫৫
৪	অক্টোবর-২৪	৬৮৩	২৪	৪১৯	২৫	১৩০	১৫	১৫	৫৫	৬৮৩	৪৭০	৪	১১৬	১৮	৫	১৭	৪৩
৫	নভেম্বর-২৫	৫৩২	২১	৩২৯	১৩	৯২	১৭	১৭	৪৩	৫৩২	৩৮৬	৬	৬৪	৮	১	১২	৪৪
৬	ডিসেম্বর-২৫	৪৬৯	২৭	২৭৮	২৩	৭৪	১১	১২	৪৪	৪৬৯	৩৬৫	৬	৩৯	২	১	১৭	২৮৭
৭	জানুয়ারী-২৬	৭৯০	২০	৩১৯	১২	৩৪৬	২৯	১৭	৪৭	৭৯০	৩৫১	৪	১১৯	৬	৬	১৭	১০৬
৮	ফেব্রুয়ারী-২৬	৪৯৫	২৮	২৩৪	১৩	১৭০	১৫	১৩	২২	৪৯৫	২৭৫	২	৮৩	৫	১১	১৭	৪৮
৯	মার্চ-২৬	৪৫০	৩৪	২৮১	১৩	৭৩	১১	১৭	২১	৪৫০	৩২৮	৪	৫০	২	১	১৪	১৫৮
১০	এপ্রিল-২৬	৫০৯	৩১	২৪০	৪৪	১৩৮	১৪	১৪	২৮	৫০৯	২৮৬	৪	৪১	৩	৩	১৭	১২৪
১১	মে-২৬	৭৭২	৩২	৪৩২	৩২	১৮৬	২০	১৭	৫৩	৭৭২	৪৬৬	৯	৮৩	১	২	১৭	১২৪
১২	জুন-২৬									০							
	মোট	৬৪৬৬	২৮৮	৩৬৯৪	২৬৭	১৫০	১৫৭	১৫৭	৩৯৪	৬৪৬৬	৪১০১	৬৭	১৭৯	৬৪	৪৫	১৫৭	১০৫২

5.1.4 Titas Gas (Illegal Connection) May 2026

তিতাস গ্যাস ট্রান্সমিশন এন্ড ডিস্ট্রিবিউশন পিএলসি (পেট্রোবাংলার একটি কোম্পানি) ১০৫, কাজী নজরুল ইসলাম এভিনিউ, কাওরন বাজার বা/এ, ঢাকা-১২১৫																		
বিষয়: অষ্টম গ্যাস সংযোগ বিচ্ছিন্নকরণ ও পাইপ লাইন অপসারণ কার্যক্রম সংক্রান্ত প্রতিবেদন।																		
মাসের নাম	আবিবি	স্পট সংখ্যা	সংযোগ বিচ্ছিন্নকৃত পাইপ লাইনের দৈর্ঘ্য (কি.মি.)	অপসারণকৃত পাইপের পরিমাণ (ফুট)	মোবাইল কোর্টসহ অভিযানের সংখ্যা	মোবাইল কোর্ট ব্যক্তিগত অভিযানের সংখ্যা	মোট অভিযান	জরিমানা আদায় (টাকা)	বিচ্ছিন্নকৃত আবাসিক সংযোগ সংখ্যা	বিচ্ছিন্নকৃত আবাসিক বার্নার সংখ্যা	বিচ্ছিন্নকৃত বাণিজ্যিক গ্রাহকের সংখ্যা	বিচ্ছিন্নকৃত বাণিজ্যিক বার্নার সংখ্যা	বাণিজ্যিক ঘণ্টাপ্রতি লোড	বিচ্ছিন্নকৃত শিল্প গ্রাহকের সংখ্যা	বিচ্ছিন্নকৃত শিল্প বার্নার সংখ্যা	শিল্প ঘণ্টাপ্রতি লোড	মামলা	জেল (দিন)
মে-২৬	মোটারবি-১	৭	০.০০	৪০০	২	১	৩	০	২	২৭	৫	১১	১,৮৫০	১	৩	৭৯.৫৬৫	০	০
	মোটারবি-২	৪	০.০০	৩৫০	২	০	২	২০০,০০০	২	২৩	১	৯	২০৫	৪	২২	৭.৮০০	০	০
	মোটারবি-৩	৭	০.০০	১০	১	১	২	১০০,০০০	২১	৮৩	১	০	৯৪২	০	০	০	০	০
	মোটারবি-৪	৬	০.০০	৭০০	২	০	২	০	২০১	৮০১	০	০	০	০	১৮	৮.২০০	০	০
	মোটারবি-৫	৩	০.০০	৩৬০	২	০	২	১৫০,০০০	১২৫	৩৫৮	০	০	০	১	৮	২.৯০০	০	৫
	মোটারবি-৬	১৭	০.০০	১০১৫	১	২	৩	৫০,০০০	৫৪	১৭৩	৪	৫৬	২,০৮২	৩	৩১	৮.৮৮২	৫	০
	আবিবি-জয়দেবপুর	৯	০.০০	৫০০	১	২	৩	৩৯৫,০০০	১৪৩	৪৬৯	০	০	০	০	০	০	০	০
	আবিবি-চন্ডা	১১	০.০০	১০৪০	১	৫	৬	২৮৭,০০০	৬২	২৮৭	০	০	০	০	০	০	০	০
	আবিবি-সাকরা	১৬	৫.২০	৬৬৭৭	৩	৫	৮	৭০,০০০	৫৫০	১,৭৫৮	৬	২৭	১.৮৭৫	৬	৫১	৬২,০০০	১	০
	আবিবি-নারায়ণগঞ্জ	১৮	০.০০	৪০০	৩	১	৪	১৮০,০০০	২,৩৭৭	৫৫২০	৪	১০	১,৬৪০	০	০	০	০	০
	আবিবি-সোনারগাঁও	২৫	৫.০০	৪৬৩১	৮	০	৮	৭৮,০০০	১,২০০	৩,৯২০	১	৫	৩৭৫	২১	২৯	৫৬.৮০০	০	২১
	আবিবি-বুপ্পগঞ্জ	২৩	০.০০	১৪০০	৪	০	৪	০	৫৫১	৬৫৭	১১	৪০	১,৩৭৪	১	৩	১,০০০	০	০
	আবিবি-নরসিংদী	৭	০.০০	০	৩	০	৩	২৫০,০০০	৫২২	১,৩৭২	১	৩	৯৫	১	০	৫.১৭৬	০	০
	আবিবি-ময়মনসিংহ	৪	০.০০	০	০	৪	৪	০	১১	৪২	০	০	০	০	০	০	০	০
	আবিবি-ভালুকা	১	০.০০	০	০	১	১	০	৬	১৩	০	০	০	০	০	০	০	০
	সেন্ট্রাল ডিজিলাপ	২৯	০.০০	০	০	১৮	১৮	০	১৭	১৫৮	৬	২৩	৫৮৭	০	০	০	০	০
	ডিজিলাপ-গাজীপুর	৩১	০.০০	০	০	১০	১০	০	০	০	০	০	০	৩	৩	১১.৪৬০	০	০
	ডিজিলাপ-নারায়ণগঞ্জ	২৭	০.০০	০	০	৬	৬	০	০	০	০	০	০	২	০	৪৭.৬৪৫	০	০
	ডিজিলাপ-ময়মনসিংহ	১৮	০.০০	০	০	৬	৬	০	০	০	০	০	০	০	০	০	০	০
	সেন্ট্রাল রাজশ	২৬১	০.০০	০	০	১০৭	১০৭	০	৪০	১৬৬	০	০	০	০	০	০	০	০
	রাজশ-গাজীপুর	৬১	০.০০	০	০	৩৭	৩৭	০	৭৬	৫১৯	০	০	০	০	০	০	০	০
	রাজশ-নারায়ণগঞ্জ	৩০	০.০০	০	০	২৩	২৩	০	৪৯	৩৯০	০	০	০	০	০	০	০	০
	রাজশ-ময়মনসিংহ	১৭৩	০.০০	০	০	২৭	২৭	০	২৪	৬৫	০	০	০	০	০	০	০	০
	মোট	৭৮৮	১০.২০	১৭৪৮৩	৩৩	২৫৬	২৮৯	১৭৬৩০০০	৫৮৯৩	১৬৮০১	৪০	১৮৪	১১০২৫	৪৬	১৬৮	২১২৪২৮	৬	২৬



[FREQUENTLY ASKED QUESTIONS](#) | [NATURAL HAZARDS](#)

What can I do to be prepared for an earthquake?

There are four basic steps you can take to be more prepared for an earthquake:

- Step 1:**
[Secure your space](#) by identifying hazards and securing moveable items.
- Step 2:**
[Plan to be safe](#) by creating a disaster plan and deciding how you will communicate in an emergency.
- Step 3:**
[Organize disaster supplies](#) in convenient locations.
- Step 4:**
[Minimize financial hardship](#) by organizing important documents, strengthening your property, and considering insurance.

From crisis to collaboration: pathways for resilience

All 3 professors agree: academic and scientific collaboration is the backbone of earthquake diplomacy. Universities offer a neutral ground, spaces where dialogue can continue even when politics falter. Structured partnerships anchor expertise within risk-informed decision-making, disaster preparedness, response and recovery.

Regular exchanges, joint standards, and evidence-based assessments, especially in times of calm, are essential to turn fleeting solidarity into long-term regional resilience.

Earthquake diplomacy reminds us that beneath political divides lie shared vulnerabilities. In such moments, human connection can turn common risk into collective action, transforming shared shocks into shared solutions, and uncertainty into resolve.



গণপ্রজাতন্ত্রী বাংলাদেশ সরকার

দুর্যোগ বিষয়ক স্থায়ী আদেশাবলি ২০১৯

দুর্যোগ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয়

৩.১.৪ ভূমিকম্প প্রস্তুতি ও সচেতনতা বৃদ্ধি কমিটি

ভূমিকম্পঝুঁকি ব্যবস্থাপনার জন্য ভূমিকম্প প্রস্তুতি ও সচেতনতা বৃদ্ধি কমিটির গঠন, সভা এবং দায়িত্ব ও কার্যাবলি নিম্নরূপ:

১	মন্ত্রী, দুর্ঘোণ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয়	সভাপতি
২	সচিব, দুর্ঘোণ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয়	সহ-সভাপতি
৩	সচিব, জনপ্রশাসন মন্ত্রণালয়	সদস্য
৪	সচিব, কৃষি মন্ত্রণালয়	সদস্য
৫	সচিব, অর্থ বিভাগ	সদস্য
৬	সচিব, অর্থনৈতিক সম্পর্ক বিভাগ	সদস্য
৭	সচিব, গৃহায়ন ও গণপূর্ত মন্ত্রণালয়	সদস্য
৮	সচিব, জ্বালানি ও খনিজসম্পদ বিভাগ	সদস্য
৯	সচিব, তথ্য মন্ত্রণালয়	সদস্য
১০	সচিব, পররাষ্ট্র মন্ত্রণালয়	সদস্য
১১	সচিব, প্রতিরক্ষা মন্ত্রণালয়	সদস্য
১২	সচিব, প্রাথমিক ও গণশিক্ষা মন্ত্রণালয়	সদস্য
১৩	সচিব, মাধ্যমিক ও উচ্চ শিক্ষা বিভাগ	সদস্য
১৪	সচিব, কারিগরি ও মাদ্রাসা শিক্ষা বিভাগ	সদস্য
১৫	সচিব, বিদ্যুৎ বিভাগ	সদস্য
১৬	সচিব, রেলপথ মন্ত্রণালয়	সদস্য
১৭	সচিব, সড়ক পরিবহন ও মহাসড়ক বিভাগ	সদস্য
১৮	সচিব, সেতু বিভাগ	সদস্য
১৯	সচিব, সমাজকল্যাণ মন্ত্রণালয়	সদস্য
২০	সচিব, মহিলা ও শিশু-বিষয়ক মন্ত্রণালয়	সদস্য
২১	সচিব, সুরক্ষা সেবা বিভাগ	সদস্য
২২	সচিব, জননিরাপত্তা বিভাগ	সদস্য
২৩	সচিব, স্থানীয় সরকার বিভাগ	সদস্য
২৪	সচিব, স্বাস্থ্য সেবা বিভাগ	সদস্য
২৫	সচিব, স্বাস্থ্য শিক্ষা ও পরিবার কল্যাণ বিভাগ	সদস্য
২৬	প্রতিনিধি, মন্ত্রিপরিষদ বিভাগ	সদস্য
২৭	মহাপরিচালক, দুর্ঘোণ ব্যবস্থাপনা অধিদপ্তর	সদস্য
২৮	মহাপরিচালক, স্বাস্থ্য অধিদপ্তর	সদস্য
২৯	মহাপরিচালক, বাংলাদেশ ফায়ার সার্ভিস ও সিভিল ডিফেন্স অধিদপ্তর	সদস্য
৩০	মহাপরিচালক, বাংলাদেশ ভূতাত্ত্বিক জরিপ অধিদপ্তর	সদস্য
৩১	মহাপরিচালক, এনজিও-বিষয়ক ব্যুরো	সদস্য
৩২	মহাপরিচালক, মহিলা-বিষয়ক অধিদপ্তর	সদস্য
৩৩	ব্যবস্থাপনা পরিচালক, জাতীয় প্রতিবন্ধী উন্নয়ন ফাউন্ডেশন	সদস্য
৩৪	চেয়ারম্যান, রাজধানী উন্নয়ন কর্তৃপক্ষ	সদস্য

৩৫	বিভাগীয় কমিশনার, ঢাকা	সদস্য
৩৬	বিভাগীয় কমিশনার, চট্টগ্রাম	সদস্য
৩৭	বিভাগীয় কমিশনার, রাজশাহী	সদস্য
৩৮	বিভাগীয় কমিশনার, খুলনা	সদস্য
৩৯	বিভাগীয় কমিশনার, বরিশাল	সদস্য
৪০	বিভাগীয় কমিশনার, সিলেট	সদস্য
৪১	বিভাগীয় কমিশনার, রংপুর	সদস্য
৪২	বিভাগীয় কমিশনার, ময়মনসিংহ	সদস্য
৪৩	মুখ্যসচিব (মাঠ প্রশাসন), মন্ত্রিপরিষদ বিভাগ	সদস্য
৪৪	মুখ্যসচিব (দুর্যোগ ব্যবস্থাপনা), দুর্যোগ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয়	সদস্য
৪৫	প্রধান নির্বাহী কর্মকর্তা, ঢাকা উত্তর সিটি কর্পোরেশন	সদস্য
৪৬	প্রধান নির্বাহী কর্মকর্তা, ঢাকা দক্ষিণ সিটি কর্পোরেশন	সদস্য
৪৭	ব্যবস্থাপনা পরিচালক, ঢাকা পানি সরবরাহ ও পয়োনিজ্ঞাশন কর্তৃপক্ষ (ঢাকা ওয়াসা)	সদস্য
৪৮	প্রধান প্রকৌশলী, গণপূর্ত অধিদপ্তর	সদস্য
৪৯	প্রধান প্রকৌশলী, সড়ক ও জনপথ অধিদপ্তর	সদস্য
৫০	প্রধান প্রকৌশলী, স্থানীয় সরকার প্রকৌশল অধিদপ্তর	সদস্য
৫১	প্রধান প্রকৌশলী, শিক্ষা প্রকৌশল অধিদপ্তর	সদস্য
৫২	প্রধান প্রকৌশলী, ঢাকা ইলেকট্রিক সাপ্লাই কোম্পানি লিমিটেড (DESCO)	সদস্য
৫৩	প্রধান প্রকৌশলী, ঢাকা পাওয়ার ডিস্ট্রিবিউশন কোম্পানি লিমিটেড (DPDC)	সদস্য
৫৪	পরিচালক, হাউজিং অ্যান্ড বিল্ডিং রিসার্চ ইনস্টিটিউট (HBRI)	সদস্য
৫৫	চেয়ারম্যান, দুর্যোগ বিজ্ঞান ও ব্যবস্থাপনা বিভাগ, ঢাকা বিশ্ববিদ্যালয়	সদস্য
৫৬	চেয়ারম্যান, ভূতত্ত্ব বিভাগ, ঢাকা বিশ্ববিদ্যালয়	সদস্য
৫৭	চেয়ারম্যান, রেড ক্রিসেন্ট সোসাইটি	সদস্য
৫৮	জাতীয় প্রতিবন্ধী ফোরামের প্রতিনিধি	সদস্য
৫৯	প্রতিনিধি, হযরত শাহজালাল আন্তর্জাতিক বিমানবন্দর, ঢাকা	সদস্য
৬০	প্রতিনিধি, নগর উন্নয়ন অধিদপ্তর	সদস্য
৬১	প্রতিনিধি, সিভিল ইঞ্জিনিয়ারিং বিভাগ, বাংলাদেশ প্রকৌশল বিশ্ববিদ্যালয়	সদস্য
৬২	প্রতিনিধি, ভূগোল ও পরিবেশবিদ্যা বিভাগ, চট্টগ্রাম বিশ্ববিদ্যালয়	সদস্য
৬৩	প্রতিনিধি, নগর ও আঞ্চলিক পরিকল্পনা বিভাগ, জাহাঙ্গীরনগর বিশ্ববিদ্যালয়	সদস্য
৬৪	প্রতিনিধি, সিভিল ও পরিবেশ প্রকৌশল বিভাগ, শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়	সদস্য
৬৫	জাতীয় কমিশনার, বাংলাদেশ স্কাউটস	সদস্য
৬৬	প্রতিনিধি, সশস্ত্র বাহিনী বিভাগ	সদস্য
৬৭	প্রতিনিধি, বাংলাদেশ পুলিশ	সদস্য
৬৮	প্রতিনিধি, র‍্যাপিড অ্যাকশন ব্যাটালিয়ন	সদস্য
৬৯	ব্যবস্থাপনা পরিচালক, নিউক্লিয়ার পাওয়ার প্ল্যান্ট কোম্পানি বাংলাদেশ লিমিটেড	সদস্য
৭০	পরিচালক, বাংলাদেশ আবহাওয়া অধিদপ্তর	সদস্য
৭১	উন্নয়ন সহযোগী প্রতিষ্ঠানের প্রতিনিধি (সরকার কর্তৃক মনোনীত)	সদস্য
৭২	ইউএনআরসি কর্তৃক মনোনীত প্রতিনিধি	সদস্য
৭৩	দুর্যোগ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয় কর্তৃক মনোনীত, জাতীয় পর্যায়ে কার্যক্রম রয়েছে, এরূপ বেসরকারি সংস্থার (এনজিও) ৩ (তিন) জন প্রতিনিধি	সদস্য
৭৪	দুর্যোগ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয় কর্তৃক মনোনীত, আন্তর্জাতিক পর্যায়ে কার্যক্রম রয়েছে, এরূপ আন্তর্জাতিক বেসরকারি আইএনজিওর ২ (দুই) জন প্রতিনিধি	সদস্য
৭৫	অতিরিক্ত সচিব/মুখ্যসচিব (দুর্যোগ ব্যবস্থাপনা), দুর্যোগ ব্যবস্থাপনা ও ত্রাণ মন্ত্রণালয়	সদস্য-সচিব

ভূমিকম্প-সচেতনতা বৃদ্ধি কমিটির সভা

- (১) বছরে অন্তত দুই বার সভা অনুষ্ঠিত হবে, তবে প্রয়োজন হলে কমিটি যেকোনো সময় সভা পারবে;
- (২) অন্যান্য এক-তৃতীয়াংশ সদস্যের উপস্থিতিতে সভার কোরাম হবে;
- (৩) কমিটি প্রয়োজনে এ বিষয়ে সংশ্লিষ্ট যেকোনো ব্যক্তিকে এর সদস্য হিসেবে কো-অপ্ট করে এর সভায় অংশগ্রহণের জন্য আমন্ত্রণ জানাতে পারবে।

ভূমিকম্প-সচেতনতা বৃদ্ধি কমিটির দায়িত্ব ও কার্যাবলি

- (১) ভূমিকম্প প্রভুতি ও সচেতনতামূলক কর্মসূচি পর্যালোচনা করা এবং সংশ্লিষ্ট সংস্থাগুলোর জন্য করণীয় সম্পর্কে সুপারিশ প্রদান;
- (২) ভূমিকম্পের প্রভুতি, সন্ধান ও উদ্ধার, পুনর্গঠনে সক্ষমতা বৃদ্ধির কৌশল নির্ধারণে পরামর্শ প্রদান;
- (৩) ভূমিকম্পের ঝুঁকিহ্রাস এবং সন্ধান ও উদ্ধারকাজে প্রয়োজনীয় যন্ত্রপাতি/বিশেষায়িত যানবাহন/উপকরণের তালিকা পর্যালোচনাপূর্বক সুপারিশ প্রদান;
- (৪) প্রাইমারি ও সেকেন্ডারি সাড়াদানকারী ব্যক্তি/সংস্থার সক্ষমতা উন্নয়নে পরামর্শ প্রদান;
- (৫) ভূমিকম্প-উত্তর পুনর্গঠন ও পুনর্বাসন-কার্যক্রমের জন্য করণীয় বিষয়ে প্রয়োজনীয় সুপারিশ প্রণয়ন;
- (৬) বাংলাদেশ ন্যাশনাল বিল্ডিং কোড অনুসরণ করে বাড়িঘর ও স্থাপনা নির্মাণে আডভোকেসি কার্যক্রম গ্রহণ;
- (৭) জরুরি সেবা দানকারী স্থাপনা ও সরবরাহ সিস্টেমগুলো দুর্যোগ-সহনশীলকরণে পরামর্শ প্রদান;
- (৮) আপদকালীন পরিকল্পনা প্রণয়ন ও মহড়া আয়োজনের জন্য প্রয়োজনীয় পরামর্শ প্রদান;
- (৯) ভূমিকম্পঝুঁকি হ্রাস, প্রভুতি ও সাড়াদানে প্রয়োজন অনুযায়ী একাধিক উপকমিটি গঠনের সুপারিশ প্রদান;
- (১০) শিক্ষাপ্রতিষ্ঠানে ‘ভূমিকম্পের সময়ে করণীয়’ বিষয়ে সচেতনতামূলক কার্যক্রম গ্রহণের জন্য সুপারিশ প্রদান;
- (১১) বাসাবাড়ি, কর্মক্ষেত্র, ব্যবসায়কেন্দ্র ও শিক্ষাপ্রতিষ্ঠানে নিয়মিত ভূমিকম্প মহড়াসহ প্রভুতিমূলক কার্যক্রম গ্রহণে সুপারিশ প্রদান;
- (১২) জরুরি মুহূর্তে যোগাযোগের জন্য শিক্ষাপ্রতিষ্ঠান, অফিস আদালত, ব্যবসায়কেন্দ্রের দায়িত্বপ্রাপ্ত ব্যক্তির মোবাইল ফোন ও ইমেইল আইডিসহ তথ্যভান্ডার প্রণয়ন, হালনাগাদকরণ ও প্রচারে সুপারিশ প্রদান;
- (১৩) ভূমিকম্প বিষয়ে বিদেশি অভিজ্ঞতা সংগ্রহ ও কাজে লাগানো;
- (১৪) ভবন নির্মাণের ক্ষেত্রে বাংলাদেশ ন্যাশনাল বিল্ডিং কোড বাধ্যতামূলক অনুসরণ;
- (১৫) বিভিন্ন মন্ত্রণালয়ের প্রতিনিধির সমন্বয়ে নিয়মিত ভ্রাম্যমাণ আদালত পরিচালনা;
- (১৬) বহুতল ভবন নির্মাণের ক্ষেত্রে জরুরি নির্গমন ব্যবস্থা শতভাগ নিশ্চিতকরণ;
- (১৭) প্রতি তিন মাস অন্তর বা নির্দিষ্ট সময় ব্যবধানে বিভিন্ন শিক্ষাপ্রতিষ্ঠান, অফিস আদালতে দুর্ঘটনা মোকাবিলায় জন্য মহড়ার আয়োজন।

অধ্যায় ৭: জাতীয় জরুরি-দুর্যোগ সমন্বয়	৩০৯
৭.১ জাতীয় জরুরি-দুর্যোগ পরিচালন কেন্দ্র (এনইওসি)	৩০৯
৭.২ সিভিল-মিলিটারি কো-অর্ডিনেশন	৩১০
৭.৩ ক্রাস্টার পদ্ধতিতে মানবিক সাড়াদান কার্যক্রম পরিচালনা	৩১০

৫.২.৩৭.১ মাধ্যমিক ও উচ্চ শিক্ষা অধিদপ্তর

মাধ্যমিক ও উচ্চ মাধ্যমিক শিক্ষা অধিদপ্তর দুর্যোগ মোকাবিলায় নিম্নলিখিত দায়িত্ব পালন করবে:

ঝুঁকিহ্রাস কার্যক্রম

- ক) নতুন বিদ্যালয় ভবন ও অন্যান্য শিক্ষাপ্রতিষ্ঠানের নকশা প্রণয়নে আপদ ও ঝুঁকি-মানচিত্র ব্যবহার করা;
- খ) ঝুঁকিপ্রবণ শিক্ষাপ্রতিষ্ঠান চিহ্নিত করা ও প্রয়োজনে ভেঞ্চে ফেলা বা রেট্রোফিটিং করার উদ্যোগ গ্রহণ;
- গ) সকল শিক্ষাপ্রতিষ্ঠানে বছরে নিয়মিত দুর্যোগে নিরাপত্তা, স্থানান্তর এবং প্রাথমিক চিকিৎসার মহড়া আয়োজন করা;
- ঘ) সকল শিক্ষাপ্রতিষ্ঠানে নিয়মিত ভূমিকম্প মহড়ার আয়োজন করা;
- ঙ) শিক্ষকদের জন্য দুর্যোগঝুঁকি ব্যবস্থাপনা-বিষয়ক প্রশিক্ষণ প্রদানের ব্যবস্থা গ্রহণ এবং শিক্ষক প্রশিক্ষণ কলেজের মাধ্যমে এ প্রশিক্ষণ কার্যক্রম মূলধারায় সম্পৃক্তকরণ;

দুর্যোগ ঝুঁকি মোকাবেলায় রাজউকের করণীয় ও গৃহীত পদক্ষেপ

দুর্যোগ ব্যবস্থাপনা স্থায়ী আদেশাবলী ২০১৯ এর অনুচ্ছেদ ৫.২.২০.৩ অনুযায়ী রাজউকের কার্যক্রম সুনির্দিষ্ট করা হয়েছে।

(খ) রাজধানী উন্নয়ন কর্তৃপক্ষ (রাজউক), রাজশাহী উন্নয়ন কর্তৃপক্ষ (আরডিএ), খুলনা উন্নয়ন কর্তৃপক্ষ (কেডিএ), চট্টগ্রাম উন্নয়ন কর্তৃপক্ষ (সিডিএ), গাজীপুর উন্নয়ন কর্তৃপক্ষ (জিডিএ) নিম্নলিখিত দায়িত্ব পালন করবে:

১	সম্ভাব্য ঝুঁকিপূর্ণ এলাকা চিহ্নিত করার লক্ষ্যে ভূমিকম্প ও অন্যান্য দুর্যোগঝুঁকি মানচিত্র অনুসরণ করা;	ডিটেইল্ড এরিয়া প্লানের সাথে Risk Sensitive Urban Expansion Plan, Seismic Microzonation Map এবং Land Use Planning এর সমীক্ষা করা হলে এটি বাস্তবায়ন করা যাবে।
২	ঝুঁকি অত্যাধিকারিত নগর পরিকল্পনা প্রণয়ন করা;	Dhaka Metropolitan Region (DMR) -এর মাটির ভারবহন ক্ষমতা নির্ণয়পূর্বক Risk Sensitive Urban Expansion Plan, Seismic Microzonation Map এবং Land Use Planning প্রণয়ন করা হয়েছে।
৩	বাংলাদেশ ন্যাশনাল বিল্ডিং কোড (BNBC-2020) অনুসরণ করে ঝুঁকিপ্রবণ এলাকায় ভবন নির্মাণ করার ক্ষেত্রে অনুমোদিত নকশা ও নির্মাণ-নির্দেশনা মেনে চলার জন্য প্রয়োজনীয় উদ্যোগ গ্রহণ;	বাংলাদেশ ন্যাশনাল বিল্ডিং কোড পরিপালন ও বাস্তবায়ন এবং প্রচার প্রচারণার কৌশলপত্র প্রণয়ন করা হয়েছে। এছাড়াও নকশা অনুমোদনকালীন সময়ে প্রণীত স্ট্রিকচারাল ডিজাইন বিএনবিসি'র সাথে সামঞ্জস্যপূর্ণ কিনা তা নিরীক্ষা করার জন্য চেকলিস্ট প্রণয়ন করা হয়েছে।
৪	সকল ভবন এবং কাঠামোর ঝুঁকি নিরূপণে জরিপ পরিচালনা করা এবং ঝুঁকিপূর্ণ ভবনের তালিকা তৈরি করা;	রাজউক আওতাধীন এলাকায় (গাজীপুর অংশ সহ) স্থাপনার সংখ্যা প্রায় ২১,৪৭,২১৯ টি। তন্মধ্যে আরসিসি স্ট্রাকচার প্রায় ৫,১৩,৫০৭ টি (২৪%) ও ম্যাসনারী বিল্ডিং ৮,৭৭,৬২০ টি (৪১%)। রাজউক ২৭০৫ টি শিক্ষা প্রতিষ্ঠান, ২০৭ টি হাসপাতাল, ৩৬ টি থানা ও ৩০৭ টি অন্যান্য ভবনসহ মোট ৩,২৫২ টি ভবনের Rapid Visual Assessment (RVA) সম্পন্ন করেছে। উক্ত ভবনসমূহের মধ্য থেকে ৫৬৮ টি শিক্ষা প্রতিষ্ঠান, ৫ টি হাসপাতাল, ৩ টি ফায়ার স্টেশন ও অন্যান্য ৩ টি সহ মোট ৫৭৯ টি ভবনের Preliminary Engineering Assessment (PEA) সম্পন্ন করেছে। উক্ত ভবনসমূহের মধ্য থেকে ২২৯ টি ভবনের Detailed Engineering Assessment (DEA) সম্পন্ন করেছে।

In Bangladesh Geodash there are 728 layers and 53 organizations (owner of data). Since it is a public website, everyone can register by creating an account, download and view data. However, for downloading some data we need permission from the owner of data (<https://geodash.gov.bd/layers/?limit=100&offset=0>).

For example, data created by WASA and Department of Disaster Management (DDM) cannot be downloaded and for downloading the data, permission is required (https://geodash.gov.bd/layers/geonode:building_dma_1007).

- 2- Are the community's building infrastructure and basic services resilient to disaster (including being located in safe areas, using hazard-resistant construction methods and structural mitigation measures)?
- **Level 1:** Majority of housing, critical infrastructure and basic services in the community are located in unsafe areas and no hazard mitigation measures are being taken.
 - **Level 2:** Majority of housing, critical infrastructure and basic services in the community are located in unsafe areas but some hazard mitigation measures are being taken.
 - **Level 3:** Some housing, critical infrastructure and basic services in the community are located in unsafe areas and some hazard mitigation measures are being taken.
 - **Level 4:** Majority of housing, critical infrastructure and basic services in the community are located in safe areas and some hazard mitigation measures are being taken for infrastructure in unsafe locations.
 - **Level 5:** Majority of housing, critical infrastructure and basic services in the community are located in safe areas and those in unsafe locations are adequately protected, through hazard resistant construction and structural mitigation measures.

রাজউক আওতাধীন এলাকায় (গাজীপুর অংশ সহ) স্থাপনার সংখ্যা প্রায় ২১,৪৭,২১৯ টি। তন্মধ্যে আরসিসি স্ট্রাকচার প্রায় ৫,১৩,৫০৭ টি (২৪%) ও ম্যাসনারী বিল্ডিং ৮,৭৭,৬২০ টি (৪১%)।

রাজউক ২৭০৫ টি শিক্ষা প্রতিষ্ঠান, ২০৭ টি হাসপাতাল, ৩৬ টি থানা ও ৩০৭ টি অন্যান্য ভবনসহ মোট ৩,২৫২ টি ভবনের Rapid Visual Assessment (RVA) সম্পন্ন করেছে। উক্ত ভবনসমূহের মধ্য থেকে ৫৬৮ টি শিক্ষা প্রতিষ্ঠান, ৫ টি হাসপাতাল, ৩ টি ফায়ার স্টেশন ও অন্যান্য ৩ টি সহ মোট ৫৭৯ টি ভবনের Preliminary Engineering Assessment (PEA) সম্পন্ন করেছে।

উক্ত ভবনসমূহের মধ্য থেকে ২২৯ টি ভবনের Detailed Engineering Assessment (DEA) সম্পন্ন করেছে। ২২৯ টি ভবনের মধ্যে ৯৪ টি স্কুল, ৭৮ টি কলেজ, ১১ টি বিশ্ববিদ্যালয় ও ৪ টি হাসপাতাল ভবনসহ মোট ১৮৭ টি ভবন রেট্রোফিটিং এর সুপারিশ করেছে। ৪২ টি ভবন আপসারণের সুপারিশ করেছে।

রাজউক ভবনের ঝুঁকি নিরূপণের জন্য প্রয়োজনীয় সরঞ্জাম সংগ্রহ করেছে। যন্ত্রপাতির এক্সেসরিজ সংগ্রহ ও কর্মকর্তাদের (বিল্ডিং ইন্সপেক্টরদের) সংক্ষিপ্ত প্রশিক্ষণ প্রদান করে Rapid Visual Assessment (RVA) এর

Gas Sector Efficiency Improvement & Carbon Abatement Project (Installation of Smart Prepaid Gas meter for TGTDC)

বিশ্বব্যাংক, গণপ্রজাতন্ত্রী বাংলাদেশ সরকার, ও তিতাল গ্যাস ট্রান্সমিশন এন্ড ডিস্ট্রিবিউশন পিএলসি এর অর্থায়নে বাস্তবায়নাধীন “Gas Sector Efficiency Improvement & Carbon Abatement Project (Installation of Smart Prepaid Gas Meter for TGTDC)” শীর্ষক প্রকল্পের আওতায় কোম্পানির মেট্রো ঢাকা বিপণন ডিভিশন (উত্তর), আঞ্চলিক বিপণন ডিভিশন (গাজীপুর) এবং আঞ্চলিক বিপণন ডিভিশন (ময়মনসিংহ) এলাকার আবাসিক শ্রেণির মিটারবিহীন গ্রাহকদের আভিলাষ মোট ১১ লক্ষ স্মার্ট প্রিপেইড গ্যাস মিটার স্থাপন করা হবে। প্রকল্পটি জানুয়ারি-২০২৪ হতে ডিসেম্বর-২০২৮ মেয়াদে বাস্তবায়নের লক্ষ্যে গত ৩১/১০/২০২৩ তারিখে অনুষ্ঠিত জাতীয় অর্থনৈতিক পরিষদের নির্বাহী কমিটি (একনেক)-এর সভায় অনুমোদিত হয়। প্রকল্পের মোট প্রাক্কলিত ব্যয় ৩৫৪২.৭১ কোটি টাকা। প্রকল্পের মেয়াদ ডিসেম্বর-২০২৮ এর মধ্যে ১১ লক্ষ স্মার্ট প্রিপেইড গ্যাস মিটার স্থাপন সম্পন্ন হবে বলে আশা করা যাচ্ছে।



জাপানিজ ইকোনমিক জোন, আড়াইহাজার, দারাবগঞ্জ, এলাকার সিজিএস নির্মাণ



প্রিপেইড গ্যাস মিটার স্থাপন প্রকল্প

Smart Metering Energy Efficiency Improvement Project (Installation of Prepaid Gas Meter for TGTDC)

এশিয় উন্নয়ন ব্যাংক (ADB), গণপ্রজাতন্ত্রী বাংলাদেশ সরকার ও তিতাল গ্যাস ট্রান্সমিশন এন্ড ডিস্ট্রিবিউশন পিএলসি (টিজিটিডিপিএলসি) এর অর্থায়নে বাস্তবায়নাধীন “Smart Metering Energy Efficiency Improvement Project (Installation of Prepaid Gas Meter for TGTDC)” শীর্ষক প্রকল্পের আওতায় কোম্পানির মেট্রো ঢাকা বিপণন ডিভিশন (দক্ষিণ) এবং আঞ্চলিক বিপণন ডিভিশন (দারাবগঞ্জ) এলাকার গৃহস্থালী শ্রেণির মিটারবিহীন গ্রাহকদের আভিলাষ ৬.৫ লক্ষ স্মার্ট প্রিপেইড গ্যাস মিটার স্থাপন করা হবে। প্রকল্পটি জানুয়ারি-২০২৪ হতে ডিসেম্বর-২০২৮ মেয়াদে বাস্তবায়নের লক্ষ্যে গত ৩১/১০/২০২৩ তারিখে অনুষ্ঠিত জাতীয় অর্থনৈতিক পরিষদের নির্বাহী কমিটি (একনেক)-এর সভায় অনুমোদিত হয়। প্রকল্পের মোট প্রাক্কলিত ব্যয় ৩৫৪২.৭১ কোটি টাকা। প্রকল্পের মেয়াদ ডিসেম্বর-২০২৮ এর মধ্যে ১১ লক্ষ স্মার্ট প্রিপেইড গ্যাস মিটার স্থাপন সম্পন্ন হবে বলে আশা করা যাচ্ছে।

ভিত্তাস গ্যাস ট্রান্সমিশন এন্ড ডিস্ট্রিবিউশন পিএলসি
১০৫, কাজী নজরুল ইসলাম এভিনিউ, কাওরান বাজার বা/এ, ঢাকা-১২১৫।

অবৈধ গ্যাস সংযোগ বিচ্ছিন্নকরণ ও পাইপ লাইন অপসারণ কার্যক্রম সংক্রান্ত মাসিক প্রতিবেদন (মে-২০২৬)
(অবৈধ বিতরণ পাইপ লাইন ও সংযোগ বিচ্ছিন্ন)

ক্র.সং	এলাকার নাম (জেলা)	চিহ্নিত অবৈধ গ্রাহক শ্রেণী	অবৈধ লাইন বিচ্ছিন্নকরণের সংখ্যা		
			কার্যের সংখ্যা	লম্বা (কি.মি.)	
১	২	৩	৪	৫	৬
১।	ঢাকা	আবাসিক	১৭৮৯	০.০০	মে ২০২৬ মাসে মোট ৭৮৮ টি স্পটে ২৮৯ টি অভিযান পরিচালনা করা হয়েছে (মোবাইল কোর্ট-৩৩ টি, রাজস্ব ও বিদ্যে বিভাগের সৈনিক অভিযান-২১৬ টি, কেন্দ্রীয় ডিজিটাল কর্তৃক অভিযান-১৮টি এবং আঞ্চলিক ডিজিটাল কর্তৃক অভিযান-২২টি)।
২।	গাজীপুর	আবাসিক	৩০০৩	৫.২০	
৩।	নারায়ণগঞ্জ	আবাসিক	১১,৮৫৯	৫.০০	
৫।	মুন্সীগঞ্জ	আবাসিক	-	০	
৬।	নারসিংদী	আবাসিক	-	০.০০	
৭।	ময়মনসিংহ	আবাসিক	১২০	০	
মোট			১৬,৮০১	১০.২০	

মে ২০২৬ মাসে মোবাইল কোর্টের ও মোবাইল কোর্ট দ্বারা উচ্ছেদ অভিযান পরিচালনা করে অবৈধ ১৬,৮০১ টি আবাসিক কার্গারের পাশাপাশি অবৈধ কার্যকরণ এর জন্য ৪৬ টি শিল্প ও ৪০ টি বাণিজ্যিক সংযোগ বিচ্ছিন্ন করা হয়েছে। ৩৩ টি মোবাইল কোর্ট কর্তৃক আদায়কৃত পরিমাণের পরিমাণ ১৭,৬০,০০০/- এবং সৈনিক প্রায় ০৬ (ছয়) মিলিয়ন ঘনফুট গ্যাস সাবায় হয়েছে।

স্বাক্ষর
২৭/০৬/২০২৬
উপসহকারী (ডাটাবেস ও ইনস্টলেশন শাখা)
প্রকৌ. মো. শাহজাহান হাফিজ
উপসহকারী
ডাটাবেস ও ইনস্টলেশন শাখা
ডাটাবেস ও ইনস্টলেশন বিভাগ
ভিত্তাস গ্যাস টি এন্ড ডি পিএলসি

স্বাক্ষর
২৭/০৬/২০২৬
ব্যবস্থাপক (ডাটাবেস ও ইনস্টলেশন শাখা)
প্রকৌ. মো. মাহমুদ হাসান
ব্যবস্থাপক
ডাটাবেস ও ইনস্টলেশন শাখা
ডাটাবেস ও ইনস্টলেশন বিভাগ
ভিত্তাস গ্যাস টি এন্ড ডি পিএলসি

উপসহকারী (ডাটাবেস ও ইনস্টলেশন)

স্বাক্ষর
২৭/০৬/২০২৬
উপসহকারী (ডিজিটাল)

প্রকৌ. মো. শাহজাহান ফারাজী
উপসহকারী
ডিজিটাল বিভাগ
ভিত্তাস গ্যাস টি এন্ড ডি পিএলসি

অবৈধ গ্যাস সংযোগ বিচ্ছিন্নকরণ/পাইপলাইন অপসারণ:

গ্যাস কারচুপি ও অবৈধ গ্যাস ব্যবহার রোধকল্পে কেন্দ্রীয় ডিজিটাল বিভাগ, আঞ্চলিক ডিজিটাল বিভাগ ও রাজস্ব জোনসমূহ কর্তৃক শিল্প, সিএনজি, ক্যাপটিভ পাওয়ার, বাণিজ্যিক ও আবাসিক শ্রেণীর গ্রাহক আভিনা নিয়মিত পরিদর্শন এবং পরিদর্শনকালে কোন অসঙ্গতি পাওয়া গেলে তাৎক্ষণিকভাবে গ্যাস সংযোগ বিচ্ছিন্ন করা হয়। ২০২৪-২৫ অর্থবছরে ২০৬টি মোবাইল কোর্ট এবং ৩৭২টি কোম্পানির নিজস্ব অভিযানে উৎসমুখ চিহ্নিত ১২৬৭টি পয়েন্টে প্রায় ২১০ কিলোমিটার অবৈধ পাইপলাইনের গ্যাস সরবরাহ বন্ধ করা হয়েছে এবং এতে প্রায় ১,০৬,৪৫৯ টি বার্নারের অবৈধ গ্যাস সংযোগ বিচ্ছিন্ন হয়েছে। এছাড়া, পরিদর্শনকালে গ্রাহক আভিনায় বিভিন্ন ধরনের অসঙ্গতির কারণে আলোচ্য অর্থবছরে ৩৪১টি শিল্প, ৭৩টি ক্যাপটিভ, ১৫১টি বাণিজ্যিক ও ১১টি সিএনজি গ্রাহকের সংযোগ বিচ্ছিন্ন করা হয়।